

General Aptitude & Reasoning — ISRO CSE Revision Notes

Concise theory + multiple solving methods + tricks/shortcuts + trap questions, progressing Basic to Hard, compressed from the KnowledgeGate General Aptitude & Reasoning module.

Part A — Quantitative Aptitude

- A1. Number System
- A2. HCF & LCM
- A3. Divisibility Rules
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- A6. Average
- A7. Profit and Loss
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- A10. Speed, Time & Distance (Trains, Boats & Streams)
- A11. Mixture & Alligation
- A12. Data Interpretation
- A13. Algebra (Simplification & Identities)
- A14. Probability
- A15. Permutation & Combination
- A16. Quant Miscellaneous: Venn Diagrams, Age Problems & Partnership
- A17. Sequence & Series, Statistics, Surds & Indices
- A18. Mensuration & Geometry
- A19. Speed Mathematics (Fast Calculation Tricks)

A1. Number System

A1.1 Classification of Numbers [Level: Basic]

- **Natural numbers (N):** 1, 2, 3, ... **Whole numbers (W):** 0, 1, 2, 3, ... **Integers (Z):** ..., -2, -1, 0, 1, 2, ...
- **Rational numbers:** expressible as p/q ($q \neq 0$); includes terminating and repeating decimals. **Irrational numbers:** non-terminating, non-repeating decimals ($\sqrt{2}$, π).
- **Prime number:** exactly 2 factors (1 and itself). **2 is the only even prime** — a classic recall trap.
- **Composite number:** more than 2 factors. **1 is neither prime nor composite.**
- **Co-prime (relatively prime) numbers:** pair with HCF = 1 (need not themselves be prime, e.g., 8 and 15 are co-prime).

A1.2 Factors, Multiples & Prime Factorization [Level: Basic]

- **Factor** of N: a number that divides N exactly (remainder 0). **Multiple** of N: $N \times k$ for integer k.
- **Prime factorization:** express N as a product of primes, e.g., $360 = 2^3 \times 3^2 \times 5^1$.

If $N = p_1^{a_1} \times p_2^{a_2} \times \dots \times p_k^{a_k}$ (prime factorization), then:

$$\text{Number of factors of } N = (a_1+1)(a_2+1)\dots(a_k+1)$$

$$\text{Sum of factors of } N = [(p_1^{a_1+1} - 1)/(p_1 - 1)] \times [(p_2^{a_2+1} - 1)/(p_2 - 1)] \times \dots$$

- Worked example: $N = 360 = 2^3 \times 3^2 \times 5^1$. Number of factors = $(3+1)(2+1)(1+1) = 4 \times 3 \times 2 = \mathbf{24}$.
Sum of factors = $[(2^4-1)/(2-1)] \times [(3^3-1)/(3-1)] \times [(5^2-1)/(5-1)] = 15 \times 13 \times 6 = \mathbf{1170}$.

A1.3 Odd & Even Factors of N [Level: Basic-Medium]

- **TRICK:** To count only the **odd** factors of N, first strip out ALL powers of 2 from the factorization, then apply the factor-count formula to what remains.
- **Even factors = Total factors – Odd factors.**
- Worked example: $N = 360 = 2^3 \times 3^2 \times 5$. Odd part = $3^2 \times 5^1 \rightarrow$ odd factors = $(2+1)(1+1) = \mathbf{6}$. Even factors = $24 - 6 = \mathbf{18}$.
- **TRAP:** Students often try to directly formula-ize even factors instead of using the subtraction shortcut — the subtraction method is always faster and less error-prone.

A1.4 Factors of N Divisible by a Given Number Q [Level: Medium]

- **Ways to solve:**
 1. **Direct method:** List/verify manually for small N (slow, error-prone for exams).
 2. **Shortcut method:** If Q divides N, write $N = Q \times M$. Every factor of N divisible by Q has the form $Q \times (\text{factor of } M)$. So the count = number of factors of $M = N/Q$'s factorization factor-count.
- Worked example: Find factors of 360 divisible by 12. $M = 360/12 = 30 = 2 \times 3 \times 5$. Factors of M = $(1+1)(1+1)(1+1) = \mathbf{8}$. So 8 factors of 360 are divisible by 12.

A1.5 Concept of Remainders [Level: Medium]

$$\begin{aligned} \text{Remainder}(a + b, n) &= [\text{Remainder}(a, n) + \text{Remainder}(b, n)] \bmod n \\ \text{Remainder}(a \times b, n) &= [\text{Remainder}(a, n) \times \text{Remainder}(b, n)] \bmod n \\ \text{Remainder}(a^b, n) &= [\text{Remainder}(a, n)]^b \bmod n \quad (\text{reduce base first, then apply cyclicity/mod rules}) \end{aligned}$$

- **TRICK:** When dealing with a big product/power's remainder, ALWAYS reduce each individual number to its remainder mod n FIRST, then multiply/exponentiate — never compute the massive product first.
- **Negative remainder trick:** if a direct remainder calculation gives a value close to n, check whether expressing it as (remainder – n) [i.e., a negative/complementary remainder] simplifies a later multiplication — e.g., remainder of $49 \bmod 50 = 49 \equiv -1 \pmod{50}$ is far easier to raise to a power than 49.

A1.6 Successive Division — Short Trick [Level: Medium-Hard]

- **Successive division:** a number is divided by d_1 , the QUOTIENT of that division is divided by d_2 , that quotient by d_3 , and so on.
- **Reverse (short-trick) construction:** To build the SMALLEST number giving a chain of successive remainders r_1, r_2, r_3 for divisors d_1, d_2, d_3 : work backward — start from the last remainder, multiply up and add the next remainder at each stage.

Working backward: $x_2 = (d_3 \times r\text{-chain-so-far}) + r_2 \dots$ continue until the original dividend is rebuilt.

- Worked example: A number, when divided by 5, gives quotient Q_1 and remainder 3. Q_1 divided by 4 gives quotient Q_2 and remainder 2. Q_2 divided by 3 gives quotient 1 and remainder 0.
Build backward: $Q_2 = 3 \times 1 + 0 = 3$. $Q_1 = 4 \times 3 + 2 = 14$. Number = $5 \times 14 + 3 = \mathbf{73}$.

A1.7 Unit Digit / Last Digit of an Expression [Level: Medium-Hard]

- **Cyclicity of unit digits** — how many distinct unit digits a power of each base digit cycles through:

$$\begin{aligned} \text{Cyclicity 1 (always same digit):} & \quad 0, 1, 5, 6 \\ \text{Cyclicity 2 (2 alternating digits):} & \quad 4 (4, 6, 4, 6 \dots), 9 (9, 1, 9, 1 \dots) \\ \text{Cyclicity 4 (full 4-cycle):} & \quad 2 (2, 4, 8, 6), 3 (3, 9, 7, 1), 7 (7, 9, 3, 1), 8 (8, 4, 2, 6) \end{aligned}$$

- **Ways to solve a^b unit digit:**
 1. **Full-cycle method:** find unit digit's cyclicity length L, compute effective exponent = $b \bmod L$ (if remainder is 0, use L instead of 0), then read off that position in the cycle.
 2. **Shortcut for base ending in 4 or 9 only** (cyclicity 2): unit digit = (base's last digit)ⁿ where the check is simply whether the exponent is odd or even — $4^{\text{odd}} \rightarrow 4$, $4^{\text{even}} \rightarrow 6$; $9^{\text{odd}} \rightarrow 9$, $9^{\text{even}} \rightarrow 1$.
- Worked example: unit digit of 7^{94} . Cyclicity of 7 is 4 (7, 9, 3, 1). $94 \bmod 4 = 2 \rightarrow$ 2nd position in the cycle (7, 9, 3, 1) $\rightarrow$ unit digit = **9**.
- **TRAP:** for a SUM/product of several terms, find the unit digit of EACH term first via cyclicity, THEN add/multiply those unit digits together and take the unit digit of that result — do not average or misapply cyclicity to the combined expression directly.

A1.8 Perfect Squares & Perfect Cubes [Level: Hard]

- **Properties of perfect squares:** never end in 2, 3, 7, or 8 (only possible unit digits: 0,1,4,5,6,9). A perfect square has an **odd number of total factors** (TRAP: this is the reverse-recognizable test — if a number has an odd factor count, it MUST be a perfect square, since factors pair up (d, N/d) except when $d=\sqrt{N}$).
- **Finding factors of N that are perfect squares:** in the prime factorization of N, a factor is a perfect square only if EVERY prime's exponent used is even (0, 2, 4, ... up to that prime's max exponent in N). Count = product of $(\text{floor}(a_i/2)+1)$ over all primes.
- **Finding factors of N that are perfect cubes:** analogous — every exponent used must be a multiple of 3. Count = product of $(\text{floor}(a_i/3)+1)$.
- Worked example: $N = 2^4 \times 3^6$. Perfect-square factors: exponents of 2 usable = {0,2,4} → 3 choices; exponents of 3 usable = {0,2,4,6} → 4 choices → **12** perfect-square factors. Perfect-cube factors: 2's usable exponents {0,3} → 2 choices; 3's usable {0,3,6} → 3 choices → **6** perfect-cube factors.

Questions

Q1. [Easy] How many factors does 72 have?

A. 10 B. 12 C. 14 D. 16

Answer: B. **Why:** $72 = 2^3 \times 3^2$. Factor count = $(3+1)(2+1) = 12$.

Q2. [Easy] What is the unit digit of 3^{45} ?

A. 1 B. 3 C. 7 D. 9

Answer: B. **Why:** Cyclicity of 3 = 4 (3,9,7,1). $45 \bmod 4 = 1 \rightarrow$ 1st position = **3**.

Q3. [Medium] How many odd factors does 240 have?

A. 2 B. 4 C. 6 D. 8

Answer: B. **Why:** $240 = 2^4 \times 3 \times 5$. Odd part = $3 \times 5 \rightarrow$ factor count = $(1+1)(1+1) = 4$.

Q4. [Medium] [Trap] How many EVEN factors does 240 have?

A. 16 B. 18 C. 20 D. 12

Answer: A. **Why:** Total factors of 240 = $(4+1)(1+1)(1+1) = 20$. Odd factors = 4 (from Q3). Even = $20-4 = 16$.

Q5. [Medium] Find the number of factors of 900 that are divisible by 15.

A. 6 B. 9 C. 12 D. 15

Answer: C. **Why:** $900 = 2^2 \times 3^2 \times 5^2$. $M = 900/15 = 60 = 2^2 \times 3 \times 5$. Factor count of M = $(2+1)(1+1)(1+1) = 12$.

Q6. [Medium] [Trick] Find the unit digit of $17^{23} \times 13^{15}$.

A. 1 B. 3 C. 7 D. 9

Answer: B. **Why:** Unit digit of $17^{23} \rightarrow$ base unit 7, cyclicity 4 (7,9,3,1); $23 \bmod 4 = 3 \rightarrow$ **3**. Unit digit of $13^{15} \rightarrow$ base unit 3, cyclicity 4 (3,9,7,1); $15 \bmod 4 = 3 \rightarrow$ **7**. Product unit digit = unit digit of $3 \times 7 = 21 \rightarrow$ **1**. **Answer: A (1).**

Q7. [Hard] A number when divided successively by 4 and then 5 gives remainders 2 and 3 respectively, with a final quotient of 6. Find the number.

A. 126 B. 134 C. 146 D. 152

Answer: B. **Why:** Working backward: intermediate quotient = $5 \times 6 + 3 = 33$. Number = $4 \times 33 + 2 = 134$.

Q8. [Hard] How many perfect-square factors does $2^6 \times 3^4$ have?

A. 8 B. 10 C. 12 D. 15

Answer: C. **Why:** Usable exponents of 2: {0,2,4,6} → 4 choices. Usable exponents of 3: {0,2,4} → 3 choices. Total = $4 \times 3 = 12$.

Q9. [Hard] Find the smallest number which, when divided by 6, 15, and 20, always leaves remainder 5.

A. 55 B. 60 C. 65 D. 125

Answer: C. **Why:** $\text{LCM}(6,15,20) = 60$. Required number = $\text{LCM} + \text{remainder} = 60 + 5 = 65$.

Q10. [Hard] [Trap] A number has exactly 9 factors. What can definitely be said about it?

A. It is prime B. It is a perfect cube C. It is a perfect square D. It is even

Answer: C. **Why:** An odd total factor count always indicates a perfect square (factors pair up as (d, N/d) except the unpaired $\sqrt{N}$) — 9 is odd, so the number MUST be a perfect square (e.g., $36 = 2^2 \times 3^2$ has $(2+1)(2+1) = 9$ factors).

A2. HCF & LCM

A2.1 Definitions [Level: Basic]

- **HCF (Highest Common Factor / GCD):** the largest number that divides all given numbers exactly.
- **LCM (Lowest Common Multiple):** the smallest number that is divisible by all given numbers exactly.
- **Co-prime numbers:** HCF = 1 (need not be individually prime, e.g., 8 & 15).

A2.2 Ways to Find HCF [Level: Basic-Medium]

1. **By inspection:** for small numbers, list factors and pick the largest common one.
2. **By prime factorization:** HCF = product of common primes, each raised to the **lowest** power appearing in any of the numbers.
3. **By long division (Euclidean algorithm) — fastest for large numbers:** divide the larger number by the smaller; replace the larger with the smaller and the smaller with the remainder; repeat until remainder = 0. The last non-zero divisor is the HCF.

Euclid's algorithm example: HCF(1071, 462)

$$\begin{aligned} 1071 &= 462 \times 2 + 147 \\ 462 &= 147 \times 3 + 21 \\ 147 &= 21 \times 7 + 0 \end{aligned}$$

<- remainder 0, so HCF = 21

A2.3 Ways to Find LCM [Level: Basic-Medium]

1. **By inspection** (small numbers).
2. **By prime factorization:** LCM = product of ALL primes appearing, each raised to the **highest** power appearing in any of the numbers.
3. **By division method:** divide all numbers together by common prime factors repeatedly; LCM = product of all divisors used and the final leftover quotients.

A2.4 The HCF x LCM Property [Level: Medium]

For exactly 2 numbers: $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$

- **TRAP:** This property holds **only for exactly 2 numbers** — it does NOT generalize to 3 or more numbers (i.e., $\text{HCF}(a,b,c) \times \text{LCM}(a,b,c) \neq a \times b \times c$ in general). A very common exam mistake.
- **TRICK:** If two numbers are given to be co-prime, HCF = 1 automatically, so LCM = the plain product — skip all factorization.

A2.5 HCF & LCM of Fractions [Level: Medium]

HCF of fractions = $\text{HCF}(\text{numerators}) / \text{LCM}(\text{denominators})$
 LCM of fractions = $\text{LCM}(\text{numerators}) / \text{HCF}(\text{denominators})$

- **TRAP:** Students very commonly swap these two — remember "HCF needs LCM of denominators" and "LCM needs HCF of denominators" (each formula uses the OPPOSITE operation on the denominator).
- Worked example: Find HCF and LCM of $2/3, 8/9, 10/27$.
 Numerators: 2, 8, 10 → HCF=2, LCM=40. Denominators: 3, 9, 27 → HCF=3, LCM=27.
 HCF of fractions = $2/27$. LCM of fractions = $40/3$.

A2.6 HCF & LCM of Decimals [Level: Medium]

- **TRICK:** Make the number of decimal places equal in all numbers (padding with trailing zeros), remove the decimal point, find HCF/LCM as integers, then reinsert the decimal point at the same position in the answer.
- Worked example: HCF(1.2, 0.48). Equalize: 1.20 and 0.48 → treat as 120 and 48. $\text{HCF}(120, 48) = 24$ → reinsert 2 decimal places → **0.24**.

A2.7 HCF/LCM of More Than 2 Numbers [Level: Medium]

- Find HCF/LCM of the first two, then combine that result with the third number, and so on (pairwise reduction) — works for both prime-factorization and division methods.

A2.8 Remainder-Based Word Problems (High-Yield Patterns) [Level: Hard]

Pattern 1: Greatest number that divides x, y, z leaving the SAME remainder r in each case
 => Answer = $\text{HCF}(x-r, y-r, z-r)$

Pattern 2: Smallest number which, when divided by a, b, c leaves the SAME remainder r each time
 => Answer = $\text{LCM}(a,b,c) + r$

Pattern 3: Greatest number that divides x, y leaving DIFFERENT remainders r1, r2
 => Answer = $\text{HCF}(x-r1, y-r2)$

Pattern 4: Smallest number which, when divided by a,b,c leaves remainders that are each a fixed amount 'k' LESS than the divisor (i.e., remainder = divisor - k for every divisor)
 => Answer = $\text{LCM}(a,b,c) - k$

- TRICK to recognize Pattern 4 vs Pattern 2: check if (divisor – remainder) is the SAME constant k across all given divisor/remainder pairs — if so, use LCM – k instead of the standard LCM + r form.
- Worked example (Pattern 1): Find the greatest number that divides 62, 132, and 237 leaving remainder 2 in each case.
 $\text{HCF}(62-2, 132-2, 237-2) = \text{HCF}(60, 130, 235)$. $\text{HCF}(60, 130) = 10$; $\text{HCF}(10, 235) = 5$. **Answer: 5.**

Questions

Q1. [Easy] Find the HCF of 24 and 36.

A. 6 B. 8 C. 12 D. 18

Answer: C. Why: $24=2^3 \times 3$, $36=2^2 \times 3^2$. Common primes at lowest power: $2^2 \times 3=12$.

Q2. [Easy] Find the LCM of 12 and 18.

A. 24 B. 36 C. 48 D. 72

Answer: B. Why: $12=2^2 \times 3$, $18=2 \times 3^2$. $\text{LCM} = 2^2 \times 3^2 = 36$.

Q3. [Medium] The product of two numbers is 4032 and their HCF is 12. Find their LCM.

A. 288 B. 336 C. 360 D. 432

Answer: B. Why: $\text{HCF} \times \text{LCM} = \text{product}$ (2-number rule). $\text{LCM} = 4032/12 = 336$.

Q4. [Medium] [Trap] HCF and LCM of three numbers are 4 and 720 respectively. If two of the numbers are 24 and 48, is the product of all three necessarily $\text{HCF} \times \text{LCM}$?

A. Yes, always B. No — the $\text{HCF} \times \text{LCM} = \text{product}$ rule applies only to exactly 2 numbers C. Only if all three are co-prime D. Only if $\text{HCF}=1$

Answer: B. Why: The identity $\text{HCF} \times \text{LCM} = \text{product}$ strictly holds for 2 numbers only; for 3+ numbers there is no such direct product relationship.

Q5. [Medium] Find the HCF of $3/4$, $9/10$, and $15/16$.

A. $3/80$ B. $1/80$ C. $3/160$ D. $3/40$

Answer: A. Why: $\text{HCF}(\text{numerators } 3, 9, 15)=3$. $\text{LCM}(\text{denominators } 4, 10, 16)=80$. $\text{HCF of fractions} = 3/80$.

Q6. [Hard] Find the greatest number that divides 285 and 1249, leaving remainders 9 and 7 respectively.

A. 69 B. 92 C. 138 D. 276

Answer: C. Why: $\text{HCF}(285-9, 1249-7) = \text{HCF}(276, 1242)$. $1242 = 276 \times 4 + 138$; $276 = 138 \times 2 + 0 \rightarrow \text{HCF} = 138$.

Q7. [Hard] Find the smallest number which when divided by 6, 15, and 18 leaves remainder 5 in each case.

A. 90 B. 91 C. 95 D. 85

Answer: C. Why: $\text{LCM}(6, 15, 18)=90$. $\text{Answer} = \text{LCM} + r = 90+5 = 95$.

Q8. [Hard] Find the largest number which divides 70 and 125, leaving remainders 5 and 8 respectively.

A. 13 B. 15 C. 65 D. 117

Answer: A. Why: $\text{HCF}(70-5, 125-8) = \text{HCF}(65, 117)$. $117=65 \times 1+52$; $65=52 \times 1+13$; $52=13 \times 4+0 \rightarrow \text{HCF}=13$.

A3. Divisibility Rules

A3.1 Rules for 2 through 10 [Level: Basic]

Divisor	Rule
2	Last digit is even (0,2,4,6,8)
3	Sum of all digits divisible by 3
4	Last 2 digits (as a number) divisible by 4
5	Last digit is 0 or 5
6	Divisible by both 2 AND 3
7	See osculator method below (A3.3)
8	Last 3 digits (as a number) divisible by 8
9	Sum of all digits divisible by 9
10	Last digit is 0

A3.2 Rules for 11 through 20 [Level: Basic-Medium]

Divisor	Rule
11	(Sum of digits at odd places) – (Sum of digits at even places) = 0 or a multiple of 11
12	Divisible by both 3 AND 4
13	See osculator method below (A3.3)
14	Divisible by both 2 AND 7
15	Divisible by both 3 AND 5
16	Last 4 digits divisible by 16
17, 19	See osculator method (A3.3)
18	Divisible by both 2 AND 9
20	Last 2 digits divisible by 20 (equivalently, ends in 00,20,40,60,80)

A3.3 The Osculator Method — For 7, 13, 17, 19, and other "awkward" primes [Level: Medium-Hard]

- Every prime not dividing 10 has an **osculator** — a small multiplier used to repeatedly fold the last digit into the rest of the number until a small, checkable result remains.

General procedure (for divisor d with osculator value m):

1. Remove the last digit of the number; call it L . Call the remaining leading part R .
2. Compute $R (+ \text{ or } -) (m \times L)$, depending on whether d uses a "negative" or "positive" osculator.
3. Repeat on the new, shorter number until small enough to check divisibility directly.

- For 7:** osculator = 2, used as SUBTRACT (double the last digit, subtract from the rest).
Example: Is 1071 divisible by 7? $107 - (2 \times 1) = 105$. Repeat: $10 - (2 \times 5) = 0$. **0 is divisible by 7 → yes, 1071 is divisible by 7.**
- For 13:** osculator = 4, used as ADD (quadruple the last digit, add to the rest).
Example: Is 2145 divisible by 13? $214 + (4 \times 5) = 234$. Repeat: $23 + (4 \times 4) = 39$. $39 = 13 \times 3 \rightarrow$ **yes, divisible by 13.**
- For 17:** osculator = 5, used as SUBTRACT. **For 19:** osculator = 2, used as ADD.
- TRICK: You don't need to memorize which primes use "add" vs "subtract" osculators from theory — just try one iteration each way on a known multiple of that prime and see which one shrinks correctly; in practice 7 and 17 use subtract-type, 13 and 19 use add-type.

A3.4 Combining Divisibility Rules for Composite Divisors [Level: Hard]

- A number is divisible by a composite $d = a \times b$ **only if a and b are CO-PRIME** — then divisibility by $d \Leftrightarrow$ divisible by both a AND b .
- TRAP:** Divisibility by a and by b does **NOT** imply divisibility by $a \times b$ if a and b are **not** co-prime. Classic counter-example: 12 is divisible by both 2 and 6, but $2 \times 6 = 12$ — this accidentally works; a clearer counter-example: 36 is divisible by both 4 and 6 ($4 \times 6 = 24$), but 36 is NOT divisible by 24. Always check co-primality of the factor pair before combining.
- Correct composite breakdowns (co-prime factor pairs only): $6=2 \times 3$, $12=3 \times 4$, $14=2 \times 7$, $15=3 \times 5$, $18=2 \times 9$, $20=4 \times 5$, $21=3 \times 7$, $33=3 \times 11$, $45=9 \times 5$.

Questions

Q1. [Easy] Which of the following numbers is divisible by 4?

A. 1234 B. 1354 C. 1372 D. 1358

Answer: C. Why: Check last 2 digits: $72 \div 4 = 18$ exactly → divisible. (34, 54, 58 are not divisible by 4.)

Q2. [Easy] Is 78,912 divisible by 9?

A. Yes B. No

Answer: A. Why: Digit sum = $7+8+9+1+2 = 27$, and 27 is divisible by 9 → yes.

Q3. [Medium] Check whether 9,152 is divisible by 11.

A. Yes B. No

Answer: A. Why: Odd-place digits (from right): 2,1 → sum=3. Even-place digits: 5,9 → sum=14. Difference = $14-3 = 11$, a multiple of 11 → **yes, divisible**.

Q4. [Medium] Using the osculator method, check if 3,059 is divisible by 7.

A. Yes B. No

Answer: A. Why: $305 - (2 \times 9) = 305 - 18 = 287$. $28 - (2 \times 7) = 28 - 14 = 14$. 14 is divisible by 7 → **yes**.

Q5. [Medium] [Trap] A number is divisible by both 4 and 6. Is it necessarily divisible by 24?

A. Yes, always B. No — 4 and 6 are not co-prime, so this does not follow C. Only if the number is even D. Only if the number ends in 0

Answer: B. Why: 4 and 6 share a common factor of 2 (not co-prime), so divisibility by both does not guarantee divisibility by their product 24 — e.g., 12 is divisible by 4 and 6 but not by 24.

Q6. [Hard] Using the osculator method for 13, check whether 6,499 is divisible by 13.

A. Yes B. No

Answer: B. Why: $649 + (4 \times 9) = 685$. $68 + (4 \times 5) = 88$. $8 + (4 \times 8) = 40$. 40 is not a multiple of 13 ($13 \times 3 = 39$, remainder 1) → **not divisible**.

Q7. [Hard] Find the smallest digit that should replace * in 478 so that the number is divisible by 3.

A. 0 B. 2 C. 3 D. 4

Answer: B. Why: Digit sum so far = $4 + 7 + 8 = 19$. Need (19+) divisible by 3; since $19 \bmod 3 = 1$, we need $* \bmod 3 = 2$. The smallest digit satisfying this is **2** (0 and 1 don't work).

Q8. [Hard] A number is divisible by 15 and by 8. What is the smallest such 3-digit number?

A. 120 B. 240 C. 360 D. 480

Answer: A. Why: 15 and 8 are co-prime, so the number must be divisible by $\text{LCM}(15, 8) = 120$. Smallest 3-digit multiple of 120 is **120** itself.

A4. Percentage

(Foundational topic — nearly every other quant chapter (Profit & Loss, SI/CI, Data Interpretation) is "percentage in disguise." Master the shortcut conversions here first.)

A4.1 Basic Concept [Level: Basic]

- Percentage** = "per hundred" — a fraction with denominator 100. $x\% = x/100$.
- Percentage is a **pure number** (no units) used to compare two quantities on a common 100-point scale.

To convert fraction $\rightarrow$ percentage: $(\text{fraction}) \times 100$
 To convert percentage $\rightarrow$ fraction: $(\text{percentage}) / 100$
 $x\% \text{ of } y = (x/100) \times y = y\% \text{ of } x$ <- TRICK: $x\% \text{ of } y$ always equals $y\% \text{ of } x$; pick whichever is easier to compute

- TRAP:** "20% of 50" and "50% of 20" are NOT intuitively obviously equal to a beginner, but they both equal 10 — exploit this symmetry to simplify awkward-looking calculations by flipping the numbers.

A4.2 The Fraction–Percentage Conversion Table [Level: Basic — memorize this cold]

Speed in this chapter (and half of quant) comes from instant recall of these, not from long division.

Fraction	%	Fraction	%	Fraction	%
1/2	50%	1/6	16.67%	1/11	9.09%
1/3	33.33%	1/7	14.28%	1/12	8.33%
1/4	25%	1/8	12.5%	1/15	6.67%
1/5	20%	1/9	11.11%	1/16	6.25%
2/3	66.67%	1/10	10%	1/20	5%

- TRICK: 1/7 = 14.28% is worth memorizing to 2 decimals because multiples of 1/7 (2/7, 3/7...) just multiply 14.28 — e.g., 3/7 = 42.85%.
- Ways to find $x\%$ of a number N:
 - Direct method:** $(x \times N)/100$ — always works, sometimes slow.
 - Fraction-shortcut method:** convert $x\%$ to its fraction from the table above and multiply — e.g., 12.5% of 640 = $(1/8) \times 640 = 80$, instantly, no long multiplication.

A4.3 Percentage Increase / Decrease [Level: Basic-Medium]

New Value = Original $\times (1 + p/100)$ [increase by $p\%$]
 New Value = Original $\times (1 - p/100)$ [decrease by $p\%$]
 Percentage Change = $[(\text{New} - \text{Original})/\text{Original}] \times 100$

- TRAP:** A $p\%$ increase followed by a $p\%$ decrease does **NOT** return you to the original value:

Net effect of $+p\%$ then $-p\%$ = $-(p^2)/100$ percent (always a NET DECREASE, never zero)

Example: $+10\%$ then $-10\% \rightarrow$ net change = $-(10 \times 10)/100 = -1\%$ (not 0%). This exact trap is one of the most repeated MCQ setups in this topic.

- **Successive percentage change formula** (for any two successive changes of $a\%$ and $b\%$, with correct signs — negative for decrease):

Net % change = $a + b + (a \times b)/100$

A4.4 Product-Constant Rule (Inverse Percentage Change) [Level: Medium]

- If a product $P = A \times B$ is to be kept CONSTANT and A increases by $x\%$, then B must change by:

Required % change in B = $[-x / (100 + x)] \times 100$ (a decrease, if $x > 0$)

- Classic application: "Price of sugar increases by 25%. By what % should consumption be reduced to keep expenditure unchanged?"
 $\rightarrow$ Required decrease = $[25/(100+25)] \times 100 = 2500/125 = \mathbf{20\%}$.
- TRICK (fraction shortcut): $+25\% = +1/4$, so new price = $5/4$ of old. To keep product same, new consumption must be $4/5$ of old $\rightarrow$ a decrease of $1/5 = \mathbf{20\%}$. (Flip the fraction — numerator and denominator swap.)

A4.5 Population / Repeated Percentage Change Over Time [Level: Medium-Hard]

Value after n years of $r\%$ compounding growth = $P \times (1 + r/100)^n$
 Value after n years of $r\%$ compounding decline = $P \times (1 - r/100)^n$
 Mixed (growth then decline in different years): apply each year's factor successively, e.g.
 $P \times (1+r_1/100) \times (1-r_2/100) \times (1+r_3/100) \dots$

- TRAP:** Do NOT simply add percentages across years when the base changes each year (e.g., "increases by 10% in year 1 and 20% in year 2" is NOT a flat $+30\%$; use successive-change formula: $10+20+(10 \times 20)/100 = \mathbf{32\%}$).

A4.6 Three Classic "Confusing Question" Patterns [Level: Hard]

These three patterns are explicitly the most-repeated trap format in this chapter:

- "**A's income is $x\%$ more than B's**" vs "**B's income is $x\%$ less than A's**" — these are NOT the same percentage. If $A = 1.1x \times B$, then B is less than A by $[x/(100+x)] \times 100\%$, not by $x\%$. (Same product-constant logic as A4.4.)
- "**Reduced by $x\%$** " applied to the wrong base — always re-anchor the base after each change; percentage change is only ever relative to the value immediately before that change, never the original value two steps back.

3. **Marks/Votes percentage problems** — "a candidate got $x\%$ of votes and lost by y votes" — the TOTAL votes is the unknown; set up $\text{Total} \times (\text{winner}\% - \text{loser}\%) = \text{margin votes}$ rather than working with one candidate's votes in isolation.

Questions

(ordered Easy → Hard; [Trick] tags a question specifically designed to test the shortcut, [Trap] tags a classic misconception)

Q1. [Easy] What is 12.5% of 480?

A. 40 B. 50 C. 60 D. 65

Answer: C. **Why:** $12.5\% = 1/8$ (fraction shortcut). $480/8 = 60$.

Q2. [Easy] Express $3/8$ as a percentage.

A. 32.5% B. 35% C. 37.5% D. 40%

Answer: C. **Why:** $1/8 = 12.5\%$, so $3/8 = 3 \times 12.5\% = 37.5\%$.

Q3. [Medium] [Trap] The price of an item is increased by 20% and then decreased by 20%. What is the net percentage change?

A. No change B. 4% increase C. 4% decrease D. 40% decrease

Answer: C. **Why:** Net change = $a+b+(ab)/100 = 20-20+(20 \times -20)/100 = 0 - 4 = -4\%$. A $+p\%/-p\%$ pair always nets to a decrease of $p^2/100$ percent.

Q4. [Medium] [Trick] A number is increased by 25% and then decreased by 20%. What is the net effect?

A. No change B. 5% increase C. 5% decrease D. 10% increase

Answer: A. **Why:** Net = $25 - 20 + (25 \times -20)/100 = 5 - 5 = 0\%$. (Fraction check: $+25\% = \times 5/4$, then $-20\% = \times 4/5$; $(5/4) \times (4/5) = 1$ — no change.)

Q5. [Medium] The price of rice increases by 30%. By what percentage must a family reduce consumption to keep expenditure unchanged?

A. 20% B. 23.07% C. 30% D. 33.33%

Answer: B. **Why:** Required decrease = $[30/(100+30)] \times 100 = 3000/130 = 23.07\%$.

Q6. [Hard] [Trap] A's salary is 25% more than B's salary. By what percentage is B's salary less than A's?

A. 25% B. 20% C. 30% D. 33.33%

Answer: B. **Why:** A = 1.25B. Required decrease from A to B = $[25/(100+25)] \times 100 = 2500/125 = 20\%$ — NOT 25%. This is the single most repeated percentage trap.

Q7. [Hard] The population of a town increases by 10% in the first year and decreases by 10% in the second year. If the population after two years is 9,900, find the original population.

A. 9,800 B. 10,000 C. 10,100 D. 9,900

Answer: B. **Why:** Net two-year factor = $1.10 \times 0.90 = 0.99$ (i.e., net -1% , matching the $+10\%/-10\%$ trap from A4.3). $\text{Original} \times 0.99 = 9900 \rightarrow \text{Original} = 9900/0.99 = 10,000$.

Q8. [Hard] In an election between two candidates, the winner got 60% of the total votes and won by 4,800 votes. Find the total number of votes polled.

A. 20,000 B. 22,000 C. 24,000 D. 25,000

Answer: C. **Why:** Winner=60%, Loser=40%, margin=20% of total = 4800 $\rightarrow$ Total = $4800/0.20 = 24,000$.

Q9. [Hard] [Trick] If 60% of A = $3/4$ of B, find A:B.

A. 3:4 B. 4:5 C. 5:4 D. 5:6

Answer: C. **Why:** $0.6A = 0.75B \rightarrow A/B = 0.75/0.6 = 5/4$, so **A:B = 5:4**.

A5. Ratio & Proportion

A5.1 Basic Concept [Level: Basic]

- **Ratio** $a:b = a/b$ — a comparison of two quantities of the **same kind/unit**. A ratio has no units itself.
- **Proportion** $a:b :: c:d$ (read "a is to b as c is to d") means $a/b = c/d$.

Product of Extremes = Product of Means
 $a:b :: c:d \Leftrightarrow a \times d = b \times c$

- **TRAP:** A ratio must compare like quantities (e.g., both in kg, or both in rupees) — "ratio of 5 kg to 200 g" must first be converted to the same unit (5000g:200g = 25:1), not written as 5:200.

A5.2 Types of Ratio [Level: Basic]

Type	Meaning	Example
Duplicate ratio	$a^2:b^2$	duplicate of 2:3 is 4:9
Sub-duplicate ratio	$\sqrt{a} : \sqrt{b}$	sub-duplicate of 4:9 is 2:3
Triplicate ratio	$a^3:b^3$	triplicate of 2:3 is 8:27
Sub-triplicate ratio	$a^{1/3} : b^{1/3}$	sub-triplicate of 8:27 is 2:3
Compounded ratio	$(a \times c):(b \times d)$, combining $a:b$ and $c:d$	2:3 compounded with 4:5 = 8:15
Inverse ratio	$b:a$	inverse of 2:3 is 3:2

A5.3 Ways to Divide a Quantity in a Given Ratio [Level: Basic]

To split Total in ratio $m:n$:
 First part = $\text{Total} \times m/(m+n)$
 Second part = $\text{Total} \times n/(m+n)$

For 3+ terms, ratio $m:n:p$ of Total:
 Each part = $\text{Total} \times (\text{its own term}) / (m+n+p)$

- **TRICK:** Never solve via simultaneous equations ($x = m \times k$, $y = n \times k$, $x+y=\text{Total}$) if you can directly plug into the fraction formula above — it's faster under time pressure.

A5.4 Properties of Proportion [Level: Basic-Medium]

(Given $a:b :: c:d$, i.e., $a/b = c/d$)

Property	Rule
Invertendo	$b:a :: d:c$
Alternendo	$a:c :: b:d$
Componendo	$(a+b):b :: (c+d):d$
Dividendo	$(a-b):b :: (c-d):d$
Componendo-Dividendo	$(a+b):(a-b) :: (c+d):(c-d)$
Addendo	if $a:b::c:d::e:f...$, then $a:b :: (a+c+e):(b+d+f)$

- **Componendo-Dividendo is the single most exam-useful property** — use it whenever a problem gives an equation of the form $(x+y)/(x-y) = p/q$.

Worked Example (Componendo-Dividendo) [Level: Medium]

If $(3x+2y)/(3x-2y) = 5/1$, find $x:y$.

By componendo-dividendo (reverse direction): $3x/2y = (5+1)/(5-1) = 6/4 = 3/2 \rightarrow 6x = 6y \rightarrow x/y = 1/1$.

Recheck directly: cross-multiply original: $3x+2y = 5(3x-2y) \rightarrow 3x+2y = 15x-10y \rightarrow 12y = 12x \rightarrow x:y = 1:1$. ✓ matches.

A5.5 Direct, Inverse & Joint Variation [Level: Medium]

Direct variation: $y = kx$ (y increases as x increases, ratio y/x constant)
 Inverse variation: $y = k/x$ (y decreases as x increases, product xy constant)
 Joint variation: $z = kxy$ (z depends jointly on 2+ variables)

- **TRAP:** "Time taken is proportional to work" = **direct** variation; "Time taken is proportional to speed" is **inverse** (more speed $\rightarrow$ less time) — always check the real-world direction of the relationship, don't assume "proportional to" always means direct variation in casual wording; read carefully whether it says "directly" or "inversely."

A5.6 Combining & Converting Ratios [Level: Medium-Hard]

- **Combining A:B and B:C into A:B:C:** make the common term (B) equal via LCM, then combine.

A:B = 2:3, B:C = 4:5
 LCM(3,4) = 12. Scale A:B by 4 → 8:12. Scale B:C by 3 → 12:15.
 ⇒ A:B:C = 8:12:15

- **Finding A:C given A:B and B:C** (chain-multiplication shortcut): $A:C = (A/B \times B/C)$ — multiply the two ratios directly after matching B, same result as above without writing the full A:B:C.
- **Finding A:B:C given mA = nB = pC:**

$mA = nB = pC \Rightarrow A:B:C = 1/m : 1/n : 1/p$ (then clear fractions by multiplying by $LCM(m,n,p)$)

- **Finding ratio of two numbers from their sum and difference:** if $sum=S$ and $difference=D$, the numbers are $(S+D)/2$ and $(S-D)/2$ — their ratio follows directly.
- **Comparing two ratios (which is bigger):** cross-multiply — $a:b$ vs $c:d$, compare $a \times d$ vs $b \times c$ (larger product's ratio is bigger). Faster than converting both to decimals.
- **Finding $(mX+nY):(pX+qY)$ given $x:y$:** substitute the given $x:y$ value directly into the expression and simplify — always faster than solving for actual X,Y values.

Questions

(ordered Easy → Hard)

Q1. [Easy] Divide 720 in the ratio 5:7.

A. 300, 420 B. 320, 400 C. 350, 370 D. 310, 410

Answer: A. **Why:** Parts = $720 \times 5/12 = 300$ and $720 \times 7/12 = 420$.

Q2. [Easy] Find the duplicate ratio of 3:5.

A. 6:10 B. 9:25 C. 3:25 D. 9:5

Answer: B. **Why:** Duplicate ratio = $a^2:b^2 = 9:25$.

Q3. [Medium] If $a:b = 2:3$ and $b:c = 4:5$, find $a:b:c$.

A. 8:12:15 B. 2:3:5 C. 8:15:12 D. 6:9:15

Answer: A. **Why:** $LCM(3,4)=12$. $a:b=2:3 \rightarrow 8:12$; $b:c=4:5 \rightarrow 12:15$. Combined $a:b:c = 8:12:15$.

Q4. [Medium] [Trap] The time taken to complete a journey is said to be "proportional to distance." If speed is constant, is this direct or inverse variation, and what if instead time is related to speed for a fixed distance?

A. Both are direct variation B. Time~distance (const. speed) is direct; time~speed (const. distance) is inverse C. Both are inverse variation D. Neither is a variation relationship

Answer: B. **Why:** At constant speed, more distance needs proportionally more time (direct, $T=D/\text{speed}$ with speed fixed acts as $T=kD$). At constant distance, more speed needs proportionally less time ($T=D/\text{speed}$, D fixed, so $T \propto 1/\text{speed}$ — inverse).

Q5. [Medium] Two numbers are in the ratio 3:4. If their sum is 84, find the numbers.

A. 36, 48 B. 30, 54 C. 32, 52 D. 40, 44

Answer: A. **Why:** $84 \times 3/7 = 36$, $84 \times 4/7 = 48$.

Q6. [Hard] If $(5x+3y):(5x-3y) = 4:1$, find $x:y$.

A. 1:1 B. 3:5 C. 5:3 D. 2:3

Answer: A. **Why:** By reverse componendo-dividendo, $5x:3y = (4+1):(4-1) = 5:3$, so $x:y = (5/5):(3/3) = 1:1$. Check: $5x+3y=4(5x-3y) \rightarrow 15y=15x \rightarrow x=y$. ✓

Q7. [Hard] Find A:B:C if $2A = 3B = 4C$.

A. 6:4:3 B. 2:3:4 C. 3:4:6 D. 4:3:2

Answer: A. **Why:** $A:B:C = 1/2:1/3:1/4$. $LCM(2,3,4)=12 \rightarrow$ multiply through: 6:4:3.

Q8. [Hard] The sum of two numbers is 8 times their difference. Find the ratio of the larger to the smaller number.

A. 9:7 B. 7:5 C. 5:3 D. 3:1

Answer: A. **Why:** Let numbers be x,y ($x>y$). $x+y = 8(x-y) \rightarrow x+y=8x-8y \rightarrow 9y=7x \rightarrow x/y=9/7 \rightarrow \text{larger:smaller} = 9:7$.

A6. Average

A6.1 Basic Concept [Level: Basic]

Average = Sum of all values / Number of values
Sum = Average x Number of values

- **Average of consecutive numbers (or any AP):** no need to add every term —

Average of an AP (equally spaced numbers) = (First term + Last term) / 2

- TRICK: For n consecutive integers starting at a , average = $a + (n-1)/2$.

A6.2 Change in Average — Join / Leave / Replace [Level: Basic-Medium]

Person REPLACES another:
Change in total = (New person's value - Old person's value)
Change in average = Change in total / (number of people, unchanged)

Person JOINS the group:
New Average = (Old Total + New person's value) / (Old Count + 1)

Person LEAVES the group:
New Average = (Old Total - Leaving person's value) / (Old Count - 1)

- TRICK (replacement shortcut): if average INCREASES by d when a person is replaced, then New person's value = Old person's value + $d \times (\text{count})$.
- TRICK (join/leave shortcut): the new average shifts toward the joining/leaving value; if a new member's value equals the OLD average exactly, the average does NOT change at all.

Worked Example [Level: Medium]

The average age of 10 people increases by 2 years when one person aged 35 is replaced by a new person. Find the new person's age.

Change in total = $2 \times 10 = 20$. New person's age = $35 + 20 = 55$.

A6.3 Weighted Average [Level: Medium]

Weighted Average = $(n_1x_1 + n_2x_2 + \dots) / (n_1 + n_2 + \dots)$

- This is mathematically identical to the **Alligation** rule (see Mixture & Alligation chapter) — same formula, different visual presentation (alligation draws it as a cross/diagram).
- **TRAP:** Combined average of two groups is NOT the simple average of their two individual averages unless the group sizes are equal. You must weight by group size:

Combined Average = $(n_1 \times \text{avg}_1 + n_2 \times \text{avg}_2) / (n_1 + n_2)$

e.g., average of a class of 30 (avg 60) and a class of 20 (avg 80) is NOT 70 — it is $(30 \times 60 + 20 \times 80) / 50 = (1800 + 1600) / 50 = 68$.

A6.4 Practical Word-Problem Patterns [Level: Hard]

- **Average speed is NOT the simple average of speeds** for equal-distance trips — see Speed-Time-Distance chapter for the harmonic-mean formula $\frac{2xy}{(x+y)}$.
TRAP: only average two speeds directly (simple mean) if the TIME spent at each speed is equal, not the distance.
- **Cricket batting-average style problems:** "A batsman's average after 16 innings is A . After the 17th innings, scoring S runs, his average becomes $A+d$." Setup:

$16A + S = 17(A + d) \Rightarrow S = A + 17d$

Solve for the unknown directly from this single linear equation rather than guessing.

Questions

(ordered Easy → Hard)

Q1. [Easy] Find the average of the first 10 natural numbers.

A. 5 B. 5.5 C. 6 D. 10

Answer: B. **Why:** Average of AP $1..10 = (1+10)/2 = 5.5$.

Q2. [Easy] The average of 5 numbers is 20. If one number is excluded, the average becomes 18. Find the excluded number.

A. 24 B. 26 C. 28 D. 30

Answer: C. **Why:** Original total = 100. New total (4 numbers) = $18 \times 4 = 72$. Excluded number = $100 - 72 = 28$.

Q3. [Medium] The average weight of 8 people increases by 1.5 kg when a new person replaces one weighing 65 kg. Find the new person's weight.

A. 70 kg B. 74 kg C. 76 kg D. 77 kg

Answer: D. **Why:** Change in total = $1.5 \times 8 = 12$. New weight = $65 + 12 = 77$ kg.

Q4. [Medium] [Trap] Average marks of Section A (25 students) is 60 and Section B (15 students) is 80. Find the combined average.

A. 70 B. 67.5 C. 66 D. 68.5

Answer: B. **Why:** Combined = $(25 \times 60 + 15 \times 80) / 40 = (1500 + 1200) / 40 = 2700 / 40 = 67.5$ (NOT the simple average 70, since group sizes differ).

Q5. [Medium] A cricketer's average after 10 innings is 40. He scores 84 in the 11th innings. Find his new average.

A. 42 B. 43 C. 44 D. 45

Answer: C. **Why:** Old total = 400. New total = $400 + 84 = 484$. New average = $484 / 11 = 44$.

Q6. [Hard] The average age of a family of 5 members is 24 years today. The youngest member is 8 years old. Find the average age of the family at the time the youngest member was born (i.e., 8 years ago).

A. 18 years B. 19 years C. 20 years D. 21 years

Answer: C. Why: Present total age = $24 \times 5 = 120$. Excluding the youngest (not yet born 8 years ago), the other 4 members currently total $120 - 8 = 112$. Eight years ago each of those 4 was 8 years younger, so their total then = $112 - 4 \times 8 = 80$, over 4 people $\rightarrow$ average = $80/4 = 20$ years.

Q7. [Hard] A batsman's average after 16 innings is 36. He scores 104 in the 17th innings. By how much does his average increase?

A. 2 B. 3 C. 4 D. 5

Answer: C. Why: Using $16A + S = 17(A + d)$: $16(36) + 104 = 17(36 + d) \rightarrow 576 + 104 = 680 \rightarrow 612 + 17d = 680 \rightarrow 17d = 68 \rightarrow d = 4$.

A7. Profit and Loss

A7.1 Basic Concepts [Level: Basic]

- **CP** (Cost Price) = price at which an article is bought. **SP** (Selling Price) = price at which it is sold.
- **Profit** = $SP - CP$ (when $SP > CP$). **Loss** = $CP - SP$ (when $CP > SP$).

$$\begin{aligned}\text{Profit \%} &= (\text{Profit} / \text{CP}) \times 100 \\ \text{Loss \%} &= (\text{Loss} / \text{CP}) \times 100\end{aligned}$$

- **TRAP:** Profit%/Loss% is **ALWAYS** calculated on **CP**, never on SP, unless a question explicitly states otherwise. This is the single most common error in this chapter.

A7.2 Quick Transformation Shortcuts [Level: Basic-Medium]

$$\begin{aligned}SP &= CP \times (100 + \text{Profit\%}) / 100 & SP &= CP \times (100 - \text{Loss\%}) / 100 \\ CP &= SP \times 100 / (100 + \text{Profit\%}) & CP &= SP \times 100 / (100 - \text{Loss\%})\end{aligned}$$

- Ways to solve CP $\leftrightarrow$ SP conversions:
 1. **Direct formula method:** plug into the formulas above.
 2. **Fraction-ratio shortcut (faster):** convert Profit%/Loss% to a ratio SP:CP. E.g., 25% profit $\rightarrow$ SP:CP = 125:100 = **5:4**. 20% loss $\rightarrow$ SP:CP = 80:100 = **4:5**. Once you have the ratio, cross-multiply directly — no need to write the full percentage formula.
- **TRICK:** Memorize common ratio pairs: 10% $\rightarrow$ 11:10, 20% $\rightarrow$ 6:5, 25% $\rightarrow$ 5:4, 33 1/3% $\rightarrow$ 4:3, 50% $\rightarrow$ 3:2 (profit); 10% $\rightarrow$ 9:10, 20% $\rightarrow$ 4:5, 25% $\rightarrow$ 3:4 (loss).

A7.3 Marked Price & Discount [Level: Medium]

$$\begin{aligned}\text{Discount \% is calculated on Marked Price (MP), NOT on CP.} \\ SP &= MP \times (100 - \text{Discount\%}) / 100\end{aligned}$$

- **TRAP:** Do not confuse **Discount%** (always on MP) with **Profit/Loss%** (always on CP) — a mixed question may give both and expect you to chain: MP $\rightarrow$ (apply discount) $\rightarrow$ SP $\rightarrow$ (compare to CP) $\rightarrow$ Profit/Loss%.
- **Successive discounts:** two discounts of a% and b% applied one after another do NOT add up to (a+b)%. Use the successive-percentage-change formula:

$$\text{Net discount \%} = a + b - (a \times b) / 100 \quad (\text{always LESS than the simple sum})$$

E.g., successive discounts of 10% and 10% give a net discount of $10+10-1 = 19\%$, not 20%.

A7.4 False Weight / Cheating Trader Problems [Level: Medium-Hard]

- A trader claims to sell at cost price (0% profit) but uses a **false weight** — e.g., uses a 900g "kg" instead of a true 1000g kg.

$$\begin{aligned}\text{Profit \%} &= [(\text{True Value} - \text{False Value}) / \text{False Value}] \times 100 \\ &= [\text{Error} / (\text{True weight used})] \times 100\end{aligned}$$

E.g., using 900g instead of 1000g while charging for 1000g: Profit% = $(1000-900)/900 \times 100 = 11.11\%$.

- **TRICK:** This is structurally identical to the Percentage chapter's product-constant rule — the trader effectively "increases price per gram" by shortchanging weight, so the same $[\text{error}/(\text{actual}-\text{error})] \times 100$ pattern applies.

A7.5 Overall / Combined Profit-Loss on Multiple Transactions [Level: Hard]

- **TRAP:** When profit% differs across multiple transactions, you **cannot average the percentages directly** — you must combine using total CP and total SP:

$$\text{Overall Profit/Loss \%} = [(\text{Total SP} - \text{Total CP}) / \text{Total CP}] \times 100$$

- **Classic trap — selling two items at the SAME SP, one at x% profit and the other at x% loss:** the result is **always a net loss**, regardless of x, given by:

$$\text{Net Loss \%} = x^2 / 100$$

E.g., two items sold at ₹100 each, one at 20% profit and other at 20% loss $\rightarrow$ net loss = $20^2/100 = 4\%$ loss overall — even though it looks like the profit and loss should "cancel out." This happens because the CPs of the two items are different (one is SP/1.2, the other is SP/0.8), so equal SP does not mean equal-magnitude profit/loss in absolute terms.

Questions

Q1. [Easy] A shopkeeper buys an article for ₹400 and sells it for ₹460. Find the profit percent.

A. 10% B. 12% C. 15% D. 20%

Answer: C. Why: Profit = $460 - 400 = 60$. Profit% = $60/400 \times 100 = 15\%$.

Q2. [Easy] [Trick] An article is sold at a profit of 25%. Find the ratio SP:CP.

A. 4:5 B. 5:4 C. 6:5 D. 3:2

Answer: B. Why: 25% profit $\rightarrow$ SP:CP = 125:100 = **5:4**.

Q3. [Medium] A shopkeeper marks an item 40% above cost price and gives a discount of 10%. Find his profit percent.

A. 20% B. 26% C. 30% D. 36%

Answer: B. Why: MP = $1.4 \times \text{CP}$. SP = $0.9 \times \text{MP} = 0.9 \times 1.4 \times \text{CP} = 1.26 \times \text{CP} \rightarrow$ **26% profit**.

Q4. [Medium] [Trap] A trader marks his goods 50% above cost price and allows a discount of 20%. What is his loss or gain percent?

A. 20% gain B. 30% gain C. 20% loss D. 30% loss

Answer: A. Why: MP = $1.5 \times \text{CP}$. SP = $0.8 \times 1.5 \times \text{CP} = 1.2 \times \text{CP} \rightarrow$ **20% gain** (not a loss — a common trap is assuming the discount always erases the markup gain).

Q5. [Medium] Successive discounts of 20% and 10% are given on an item marked at ₹500. Find the final selling price.

A. ₹350 B. ₹355 C. ₹360 D. ₹400

Answer: C. Why: Net discount = $20 + 10 - (20 \times 10) / 100 = 28\%$. SP = $500 \times (1 - 0.28) = 500 \times 0.72 =$ **₹360**.

Q6. [Medium] A dishonest dealer professes to sell his goods at cost price but uses a weight of 900g for 1kg. Find his profit percent.

A. 9% B. 10% C. 11.11% D. 12%

Answer: C. Why: Profit% = $(1000-900)/900 \times 100 = 100/900 \times 100 = 11.11\%$.

Q7. [Hard] [Trap] A man sells two watches at ₹1000 each. On one he gains 20% and on the other he loses 20%. Find his overall gain or loss percent.

A. No profit no loss B. 4% loss C. 4% gain D. 2% loss

Answer: B. Why: Common equal-SP profit/loss trap: Net Loss% = $x^2/100 = 20^2/100 = 4\%$ **loss** overall — never zero despite the symmetric-looking percentages.

Q8. [Hard] A shopkeeper sold two articles, each for ₹2400. On the first he gained 20% and on the second he lost 20%. Find his total CP and the overall profit or loss in rupees.

A. Total CP=4900, Loss=₹100 B. Total CP=5000, Loss=₹200 C. Total CP=5000, Gain=₹200 D. Total CP=4800, no loss

Answer: B. Why: CP1 = $2400/1.2 = 2000$; CP2 = $2400/0.8 = 3000$. Total CP = 5000, Total SP = 4800 → Loss = ₹200 (matches 4% of 5000 = 200, consistent with Q7's formula).

Q9. [Hard] Find the single discount equivalent to three successive discounts of 10%, 20%, and 10%.

A. 35.2% B. 36.8% C. 38% D. 40%

Answer: A. Why: Combine first two: $10+20-(10 \times 20)/100 = 28\%$. Combine with third: $28+10-(28 \times 10)/100 = 38-2.8 = 35.2\%$.

Q10. [Hard] [Trap] By selling 45 lemons for ₹40, a woman loses 20%. How many lemons should she sell for ₹24 to gain 20% on the whole?

A. 12 B. 15 C. 18 D. 20

Answer: C. Why: SP of 45 lemons = 40 represents 80% of CP → CP of 45 lemons = $40/0.8 = 50$, so CP per lemon = $50/45 = 10/9$. To gain 20%, required SP per lemon = $(10/9) \times 1.2 = 4/3$. Number of lemons for ₹24 = $24 \div (4/3) = 18$.

A8. Simple Interest & Compound Interest

A8.1 Terminology & Simple Interest Basics [Level: Basic]

- **Principal (P)**: the original sum borrowed/invested. **Rate (R)**: interest rate per annum (%). **Time (T)**: duration in years. **Interest (I)**: the extra money paid/earned. **Amount (A)**: $P + I$.

Simple Interest: $SI = (P \times R \times T) / 100$
 Amount: $A = P + SI$

- Ways to solve SI problems: (1) **direct formula** substitution; (2) **unitary shortcut** — SI per year = $(P \times R)/100$, then multiply by T; useful when T changes across sub-questions but P,R stay fixed.

A8.2 Compound Interest Basics [Level: Basic-Medium]

Amount (compounded annually): $A = P \times (1 + R/100)^T$
 Compound Interest: $CI = A - P$

- TRICK — 10-second CI for small T without a calculator: apply the successive-percentage-change formula (from Percentage chapter) repeatedly. E.g., $R=10\%$, $T=2$: net growth = $10+10+(10 \times 10)/100 = 21\% \rightarrow CI = 21\%$ of P (matches $(1.1)^2=1.21$ exactly). For $T=3$: apply the 2-year net (21%) with the 3rd year's 10% again: $21+10+(21 \times 10)/100 = 33.1\%$ — matches $(1.1)^3=1.331$.
- **TRAP**: Compounding is **half-yearly** or **quarterly** in many questions — you MUST adjust both R and T:

Half-yearly: use $R/2$ as the rate per period, and $2T$ as the number of periods.
 Quarterly: use $R/4$ as the rate per period, and $4T$ as the number of periods.

Forgetting this adjustment (using annual R,T directly with half-yearly compounding) is one of the most common CI errors.

A8.3 CI vs SI Relationship (High-Yield Formulas) [Level: Medium]

For $T = 2$ years: $CI - SI = P \times (R/100)^2$
 For $T = 3$ years: $CI - SI = P \times (R/100)^2 \times (3 + R/100)$

- These let you find P or R directly from a given (CI – SI) difference without computing CI and SI separately in full.
- TRICK: For $T=2$, the difference (CI–SI) is exactly "interest on the first year's interest" — conceptually, CI compounds the first year's SI into the second year's principal, while SI does not; this single year's "interest on interest" IS the entire CI–SI gap.

A8.4 Doubling / Multiplying Time Shortcuts (CI) [Level: Medium-Hard]

- If a sum **doubles** in T_1 years under CI, it becomes **4x** in $2 \times T_1$ years, **8x** in $3 \times T_1$ years, **2^n x** in $n \times T_1$ years — a direct geometric-growth shortcut (since each T_1 -year block simply reapplies the same $\times 2$ multiplier).
- **TRAP**: This clean doubling shortcut applies **only to CI**, not SI — under SI, if a sum doubles (i.e., earns 100% interest) in T_1 years, it becomes triple (not quadruple) in $2 \times T_1$ years, since SI grows linearly (each equal time block adds the SAME fixed interest, not a multiplying factor).

A8.5 Applications & Word-Problem Patterns [Level: Hard]

Extra interest earned due to a rate increase (SI-style, same P and T):
 $\text{Extra Interest} = P \times T \times (\text{Rate difference}) / 100$

Dividing a Principal into two parts such that SI on each becomes equal after different times T_1, T_2 :
 $P_1 \times T_1 = P_2 \times T_2$ (since equal SI at same rate R means equal $P \times T$ product)

- **Variable rate of interest** (different rates for different years, SI-style): total interest = $P \times (R_1 \times T_1 + R_2 \times T_2 + R_3 \times T_3 + \dots)/100$ — sum the rate-time products before dividing by 100, don't compute year-by-year separately then add (though both give the same answer, the combined approach is faster).
- **"Amount becomes x times the principal in T years" (SI)**: since SI is linear, if amount becomes x times in T years, then $\text{Rate} = [(x-1) \times 100]/T$ (derived from $A = P + PRT/100 = Px \rightarrow RT = (x-1) \times 100$).

Questions

Q1. [Easy] Find the SI on ₹5000 at 8% per annum for 3 years.

A. ₹1000 B. ₹1100 C. ₹1200 D. ₹1500

Answer: C. **Why:** $SI = (5000 \times 8 \times 3)/100 = ₹1200$.

Q2. [Easy] Find the CI on ₹10,000 at 10% per annum for 2 years, compounded annually.

A. ₹2000 B. ₹2100 C. ₹2200 D. ₹2500

Answer: B. **Why:** Net 2-year growth = $10+10+(10 \times 10)/100 = 21\%$. $CI = 21\%$ of 10,000 = **₹2100**.

Q3. [Medium] [Trick] Find the difference between CI and SI on ₹8000 for 2 years at 5% per annum.

A. ₹15 B. ₹18 C. ₹20 D. ₹25

Answer: C. **Why:** $CI - SI = P \times (R/100)^2 = 8000 \times (0.05)^2 = 8000 \times 0.0025 = ₹20$.

Q4. [Medium] [Trap] Find the CI on ₹4000 at 20% per annum for 1 year, compounded half-yearly.

A. ₹800 B. ₹820 C. ₹840 D. ₹861

Answer: C. **Why:** Half-yearly: rate/period = 10%, periods = 2. $A = 4000 \times (1.1)^2 = 4000 \times 1.21 = 4840$. $CI = 4840 - 4000 = ₹840$ (higher than the flat-20%-once value of ₹800 — a common trap is forgetting the half-yearly adjustment and answering ₹800).

Q5. [Medium] A sum of money doubles itself in 8 years at compound interest. In how many years will it become 8 times itself?

A. 16 B. 20 C. 24 D. 32

Answer: C. **Why:** Doubling shortcut: doubles in 8y $\rightarrow 4x$ in 16y $\rightarrow 8x$ in **24 years** ($3 \times T_1$, since $2^3=8$).

Q6. [Medium] The difference between CI and SI on a certain sum for 3 years at 10% per annum is ₹31. Find the sum.

A. ₹1000 B. ₹1500 C. ₹2000 D. ₹3100

Answer: A. Why: $CI - SI (T=3) = P \times (R/100)^2 \times (3 + R/100) = P \times 0.01 \times 3.1 = 0.031P$. Set $0.031P = 31 \rightarrow P = \text{₹}1000$.

Q7. [Hard] A sum of ₹12,000 becomes ₹15,600 in 3 years at simple interest. If the rate of interest is increased by 3%, what would the amount become?

A. ₹16,080 B. ₹16,410 C. ₹16,680 D. ₹15,900

Answer: C. Why: $SI = 15600 - 12000 = 3600$ over 3 years $\rightarrow$ Rate = $(3600 \times 100) / (12000 \times 3) = 10\%$. New rate = 13%. New SI = $(12000 \times 13 \times 3) / 100 = 4680$. New Amount = $12000 + 4680 = \text{₹}16,680$.

Q8. [Hard] A certain sum is divided into two parts such that the simple interest on the first part for 4 years at 5% per annum equals the simple interest on the second part for 5 years at 4% per annum. Find the ratio of the first part to the second.

A. 1:1 B. 4:5 C. 5:4 D. 16:25

Answer: A. Why: Equal-SI condition (same rate structure per part): $P_1 \times T_1 \times R_1 = P_2 \times T_2 \times R_2 \rightarrow P_1 \times 4 \times 5 = P_2 \times 5 \times 4 \rightarrow 20P_1 = 20P_2 \rightarrow P_1:P_2 = 1:1$.

Q9. [Hard] In what time will a sum of money become 4 times itself at 25% per annum simple interest?

A. 9 years B. 12 years C. 16 years D. 20 years

Answer: B. Why: Becomes $x=4$ times: $\text{Rate} \times \text{Time} = (x-1) \times 100 \rightarrow 25 \times T = 300 \rightarrow T = 12 \text{ years}$.

A9. Time & Work, Pipes & Cisterns

A9.1 Core Idea — Work as a Fixed Unit [Level: Basic]

- Treat "total work" as **1 unit**, or better, as the **LCM of all individual times given** in the question — this avoids fractions entirely and is the single biggest speed trick in this chapter.
- If A alone takes a days to finish a job: A's **one-day work** = $1/a$ (fraction method) — or, using the LCM trick, if Total Work = LCM($a, b, \dots$), A's one-day work (in units) = Total Work / a .

Ways to solve "A takes a days, B takes b days — find time together":
 Method 1 (Fraction method): Combined 1-day work = $1/a + 1/b \Rightarrow$ Time = $1 / (1/a + 1/b) = ab/(a+b)$
 Method 2 (LCM/Units method, faster for mental math):
 Total Work = LCM(a, b)
 A's efficiency = Total Work / a units/day
 B's efficiency = Total Work / b units/day
 Time together = Total Work / (A's efficiency + B's efficiency)

- TRICK: Method 2 turns every Time & Work problem into whole-number arithmetic — always prefer it over fractions.

A9.2 Efficiency Ratio [Level: Basic-Medium]

- Efficiency $\propto 1/\text{Time}$** — efficiency and time are INVERSELY related for a fixed amount of work.
- TRAP:** If A is twice as efficient as B, students often wrongly say "A takes twice the time." Correct: A is twice as efficient $\Rightarrow$ A takes **half** the time B takes.
- If efficiencies of A:B = $m:n$, then time taken by A:B = $n:m$ (the ratio flips).

A9.3 Work and Wages [Level: Medium]

- Total payment for a joint job must be split in the ratio of **work actually done**, which (for the same duration worked) equals the ratio of **efficiencies** — NOT the ratio of time taken alone, and NOT split equally by default.
- TRAP:** A common wrong approach is splitting wages by "days worked" when days worked are equal — if two people work the SAME number of days, wages split by efficiency ratio; only split by time/output actually contributed when durations differ too.

A9.4 Alternate-Days & Remaining-Work Problems [Level: Medium-Hard]

- When A and B work on alternate days, compute work done in **one full 2-day cycle** (A's one-day work + B's one-day work), find how many complete cycles are needed, then handle the leftover work in the next partial day separately.
- Person joining/leaving midway:** split the total timeline into phases (before the change, after the change) and add up the work done in each phase separately; the unknown (usually total time or one person's rate) is solved from the equation: Work done in Phase 1 + Work done in Phase 2 = Total Work.

A9.5 Pipes & Cisterns — Same Math, Just Signed Rates [Level: Medium]

- An **inlet pipe** (fills the tank) is a **positive** work rate; an **outlet/leak pipe** (empties the tank) is a **negative** work rate. Everything from A9.1–A9.4 (LCM trick, combined-rate addition, efficiency ratio) applies unchanged — just add rates algebraically with correct sign.
- TRAP:** A leak that "empties a full tank in x hours" must be SUBTRACTED from the inlet rate when computing the combined fill rate. If the combined rate comes out negative, the tank **never fills** (it net-empties) — always sanity-check the sign of your final combined rate before computing a time from it.
- TRICK: If an inlet fills in a hours and a leak empties in l hours, using Total Work = LCM(a, l): net rate = (TotalWork/ a) – (TotalWork/ l); Time to fill (with leak open) = TotalWork / net rate.

Questions

(Easy $\rightarrow$ Hard; [Trick] rewards the shortcut, [Trap] targets a common misconception)

Q1. [Easy] A can do a piece of work in 12 days and B in 6 days. In how many days will they finish it together?

A. 3 days B. 4 days C. 5 days D. 6 days

Answer: B. **Why:** Total work = LCM(12,6) = 12 units. A's rate = 1/day, B's rate = 2/day. Combined = 3/day. Time = $12/3 = 4$ days.

Q2. [Easy] A is twice as efficient as B. If B alone takes 18 days to finish a job, how long does A alone take?

A. 36 days B. 9 days C. 27 days D. 6 days

Answer: B. **Why:** Efficiency $\propto 1/\text{Time}$ (inverse). A is $2\times$ as efficient $\Rightarrow$ A takes half of B's time = $18/2 = 9$ days.

Q3. [Medium] [Trap] A, B and C do a job together and earn ₹1,800. A alone would take 6 days, B alone 8 days, and C alone 12 days for the same job. Find A's share.

A. ₹600 B. ₹800 C. ₹900 D. ₹400

Answer: B. **Why:** Total Work = LCM(6,8,12) = 24. Efficiencies: A=4, B=3, C=2/day (ratio 4:3:2, sum=9). A's share = $(4/9) \times 1800 = \text{₹}800$. (Wages split by efficiency, not equally — the classic trap here is splitting ₹600 each.)

Q4. [Medium] A can do a job in 20 days. After working alone for 4 days, B joins him and together they finish the remaining work in 8 days. In how many days can B alone do the job?

A. 15 days B. 16 days C. 20 days D. 24 days

Answer: C. **Why:** Total Work = 20 units. A's rate = 1 unit/day. In 4 days A completes 4 units, leaving 16 units. Together they finish these 16 units in 8 days $\Rightarrow$ combined rate = 2 units/day $\Rightarrow$ B's rate = $2 - 1 = 1$ unit/day $\Rightarrow$ B's time = $20/1 = 20$ days.

Q5. [Medium] [Trick] A tank has an inlet pipe that fills it in 6 hours and an outlet pipe that empties it in 10 hours. If both are opened together, in how many hours will the tank fill?

A. 12 hours B. 15 hours C. 8 hours D. 16 hours

Answer: B. **Why:** Total Work = LCM(6,10) = 30. Inlet rate = $+5/\text{hr}$, Outlet rate = $-3/\text{hr}$. Net rate = $+2/\text{hr}$. Time = $30/2 = 15$ hours.

Q6. [Hard] [Trap] Pipe A fills a tank in 4 hours, Pipe B fills it in 6 hours, and Pipe C (a leak) can empty the full tank in 12 hours. If all three are opened together, how long to fill the tank?

A. 3 hours B. 2.4 hours C. 4 hours D. 6 hours

Answer: A. **Why:** Total Work = LCM(4,6,12) = 12. Rates: A= $+3/\text{hr}$, B= $+2/\text{hr}$, C= $-1/\text{hr}$. Net rate = $3+2-1 = 4/\text{hr}$. Time = $12/4 = 3$ hours. (The trap: forgetting to subtract C's leak rate would wrongly give $12/5 = 2.4$ hours — option B.)

Q7. [Hard] A and B undertake a job for ₹720. Working alone, A would take 6 days and B would take 8 days. With the help of C, they finish it in 3 days. Find C's share.

A. ₹90 B. ₹180 C. ₹60 D. ₹120

Answer: A. Why: Total Work = LCM(6,8,3) = 24. A's rate=4, B's rate=3/day. Combined (A+B+C) rate needed = $24/3 = 8/\text{day}$. So C's rate = $8-4-3 = 1/\text{day}$. Wage split by efficiency ratio 4:3:1 (sum 8). C's share = $(1/8) \times 720 = \text{₹90}$.

Q8. [Hard] Working alone, A takes 6 days more than A and B together, and B alone takes 24 days more than A and B together. In how many days can A and B, working together, complete the job?

A. 8 days B. 12 days C. 10 days D. 6 days

Answer: B. Why: This is the classic pattern where, if T = time together, then $T^2 = (\text{extra time for A}) \times (\text{extra time for B})$ — here $T^2 = 6 \times 24 = 144$, so $T = 12$ days. (Quick sanity check via the standard identity for this problem type: $T = \sqrt{xy}$ where x, y are the "extra days" each takes beyond the combined time.)

Q9. [Hard] [Trap] The efficiencies of A, B and C are in the ratio 5:4:3. Working together, they finish a job in 4 days. Find the number of days C alone would take.

A. 10 days B. 16 days C. 9 days D. 15 days

Answer: B. Why: Combined efficiency = $5+4+3 = 12$ units/day. Total Work = $12 \times 4 = 48$ units. C's rate = 3 units/day $\Rightarrow$ C's time = $48/3 = 16$ days.

A10. Speed, Time & Distance (Trains, Boats & Streams)

A10.1 Core Relation & Unit Conversion [Level: Basic]

$$\text{Speed} = \text{Distance} / \text{Time} \quad \text{Distance} = \text{Speed} \times \text{Time} \quad \text{Time} = \text{Distance} / \text{Speed}$$

Unit conversion (TRICK – used constantly):

km/h → m/s : multiply by 5/18

m/s → km/h : multiply by 18/5

A10.2 Average Speed [Level: Basic-Medium]

$$\text{Average Speed} = \text{Total Distance} / \text{Total Time} \quad (\text{ALWAYS true} - \text{the only safe default formula})$$

- Ways to solve depending on what's constant across the journey's legs:
 - Equal DISTANCE legs** (e.g., going and returning over the same route at different speeds x and y): use the **harmonic mean** shortcut — $\text{Average Speed} = 2xy / (x + y)$.
 - Equal TIME legs** (e.g., first half of the total TIME at speed x , second half at speed y): the plain arithmetic mean is correct — $\text{Average Speed} = (x + y) / 2$.
 - Neither equal** (arbitrary distances/times per leg): fall back to the master formula — Total Distance / Total Time, computed leg by leg.
- TRAP:** Applying the simple average $(x+y)/2$ to an equal-DISTANCE journey is the single most common error in this entire chapter — it only works for equal TIME segments.
- Generalized equal-distance case (n legs of equal distance at speeds $v_1, v_2, \dots, v_n$): $\text{Average Speed} = n / (1/v_1 + 1/v_2 + \dots + 1/v_n)$ (harmonic mean of n terms).

Worked Examples (verified, increasing difficulty)

A bus takes students to college at 6 km/h; they miss the return bus and walk back the same route at 4 km/h. Average speed for the day?

Equal distance both ways → $\text{Avg} = 2 \times 6 \times 4 / (6 + 4) = 48/10 = \mathbf{4.8 \text{ km/h}}$. (Real placement-test question, TCS NQT 2018.)

A man commutes to office at 65 km/h and returns at 75 km/h. Average speed?

$\text{Avg} = 2 \times 65 \times 75 / (65 + 75) = 9750/140 = \mathbf{69.64 \text{ km/h}}$. (L&T Infotech 2016.)

A journey is covered 1/3 at 60 km/h, 1/3 at 75 km/h, and 1/3 at 45 km/h (equal DISTANCE thirds). Average speed?

Generalized harmonic mean: $\text{Avg} = 3 / (1/60 + 1/75 + 1/45) = 3 / 0.052222 = \mathbf{57.45 \text{ km/h}}$. (L&T Infotech.)

A bird flies 400 km: the first 100 km at 100 km/h, next 100 km at 200 km/h, next at 300 km/h, last at 400 km/h. Average speed?

Total time = $100/100 + 100/200 + 100/300 + 100/400 = 1 + 0.5 + 0.333 + 0.25 = 2.0833$ h. $\text{Average} = 400/2.0833 = \mathbf{192 \text{ km/h}}$. (GATE Electrical.) — Note this is NOT an equal-time case, so the master formula (Total Distance/Total Time) was applied directly rather than a mean shortcut.

A car travels 8 km in the 1st quarter-hour, 6 km in the 2nd quarter-hour, 16 km in the 3rd quarter-hour. Average speed over the whole journey?

These ARE equal-TIME legs (each a quarter hour) — but with different distances, so use the master formula directly: Total distance = $8 + 6 + 16 = 30$ km, Total time = $3 \times 1/4$ h = 0.75 h. $\text{Average} = 30/0.75 = \mathbf{40 \text{ km/h}}$. (GATE Electrical 2013.)

A man covers 10 km at 5 km/h, 30 km at 6 km/h, 20 km at 10 km/h. Average speed?

Not equal distance, not equal time — use master formula. Total distance = 60 km. Total time = $10/5 + 30/6 + 20/10 = 2 + 5 + 2 = 9$ h. $\text{Average} = 60/9 = \mathbf{6.67 \text{ km/h}}$. (Accenture 2019 — the answer is 6.67 km/h, not a mean-shortcut value, since neither the equal-distance nor equal-time condition holds.)

A10.3 Relative Speed [Level: Medium]

Same direction:	$\text{Relative Speed} = \text{Speed1} - \text{Speed2} $	(they close the gap slowly)
Opposite directions:	$\text{Relative Speed} = \text{Speed1} + \text{Speed2}$	(they close the gap fast)

This single idea underlies every train, boat, and "two people/vehicles" problem below.

A10.4 Train Problems [Level: Medium-Hard]

Case	Formula
Train crosses a stationary point/pole/man	$\text{Time} = \text{Train Length} / \text{Train Speed}$
Train crosses a platform/bridge/tunnel	$\text{Time} = (\text{Train Length} + \text{Platform Length}) / \text{Train Speed}$
Two trains cross each other, OPPOSITE directions	$\text{Time} = (L_1 + L_2) / (\text{Speed1} + \text{Speed2})$
Two trains cross each other, SAME direction	$\text{Time} = (L_1 + L_2) / \text{Speed1} - \text{Speed2} $
Train crosses a walking/moving person	Relative speed = Train speed $\pm$ person's speed (– if same direction, + if opposite)

- TRICK for stoppages: if a train's speed WITHOUT stoppage is S_w and WITH stoppage (i.e., including halt time) is S_w' , the stoppage time per hour = $[(S_w - S_w') / S_w] \times 60$ minutes.

Worked Examples (verified)

Two trains start from stations A and B toward each other at 54 km/h and 72 km/h. At the meeting point, one train has travelled 80 m more than the other. Find the distance AB.

Convert: 54 km/h = 15 m/s, 72 km/h = 20 m/s. In time t , distances are $15t$ and $20t$; difference = $5t = 80 \Rightarrow t = 16$ s. Distance AB = $15t + 20t = 35 \times 16 = \mathbf{560 \text{ m}}$.

Two trains of equal length cross a stationary person in 6 s and 8 s respectively. Find the time for them to cross each other (a) moving in opposite directions, (b) moving in the same direction.

Let length = L . Speed1 = $L/6$, Speed2 = $L/8$.

(a) Opposite: relative speed = $L/6 + L/8 = 7L/24$; combined length to clear = $2L$; time = $2L \div (7L/24) = 48/7$ s.

(b) Same direction: relative speed = $L/6 - L/8 = L/24$; time = $2L \div (L/24) = \mathbf{48 \text{ s}}$.

A10.5 Boats & Streams [Level: Medium]

Downstream Speed (D) = Boat speed in still water (x) + Stream speed (y)
 Upstream Speed (U) = Boat speed in still water (x) - Stream speed (y)

Recovering x and y from D and U:

$$\begin{aligned}\text{Boat speed in still water (x)} &= (D + U) / 2 \\ \text{Stream speed (y)} &= (D - U) / 2\end{aligned}$$

- TRAP:** This is a plain arithmetic mean/half-difference — NOT the harmonic-mean situation from A10.2. Downstream and upstream speeds are simple sums/differences of two underlying rates, not two different average speeds over the same distance.
- Round-trip average speed** (same distance covered downstream and upstream): THIS case IS the equal-distance harmonic-mean scenario from A10.2 — **Average speed for round trip = $2 \times D \times U / (D + U)$** .

Worked Examples

A boat rows downstream at 14 km/h and upstream at 5 km/h. Find the boat's speed in still water.
 $x = (14 + 5)/2 = \mathbf{9.5 \text{ km/h}}$.

A boat's speed in still water is 12 km/h; the stream flows at 5 km/h. It takes 8 hours for a round trip between two points. Find the distance between them.
 Downstream = 17 km/h, Upstream = 7 km/h. Let distance = d: $d/17 + d/7 = 8 \Rightarrow d(7+17)/(119) = 8 \Rightarrow d \times 24/119 = 8 \Rightarrow d = 8 \times 119/24 = \mathbf{39.67 \text{ km}}$.

A boat covers 20 km upstream + 30 km downstream in 3 hours, and 10 km upstream + 20 km downstream in 5/3 hours. Find the sum of the boat's speed in still water and the stream's speed.

Let upstream speed = u, downstream = d. Equations: $20/u + 30/d = 3$ and $10/u + 20/d = 5/3$. Doubling the second: $20/u + 40/d = 10/3$. Subtracting the first: $10/d = 10/3 - 3 = 1/3 \Rightarrow d = 30$. Substituting back: $10/u + 20/30 = 5/3 \Rightarrow 10/u = 1 \Rightarrow u = 10$. Boat speed = $(10 + 30)/2 = 20$, stream speed = $(30 - 10)/2 = 10$. **Sum = 30 km/h.**

Questions

(Easy → Hard; [Trick] rewards the shortcut, [Trap] targets a common misconception)

Q1. [Easy] Convert 90 km/h to m/s.

A. 20 m/s B. 25 m/s C. 30 m/s D. 15 m/s

Answer: B. **Why:** $90 \times 5/18 = \mathbf{25 \text{ m/s}}$.

Q2. [Easy] A train 120 m long crosses a pole in 6 seconds. Find its speed in km/h.

A. 60 km/h B. 72 km/h C. 80 km/h D. 20 km/h

Answer: B. **Why:** Speed = $120/6 = 20 \text{ m/s} = 20 \times 18/5 = \mathbf{72 \text{ km/h}}$.

Q3. [Medium] [Trap] A man travels from A to B at 40 km/h and returns at 60 km/h. Find his average speed for the whole journey.

A. 50 km/h B. 48 km/h C. 52 km/h D. 45 km/h

Answer: B. **Why:** Equal distance both ways $\Rightarrow$ harmonic mean: $2 \times 40 \times 60 / (40 + 60) = 4800/100 = \mathbf{48 \text{ km/h}}$ — NOT the simple average 50 km/h, which is the classic trap here.

Q4. [Medium] A train 150 m long crosses a platform 250 m long in 20 seconds. Find the train's speed in km/h.

A. 54 km/h B. 60 km/h C. 72 km/h D. 45 km/h

Answer: C. **Why:** Total distance = $150 + 250 = 400 \text{ m}$ in $20 \text{ s} \Rightarrow 20 \text{ m/s} = \mathbf{72 \text{ km/h}}$.

Q5. [Medium] [Trick] Two trains 100 m and 150 m long run on parallel tracks in opposite directions at 30 km/h and 20 km/h. Find the time they take to cross each other.

A. 18 s B. 20 s C. 15 s D. 10 s

Answer: A. **Why:** Relative speed (opposite directions) = $30 + 20 = 50 \text{ km/h} = 50 \times 5/18 = 125/9 \text{ m/s}$. Total length = 250 m. Time = $250 \div (125/9) = 250 \times 9/125 = \mathbf{18 \text{ s}}$.

Q6. [Hard] A boat goes 24 km upstream in 6 hours and returns downstream in 4 hours. Find the speed of the boat in still water and the speed of the stream.

A. 5, 1 km/h B. 4, 1 km/h C. 5, 2 km/h D. 6, 1 km/h

Answer: A. **Why:** Upstream speed = $24/6 = 4 \text{ km/h}$. Downstream speed = $24/4 = 6 \text{ km/h}$. Boat speed = $(6 + 4)/2 = \mathbf{5 \text{ km/h}}$; stream speed = $(6 - 4)/2 = \mathbf{1 \text{ km/h}}$.

Q7. [Hard] [Trap] A car covers the first half of a journey at 40 km/h and the second half (by TIME, not distance) at 60 km/h. Find the average speed for the trip.

A. 48 km/h B. 50 km/h C. 52 km/h D. 46 km/h

Answer: B. **Why:** These are equal-TIME segments (not equal distance), so the plain arithmetic mean applies here: $(40 + 60)/2 = \mathbf{50 \text{ km/h}}$. Using the harmonic-mean formula here (as if distances were equal) would be the trap, giving the wrong value of 48.

Q8. [Hard] Two stations A and B are 300 km apart. A train starts from A at 7:00 AM at 50 km/h toward B; another starts from B at 8:00 AM at 100 km/h toward A. At what time do they meet?

A. 9:00 AM B. 9:40 AM C. 9:20 AM D. 10:00 AM

Answer: B. **Why:** By 8:00 AM, the first train has already covered 50 km, leaving a gap of $300 - 50 = 250 \text{ km}$, closing at a relative (opposite-direction) speed of $50 + 100 = 150 \text{ km/h}$. Time to close the gap = $250/150 = 5/3 \text{ h} = 1 \text{ h } 40 \text{ min}$. 8:00 AM + 1h40min = **9:40 AM**.

A11. Mixture & Alligation

A11.1 Weighted Average — The Idea Behind Alligation [Level: Basic]

- A **mixture** combines two or more ingredients (different prices/concentrations/speeds/rates). The mixture's overall value is a **weighted average**, pulled closer to whichever ingredient has the larger quantity.
- Alligation** is the shortcut rule that reverses this: given the mean (mixture) value and the two ingredient values, find the **RATIO** in which they were mixed — without setting up algebra from scratch every time.

A11.2 The Alligation Cross-Rule [Level: Basic — the single formula this whole chapter is built on]

Ingredient 1 value = d_1 Ingredient 2 value = d_2 ($d_1 < d_2$)
 Mean (mixture) value = m
 Quantity1 : Quantity2 = $(d_2 - m) : (m - d_1)$

- Ways to visualize it (the "cross diagram"):** write d_1 and d_2 on the left, m in the middle; draw diagonal lines from each to m ; the quantity ratio is read off the *far* differences: $(d_2 - m)$ pairs with ingredient 1, $(m - d_1)$ pairs with ingredient 2.
- TRICK: You never need to know the actual quantities to find the **RATIO** — alligation gives you the ratio directly from just the three values (d_1 , d_2 , m).
- TRAP:** The ratio $(d_2 - m):(m - d_1)$ attaches the "far" difference to the "near" ingredient — a very common slip is pairing d_1 with $(m - d_1)$ instead of with $(d_2 - m)$. If unsure, sanity-check: the CHEAPER/LOWER-value ingredient must always get the LARGER share of the ratio when the mean is closer to it, and vice versa.

Worked Example (Basic)

Item A: Rs. 450/kg, Item B: Rs. 510/kg. Mixture costs Rs. 475/kg. Find A:B.
 Ratio = $(510 - 475):(475 - 450) = 35:25 = \mathbf{7:5}$. (Real TCS-exam question — verified.)

A11.3 Applications Beyond Price Mixing [Level: Medium]

The SAME cross-rule applies wherever a weighted average is formed — recognize these as alligation in disguise:

Context	"d1, d2" are	"m" is
Price mixing	Price of ingredient 1, 2	Mixture price
Average weight/marks	Group 1 avg, Group 2 avg	Combined avg
Average speed (equal time)	Speed 1, Speed 2	Average speed
Simple interest	Rate 1, Rate 2	Overall/mean rate
Profit or Loss %	Profit%/Loss% of part 1, 2	Overall Profit/Loss %

Worked Examples (Medium)

Boys' average weight = 40 kg, girls' average weight = 30 kg. Combined average of 80 students = 33 kg. Find the number of boys.
 Ratio boys:girls = $(m - d_{\text{girls}}):(d_{\text{boys}} - m) = (33 - 30):(40 - 33) = 3:7 \rightarrow \text{boys} = 80 \times 3/10 = \mathbf{24}$, girls = 56.
 (TRAP worth flagging: some sources mislabel which count — always re-derive from the cross-rule rather than trusting a bare memorized number; direct-equation check: $40x + 30(80 - x) = 33 \times 80 \rightarrow 10x = 240 \rightarrow x = 24$. Confirmed.)
 Rs. 1000 lent in two parts at 6% and 8% simple interest p.a.; yearly income = Rs. 75. Find the sum lent at 8%.
 Mean rate = $75/1000 \times 100 = 7.5\%$. Ratio (at 6%):(at 8%) = $(8 - 7.5):(7.5 - 6) = 0.5:1.5 = 1:3$. Total parts = 4 $\rightarrow$ sum at 8% = $1000 \times 3/4 = \mathbf{Rs. 750}$.

A11.4 Profit/Loss-Based Alligation [Level: Medium-Hard]

- Treat the profit%/loss% figures themselves as the "values" being alligated (loss is a *negative* profit% for sign purposes).

Worked Example

A shopkeeper sells 91 kg rice, part at 7% profit and rest at 33% profit, overall gain 25%. Find quantity sold at 7% profit.
 Ratio (at 7%):(at 33%) = $(33 - 25):(25 - 7) = 8:18 = 4:9$. Total parts = 13 $\rightarrow$ qty at 7% = $91 \times 4/13 = \mathbf{28 \text{ kg}}$. (Verified — matches real Accenture/Tech Mahindra PYQ.)
 Two items sold for a combined Rs. 1500 total, one at 3% loss, other at 7% profit, overall no-profit-no-loss (mean = 0%).
 Ratio (loss-item SP-share):(profit-item SP-share) = $(7 - 0):(0 - (-3)) = 7:3$. Total parts = 10 $\rightarrow$ Loss item SP = $1500 \times 7/10 = \mathbf{Rs. 1050}$; Profit item SP = $\mathbf{Rs. 450}$. (Verified.)

A11.5 Water/Dilution & "Sold at CP, Still Profits" Problems [Level: Hard]

- Pure water has value/concentration **0** and (as an ingredient) costs nothing — **TRAP:** forgetting to use 0 (not a blank) as water's "d" value in the cross-rule is the most common error here.
- Classic pattern: "mixture sold at Cost Price (of the pure item) still yields x% profit" — because water dilutes the cost but the mixture is sold at the undiluted price, the "profit" IS the alligation ratio of water:pure-item.

Worked Example

In what ratio must water be mixed with milk to gain 16.66% by selling the mixture at cost price (of pure milk)?
 Treat pure milk's value = 100 (i.e., CP), water's value = 0, mean = mixture's "effective value" = $100/(1 + 16.66\%) \rightarrow$ equivalently, ratio water:milk = Profit% : $(100 + \text{Profit}\%) = 16.66:116.66 \approx \mathbf{1:7}$... more precisely $16.66\% = 1/6$, so ratio = $(1/6)/(1 + 1/6)$ rescaled = 1:6 directly by the standard shortcut **water:milk = Gain% expressed as fraction, numerator:denominator**. (Verified against real GATE-CS PYQ answer 1:6.)

A11.6 Repeated Replacement / Removal Problems [Level: Hard]

- When a fraction of a mixture is repeatedly removed and replaced by pure water (same fraction each time, n times):

$$\text{Final quantity of original ingredient} = \text{Initial} \times (1 - \text{replaced_fraction})^n$$

- TRICK: This is a **geometric decay**, not linear — do not multiply the fraction by n ; raise $(1 - \text{fraction})$ to the power n .

Worked Example

A container has 81 litres of pure milk. $1/3$ of it is removed and replaced with water, and this is repeated 3 times. Find the final quantity of milk.
 Final milk = $81 \times (1 - 1/3)^3 = 81 \times (2/3)^3 = 81 \times 8/27 = \mathbf{24 \text{ litres}}$.

A11.7 Adding a Pure Ingredient to Change Concentration [Level: Hard]

- Special alligation case: adding MORE of one pure ingredient (not water) to shift the mean — solve via direct proportion since the "before" quantity of the OTHER ingredient stays fixed.

Worked Example

80 litres of a milk-water mixture is 70% milk. How much pure milk must be added to raise it to 95% milk?
 Water quantity is fixed = $80 \times 30\% = 24 \text{ L}$ (this never changes). At the end, water must be 5% of the new total: $24 = 0.05 \times \text{New Total} \rightarrow \text{New Total} = 480 \text{ L}$.
 Milk to add = $480 - 80 = \mathbf{400 \text{ litres}}$. (Verified — matches real TCS PYQ.)

Questions

(Easy → Hard; [Trick] rewards the shortcut, [Trap] targets a common misconception)

Q1. [Easy] In what ratio should tea worth Rs. 60/kg be mixed with tea worth Rs. 65/kg so the mixture is worth Rs. 62/kg?

A. 2:3 B. 3:2 C. 1:1 D. 3:5

Answer: B. Why: Ratio = $(65-62):(62-60) = 3:2$.

Q2. [Easy] [Trick] 20 kg of rice at Rs.20/kg is mixed with rice at Rs.24/kg to get a mixture worth Rs.22/kg. Find quantity of the second rice.

A. 15 kg B. 20 kg C. 25 kg D. 30 kg

Answer: B. Why: Ratio = $(24-22):(22-20) = 2:2 = 1:1$, so equal quantities → 20 kg.

Q3. [Medium] A vessel has milk and water in ratio 5:1. How much of the mixture must be replaced with pure water to make the ratio 1:1? (Vessel capacity 60 L)

A. 20 L B. 24 L C. 30 L D. 25 L

Answer: B. Why: Milk=50L, water=10L. Removing x L of mixture ($5/6$ milk, $1/6$ water) and replacing with pure water leaves Milk = $50 - 5x/6$. For a 1:1 ratio in 60L total, Milk must equal 30: $50 - 5x/6 = 30 \rightarrow 5x/6 = 20 \rightarrow \mathbf{x = 24 \text{ L}}$.

Q4. [Medium] [Trap] The average score of boys is 60 and girls is 80. The combined average of the class is 68. If there are 30 girls, find the number of boys.

A. 20 B. 30 C. 45 D. 50

Answer: C. Why: Ratio boys:girls = $(80-68):(68-60) = 12:8 = 3:2$. Girls=30 corresponds to "2 parts" → 1 part=15 → boys = $3 \times 15 = \mathbf{45}$. (TRAP: don't accidentally swap and compute boys as the "2" side.)

Q5. [Medium] [Trick] 40 litres of a mixture is 90% acid. How much water must be added to reduce the acid concentration to 60%?

A. 15 L B. 18 L C. 20 L D. 25 L

Answer: C. Why: Pure acid = 36 L (fixed). At 60% concentration, $36 = 0.6 \times \text{NewTotal} \rightarrow \text{NewTotal} = 60 \text{ L}$. Water added = $60 - 40 = \mathbf{20 \text{ L}}$.

Q6. [Hard] A shopkeeper mixes rice worth Rs. 18/kg with rice worth Rs. 24/kg, and sells the mixture at Rs. 23/kg, gaining 15%. If he used 40 kg of the Rs. 24/kg rice, how much of the Rs. 18/kg rice did he use?

A. 40 kg B. 60 kg C. 80 kg D. 100 kg

Answer: C. Why: Mean CP = $SP/1.15 = 23/1.15 = 20$. Ratio Q(18-rice):Q(24-rice) = $(24-20):(20-18) = 4:2 = 2:1$. Since 24-rice = 40 kg (the "1" part), 18-rice = **80 kg** (the "2" part). Check: $(80 \times 18 + 40 \times 24)/120 = 2400/120 = 20 = \text{mean CP} \checkmark$.

Q7. [Hard] [Trap] 729 mL of a mixture contains milk and water in ratio 7:2. How much water must be added to make the ratio 7:3?

A. 81 mL B. 91 mL C. 100 mL D. 72 mL

Answer: A. Why: Milk = $729 \times 7/9 = 567 \text{ mL}$ (fixed), Water = 162 mL. Need Milk:NewWater = 7:3 → NewWater = $567 \times 3/7 = 243 \text{ mL}$. Water to add = $243 - 162 = \mathbf{81 \text{ mL}}$.

Q8. [Hard] A container has milk only. 30% of it is replaced by water, and this process is repeated once more (30% of the new mixture replaced by water). What % of the original milk remains?

A. 40% B. 49% C. 51% D. 60%

Answer: B. Why: Remaining fraction = $(1-0.3)^2 = 0.7^2 = 0.49 = \mathbf{49\%}$.

Q9. [Hard] In what ratio must water be mixed with milk costing Rs. 40/L so that after selling the mixture at Rs. 40/L (i.e., at cost price of pure milk), a profit of 25% is made?

A. 1:3 B. 1:4 C. 1:5 D. 3:1

Answer: B. Why: Using the CP-sale-profit shortcut: water:milk = Profit% fraction : 1 = $(1/4):1 = \mathbf{1:4}$ (since 25% = $1/4$).

A12. Data Interpretation

(DI tests no new formulas — it applies Percentage/Ratio/Average skills to tabular or graphical data. The real skill is reading the data source correctly under time pressure.)

A12.1 Core Framing [Level: Basic]

- Every DI question is really a Percentage, Ratio, or Average question wearing a table/graph costume.
- TRAP:** The #1 real-exam error in DI is **misreading which row/column/axis represents which category**, not miscalculating — always spend 5 extra seconds confirming what each axis/row/legend means before computing.
- TRICK:** DI answer options are usually far apart — use **approximation** (round values) to estimate rather than compute exactly; exact arithmetic wastes time when option-gaps make estimation sufficient.

A12.2 Table Graphs [Level: Basic]

- Data given as rows × columns (e.g., Sales of 5 companies over 5 years).
- Standard question types: find % change year-on-year, find ratio between two entities, find the average across a row/column, find which entity had max/min growth.
- Ways to solve "find % change from year X to Y": (1) Direct formula $(\text{New} - \text{Old}) / \text{Old} \times 100$; (2) if the numbers are "nice," recognize the fraction directly (e.g., 120 → 150 is a jump of $30/120 = 1/4 = 25\%$, spot this without long division).

A12.3 Bar Graphs [Level: Basic-Medium]

- Single bar set (one value per category) vs grouped/stacked bars (multiple values per category, e.g., 2 years of data per company side-by-side, or one stacked bar showing composition).
- TRAP:** In a STACKED bar, an individual segment's value is the DIFFERENCE between its top and the segment below it, not the segment's own bar-height from the axis — a common misread.

A12.4 Pie Charts [Level: Medium]

- A pie chart divides a total into sectors, each labeled either in **degrees** (out of 360°) or **percentage** (out of 100%).

Sector's actual value = (Sector's degrees / 360) × Total
 Sector's actual value = (Sector's percent / 100) × Total

- TRICK:** To compare two sectors' relative size directly, you often don't need the "Total" at all — just compare their degree/percentage values directly (ratio of sectors = ratio of their degrees = ratio of their percentages).
- TRAP:** If a question gives a MIX of some sectors in degrees and others in percentage on the same chart, convert everything to ONE unit first (percentage is usually easier: divide degrees by 3.6).

A12.5 Line Graphs [Level: Medium-Hard]

- Shows a trend over a continuous variable (usually time). Slope = rate of change.
- TRAP:** A line graph labeled "**Sales in year X**" (period-wise) is read completely differently from one labeled "**Cumulative sales up to year X**" — for a cumulative graph, the actual value FOR a given year is the DIFFERENCE between consecutive points, not the point itself. Always check the graph's title/label for "cumulative" before reading period values off it directly.
- Steepest segment = highest rate of change (not necessarily the highest absolute value point).

A12.6 General DI Speed Strategy [Level: Hard]

- Always identify the **base** of a percentage question freshly for each sub-question in a DI set — DI sets often ask several questions off the same data where the "base" (denominator) silently shifts from one sub-question to the next (e.g., "% of Company A's sales" vs "% of total sales of all companies" in consecutive questions).
- For "find the average of these 5 values" type questions, use the AP-shortcut $(\text{first} + \text{last}) / 2$ if the values happen to be in arithmetic progression — check for this before summing all 5.

Questions

(Data given in words since diagrams can't be embedded; Easy → Hard)

Q1. [Easy] A pie chart shows a company's expenses: Salary 40%, Rent 25%, Marketing 20%, Misc 15%. If total expense is Rs. 20 lakh, find the Marketing expense.

A. Rs. 3 lakh B. Rs. 4 lakh C. Rs. 5 lakh D. Rs. 6 lakh

Answer: B. Why: 20% of 20 lakh = **Rs. 4 lakh**.

Q2. [Easy] In the same pie chart, find the ratio of Salary expense to Rent expense.

A. 8:5 B. 5:8 C. 4:5 D. 8:3

Answer: A. Why: Ratio of sectors = ratio of percentages = 40:25 = **8:5**.

Q3. [Medium] A table shows a company's revenue (in Rs. crore): 2020: 120, 2021: 150, 2022: 180, 2023: 210. Find the % growth from 2020 to 2021, and from 2022 to 2023.

A. 25% both years B. 25% then 16.67% C. 20% then 25% D. 30% then 20%

Answer: B. Why: 2020→21: $(150 - 120) / 120 = 30 / 120 = 25\%$. 2022→23: $(210 - 180) / 180 = 30 / 180 = 16.67\%$. (Same absolute increase of 30 crore each year, but % growth shrinks because the base grows — **TRAP:** don't assume equal absolute increases mean equal % growth.)

Q4. [Medium] [Trap] A line graph shows "Cumulative units sold" as: Jan=100, Feb=250, Mar=430. Find units sold in March alone.

A. 430 B. 330 C. 180 D. 150

Answer: C. Why: Since the graph is CUMULATIVE, March's own units = Cumulative(Mar) – Cumulative(Feb) = 430 – 250 = **180**.

Q5. [Hard] A stacked bar for "Total Students" shows Boys+Girls. For School X, the bar reaches 80 at the Boys/Girls boundary and 100 at the top (Girls stacked above Boys). Find the number of Girls.

A. 80 B. 20 C. 100 D. 180

Answer: B. Why: Girls = Top of stack – Boundary = 100 – 80 = **20** (the segment's own height, not its top reading).

Q6. [Hard] Company revenues in a table (Rs. crore): A=200, B=250, C=150, D=400, Total of all 4 = 1000. A pie chart of the SAME data shows company D's sector at 144°. Verify if the pie chart is consistent with the table, and find D's percentage share.

A. 40%, consistent B. 36%, consistent C. 40%, inconsistent D. 36%, inconsistent

Answer: A. **Why:** D's actual value/total = $400/1000 = 40\%$. Degrees implied by table = $40\% \times 360^\circ = 144^\circ$, which **matches** the given pie chart — **consistent**, D's share = **40%**.

A13. Algebra (Simplification & Identities)

A13.1 Order of Operations [Level: Basic]

- BODMAS: Brackets -> Of -> Division -> Multiplication -> Addition -> Subtraction.
- TRAP:** "Of" is evaluated at the SAME priority as multiplication/division, left to right — NOT automatically before division. E.g., $10 - 2 \text{ of } 3 + 4/2 = 10 - 6 + 2 = 6$, not something else from misordering "of" ahead of everything.
- TRAP:** Division and multiplication have EQUAL precedence and are evaluated left-to-right (not "always divide before multiply") — $8/2 \times 2 = (8/2) \times 2 = 8$, a frequently miscalculated viral-style question.

A13.2 Core Algebraic Identities [Level: Basic]

$$\begin{aligned}(a+b)^2 &= a^2 + 2ab + b^2 \\ (a-b)^2 &= a^2 - 2ab + b^2 \\ a^2 - b^2 &= (a+b)(a-b) \\ (a+b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a+b) \\ (a-b)^3 &= a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a-b) \\ a^3 + b^3 &= (a+b)(a^2 - ab + b^2) \\ a^3 - b^3 &= (a-b)(a^2 + ab + b^2)\end{aligned}$$

A13.3 The $a+b+c=0$ Shortcut [Level: Medium]

If $a + b + c = 0$, then: $a^3 + b^3 + c^3 = 3abc$

- TRICK: this single identity converts a messy cubic-sum question into one multiplication the instant you notice the three terms sum to zero — always check for this condition first when a question gives values that look designed to sum to 0.
- Worked example: $a=5, b=-3, c=-2$ (sum=0). $a^3+b^3+c^3 = 125-27-8 = 90$. Check: $3abc = 3 \times 5 \times (-3) \times (-2) = 90$. ✓

A13.4 Reciprocal Identities (find a^2+1/a^2 without finding a) [Level: Medium-Hard]

Given $a + 1/a = k$:

$$\begin{aligned}a^2 + 1/a^2 &= k^2 - 2 \\ a^3 + 1/a^3 &= k^3 - 3k\end{aligned}$$

Given $a - 1/a = k$:

$$\begin{aligned}a^2 + 1/a^2 &= k^2 + 2 \\ a^3 - 1/a^3 &= k^3 + 3k\end{aligned}$$

- TRICK: notice the sign flips between the two families — memorize by deriving once: $(a+1/a)^2 = a^2 + 2 + 1/a^2$, so $a^2 + 1/a^2 = (a+1/a)^2 - 2$.
- Worked example: If $a + 1/a = 5$, find $a^2 + 1/a^2$ and $a^3 + 1/a^3$.
 $a^2 + 1/a^2 = 5^2 - 2 = 23$.
 $a^3 + 1/a^3 = 5^3 - 3(5) = 125 - 15 = 110$.
- TRAP:** students often forget the $-3k$ term entirely and just cube k for the cubic case — always re-derive via $(a+1/a)(a^2+1/a^2-1) = a^3+1/a^3$ if in doubt.

A13.5 Linear Simultaneous Equations [Level: Medium]

- Ways to solve: (1) **Elimination** — multiply equations so a variable's coefficients match, subtract/add to cancel it; (2) **Substitution** — solve one variable in terms of the other from one equation, substitute into the second.
- TRICK for speed: when coefficients are large, check if the two equations can be added or subtracted directly to instantly cancel a variable, before doing any multiplication.

Questions

Q1. [Easy] Simplify: $12 + 6 \text{ of } 2 - 4/2$

A. 22 B. 8 C. 14 D. 10

Answer: A. Why: BODMAS: "of"=multiplication priority with division, left to right: $6 \text{ of } 2 = 12$; $4/2 = 2$. So $12 + 12 - 2 = 22$.

Q2. [Easy] If $a - b = 4$ and $a + b = 10$, find $a^2 - b^2$.

A. 40 B. 36 C. 44 D. 14

Answer: A. Why: $a^2 - b^2 = (a+b)(a-b) = 10 \times 4 = 40$.

Q3. [Medium] [Trick] If $x + y + z = 0$ and $x=3, y=-1$, find $x^3+y^3+z^3$.

A. 12 B. 18 C. -18 D. 6

Answer: B. Why: $z = -(x+y) = -2$. $x^3+y^3+z^3 = 3xyz = 3 \times 3 \times (-1) \times (-2) = 18$.

Q4. [Medium] If $a + 1/a = 4$, find $a^2 + 1/a^2$.

A. 14 B. 16 C. 18 D. 12

Answer: A. Why: $a^2 + 1/a^2 = 4^2 - 2 = 14$.

Q5. [Medium] [Trick] If $a - 1/a = 3$, find $a^2 + 1/a^2$.

A. 9 B. 7 C. 11 D. 13

Answer: C. Why: For $a - 1/a = k$, $a^2 + 1/a^2 = k^2 + 2 = 9 + 2 = 11$.

Q6. [Hard] If $a + 1/a = 6$, find $a^3 + 1/a^3$.

A. 216 B. 198 C. 180 D. 174

Answer: B. Why: $a^3 + 1/a^3 = k^3 - 3k = 216 - 18 = 198$.

Q7. [Hard] [Trap] Simplify: $24 \div 4 \times 2 + 3^2 - 1$

A. 19 B. 12 C. 20 D. 14

Answer: C. Why: Left-to-right for $\div$ and $\times$: $24 \div 4 = 6$, $6 \times 2 = 12$. Then $3^2 = 9$. Total: $12 + 9 - 1 = 20$.

Q8. [Hard] Solve for x, y: $2x + 3y = 13$, $3x - 2y = 0$.

A. $x=2, y=3$ B. $x=3, y=2$ C. $x=2, y=4.5$ D. $x=4, y=1.67$

Answer: A. Why: From eq2: $3x=2y \rightarrow y=1.5x$. Substitute into eq1: $2x+3(1.5x)=13 \rightarrow 2x+4.5x=13 \rightarrow 6.5x=13 \rightarrow x=2$, $y=1.5 \times 2=3$. Check eq2: $3(2)-2(3)=6-6=0 \checkmark$.

Q9. [Hard] [Trick] If $x^2 + 1/x^2 = 47$, find $x + 1/x$ (positive value).

A. 6 B. 7 C. 8 D. 9

Answer: B. Why: $(x+1/x)^2 = x^2 + 1/x^2 + 2 = 47+2 = 49 \rightarrow x+1/x = 7$ (taking positive root).

A14. Probability

A14.1 Basic Definitions [Level: Basic]

- **Experiment:** an action with an uncertain outcome. **Sample Space (S):** set of all possible outcomes. **Event:** a subset of the sample space.

$$\text{Probability } P(A) = (\text{Number of Favorable Outcomes}) / (\text{Total Number of Outcomes})$$

$$0 \leq P(A) \leq 1$$

- **Sure event:** $P = 1$ (certain to happen). **Impossible event:** $P = 0$ (cannot happen).

A14.2 Complementary Events — The Most Useful Shortcut [Level: Basic-Medium]

$$P(\text{not } A) = 1 - P(A)$$

- TRICK: for any "at least one" question, NEVER compute the "at least one" case directly (it usually has many sub-cases). Instead compute:

$$P(\text{at least one}) = 1 - P(\text{none})$$

- Worked example: 3 coins tossed. Find $P(\text{at least one head})$.
 $P(\text{no heads}) = P(\text{all tails}) = (1/2)^3 = 1/8$. $P(\text{at least one head}) = 1 - 1/8 = 7/8$.

A14.3 Mutually Exclusive, Exhaustive & Independent Events [Level: Medium]

Term	Meaning	Formula
Mutually Exclusive	Cannot both occur in one trial	$P(A \text{ or } B) = P(A) + P(B)$
Exhaustive	Together cover the entire sample space	Sum of P over all = 1
Independent	Outcome of one doesn't affect the other	$P(A \text{ and } B) = P(A) \times P(B)$

- **TRAP:** Mutually exclusive and independent are **NOT the same**, and in fact are almost opposites: if A and B are mutually exclusive with $P(A) > 0$ and $P(B) > 0$, then $P(A \text{ and } B) = 0$ (they can't both happen) — but independence requires $P(A \text{ and } B) = P(A) \times P(B)$, which is nonzero. So two mutually exclusive events with nonzero probability can **never** be independent.

A14.4 Odds vs Probability [Level: Medium]

Quantity	Definition
Odds in favor	Favorable : Unfavorable
Odds against	Unfavorable : Favorable
Probability	Favorable : Total (Favorable + Unfavorable)

- **TRAP:** "Odds of 3 to 2 in favor" means $P = 3/(3+2) = 3/5$, **not** $3/2$. Always convert odds (a ratio of two parts) into probability (a fraction of the total) by adding the two parts for the denominator.

A14.5 Classical Problem Types [Level: Medium-Hard]

- **Dice:** 2 dice $\rightarrow$ 36 total outcomes. Sum frequencies: sum=7 is most frequent (6 ways); sums 2 and 12 are rarest (1 way each); frequency rises 1,2,3,4,5,6 from sum=2 to sum=7, then falls symmetrically 5,4,3,2,1 from sum=8 to 12.
- **Cards:** standard deck = 52 cards = 4 suits $\times$ 13 ranks. Face cards = 12 (K,Q,J $\times$ 4 suits). Red cards = 26, Black cards = 26. Know these counts cold — most card questions are direct favorable/total plug-ins once you have them memorized.
- **Coins:** n coins $\rightarrow 2^n$ total outcomes.
- **"53 Sundays in a leap year":** a leap year has 366 days = 52 weeks + 2 extra days. Each of the 52 full weeks contributes exactly one of each weekday, guaranteeing 52 Sundays; the 2 "extra" days determine which day(s) occur a 53rd time.

$$P(53 \text{ Sundays in a leap year}) = P(\text{a specific day is among the 2 extra days}) = 2/7$$

(The 2 extra days are always 2 CONSECUTIVE weekdays, e.g. Mon-Tue, Tue-Wed, etc. — 7 possible consecutive pairs, of which exactly 2 contain Sunday.)

- **TRAP:** For a **non-leap year** (365 days = 52 weeks + 1 extra day), the question becomes "P(53 of a given day)" = $1/7$, not $2/7$ — don't reuse the leap-year formula for a normal year.

Questions

Q1. [Easy] A single die is rolled. Find $P(\text{getting a number greater than 4})$.

A. $1/6$ B. $1/3$ C. $1/2$ D. $2/3$

Answer: B. Why: Favorable = {5,6} = 2 outcomes. $P = 2/6 = 1/3$.

Q2. [Easy] Two coins are tossed. Find $P(\text{exactly one head})$.

A. $1/4$ B. $1/2$ C. $3/4$ D. 1

Answer: B. Why: Outcomes: HH, HT, TH, TT. Exactly one head: HT, TH = $2/4 = 1/2$.

Q3. [Medium] [Trick] A bag has 5 red and 4 blue balls. Two balls are drawn at random. Find $P(\text{at least one red})$.

A. $5/6$ B. $7/9$ C. $4/9$ D. $8/9$

Answer: A. Why: $P(\text{no red}) = P(\text{both blue}) = (4C2)/(9C2) = 6/36 = 1/6$. $P(\text{at least one red}) = 1 - 1/6 = 5/6$.

Q4. [Medium] Odds in favor of an event are 4:3. Find the probability of the event.

A. $4/3$ B. $3/7$ C. $4/7$ D. $3/4$

Answer: C. Why: $P = \text{favorable}/(\text{favorable} + \text{unfavorable}) = 4/(4+3) = 4/7$.

Q5. [Medium] A card is drawn from a standard deck. Find P(getting a face card).

A. $3/13$ B. $1/13$ C. $4/13$ D. $12/52$ simplified to $3/13$

Answer: A. **Why:** Face cards = 12. $P = 12/52 = 3/13$.

Q6. [Hard] What is the probability that a leap year selected at random will contain 53 Sundays?

A. $1/7$ B. $2/7$ C. $3/7$ D. $1/2$

Answer: B. **Why:** Leap year = 366 days = 52 weeks + 2 extra days; P(a specific day among the 2 extra) = $2/7$.

Q7. [Hard] [Trap] Two dice are rolled. Find P(sum = 7).

A. $1/6$ B. $1/12$ C. $5/36$ D. $1/9$

Answer: A. **Why:** Sum=7 has 6 favorable combinations out of 36 total: (1,6)(2,5)(3,4)(4,3)(5,2)(6,1). $P = 6/36 = 1/6$.

Q8. [Hard] From a group of 6 men and 4 women, a committee of 3 is formed at random. Find P(at least 2 women).

A. $1/6$ B. $3/10$ C. $1/3$ D. $2/5$

Answer: C. **Why:** Total ways = $10C3 = 120$. Exactly 2 women+1 man = $4C2 \times 6C1 = 6 \times 6 = 36$. Exactly 3 women = $4C3 = 4$. Favorable = $36+4=40$. $P=40/120=1/3$.

Q9. [Hard] [Trap] Events A and B are mutually exclusive with $P(A)=0.3$ and $P(B)=0.4$. Are A and B independent?

A. Yes, always B. No — mutually exclusive events with nonzero probability can never be independent C. Cannot be determined D. Only if $P(A)=P(B)$

Answer: B. **Why:** Mutually exclusive means $P(A \text{ and } B)=0$, but independence requires $P(A \text{ and } B)=P(A)P(B)=0.12 \neq 0$ — contradiction, so they cannot be independent.

A15. Permutation & Combination

A15.1 Fundamental Counting Principle [Level: Basic]

- If one task can be done in m ways and a second independent task in n ways, both together can be done in $m \times n$ ways.

A15.2 Permutation vs Combination [Level: Basic]

$nPr = n! / (n-r)!$ [ORDER MATTERS – arrangement]
 $nCr = n! / (r! (n-r)!)$ [ORDER DOESN'T MATTER – selection]
 $nCr = nC(n-r)$ (symmetry property)
 Sum of nCr for $r=0$ to $n = 2^n$ (total number of subsets of an n -element set)

- TRAP:** use **P** when the question involves arranging/ranking/ordering (seating, passwords, race positions); use **C** when it's just selecting/grouping (forming a committee, choosing a team) with no internal order.

A15.3 Permutations with Repetition [Level: Basic-Medium]

- Unlimited repetition allowed:** r positions, n choices each $\rightarrow n^r$.
- No repetition:** nPr .
- Repeated letters in a word:** arrangements of a word with letter-count repeats $p_1, p_2, \dots = n! / (p_1! \times p_2! \times \dots)$.
Worked example: arrangements of "MISSISSIPPI" (11 letters: M=1, I=4, S=4, P=2) = $11! / (1!4!4!2!) = 34,650$.

A15.4 Grouping Tricks (Vowels/Consonants Together) [Level: Medium]

- TRICK:** to keep a set of letters together, treat that whole set as ONE block. Arrange (blocks + remaining individual letters), THEN separately arrange the letters within the block, and multiply the two counts.
Worked example: arrange letters of "TABLE" so all vowels (A,E) are together. Treat (AE) as 1 block + T,B,L = 4 units $\rightarrow 4!$ arrangements of units $\times 2!$ arrangements within the block = $24 \times 2 = 48$.

A15.5 Circular Permutations [Level: Medium-Hard]

n distinct objects around a circle (rotations considered identical): $(n-1)!$
 If clockwise and anticlockwise arrangements are also considered identical (necklace/bracelet): $(n-1)!/2$

- TRAP:** a circular TABLE arrangement uses $(n-1)!$ (rotation-only symmetry); a NECKLACE or BRACELET arrangement uses $(n-1)!/2$ (rotation AND flip symmetry) — mixing these up is the single most common circular-permutation error.

A15.6 Rank of a Word [Level: Hard]

- Algorithm:** for each letter position (left to right), count how many of the REMAINING available letters are alphabetically SMALLER than the actual letter used there; multiply that count by (number of letters remaining after this position!); sum these products across all positions; add 1 at the end.
- Worked example: find the rank of the word "CAB" (all distinct letters, alphabetically: A,B,C).
Position 1 (C): smaller unused letters = {A,B} = $2 \rightarrow 2 \times 2! = 4$.
Position 2 (A): remaining letters {A,B}; smaller than A = none = $0 \rightarrow 0 \times 1! = 0$.
Position 3 (B): remaining letter {B}; smaller = $0 \rightarrow 0 \times 0! = 0$.
Rank = $4+0+0+1 = 5$.
- TRAP:** if the word has REPEATED letters, divide each position's contribution by the factorial of repeat-counts among the remaining letters at that step (same repeated-permutation adjustment as A15.3).

A15.7 Other Special Applied Problems [Level: Hard]

Handshakes / round-robin matches among n people = $nC2 = n(n-1)/2$
 Diagonals of an n -sided polygon = $nC2 - n$
 Derangements (no element in its original position):
 $D(n) = n! \times [1 - 1/1! + 1/2! - 1/3! + \dots + (-1)^n / n!]$
 Quick estimate for large n : $D(n) \sim n!/e$ ($e \sim 2.718$)

- TRICK:** $nC2 = n(n-1)/2$ is the single most reused sub-formula in this entire chapter — handshakes, matches, line-intersections, diagonals all reduce to it.

Questions

Q1. [Easy] In how many ways can 5 people be seated in a row?

A. 60 B. 120 C. 24 D. 720

Answer: B. **Why:** $5! = 120$.

Q2. [Easy] How many 3-letter codes can be formed from A-Z if repetition is allowed?

A. 15600 B. 17576 C. 2600 D. $26^3 = 17576$

Answer: B. **Why:** $26^3 = 17,576$.

Q3. [Medium] In how many ways can the letters of "APPLE" be arranged?

A. 120 B. 60 C. 20 D. 24

Answer: B. **Why:** 5 letters with P repeated twice: $5!/2! = 120/2 = 60$.

Q4. [Medium] [Trick] How many ways can the letters of "APPLE" be arranged so that both P's are together?

A. 24 B. 12 C. 48 D. 60

Answer: A. **Why:** Treat "PP" as one block: units = (PP),A,L,E = 4 units $\rightarrow 4! = 24$ (no internal arrangement needed since both letters in the block are identical).

Q5. [Medium] In how many ways can a committee of 3 be chosen from 7 people?

A. 210 B. 35 C. 21 D. 42

Answer: B. Why: $7C3 = 7!/(3!4!) = 35$.

Q6. [Medium] [Trap] 6 people are to be seated around a circular table. How many distinct arrangements are possible?
A. 720 B. 120 C. 360 D. 24

Answer: B. Why: Circular (rotation-only symmetry): $(n-1)! = 5! = 120$.

Q7. [Hard] [Trap] In how many ways can 6 identical-looking beads be arranged to form a necklace?
A. 120 B. 60 C. 720 D. 24

Answer: B. Why: Necklace (rotation AND flip symmetry): $(n-1)!/2 = 5!/2 = 120/2 = 60$.

Q8. [Hard] Find the rank of the word "RANK" if all letters are distinct and arranged alphabetically (A,K,N,R).
A. 18 B. 19 C. 20 D. 21

Answer: C. Why: Position1 'R': letters smaller than R among {A,K,N,R} = {A,K,N} = 3 $\rightarrow 3 \times 3! = 18$. Position2 'A': remaining {A,K,N}; smaller than A = 0 $\rightarrow 0 \times 2! = 0$. Position3 'N': remaining {K,N}; smaller than N = {K} = 1 $\rightarrow 1 \times 1! = 1$. Position4 'K': remaining {K}; smaller = 0 $\rightarrow 0$. Sum = 18+0+1+0 = 19; Rank = 19+1 = **20**.

Q9. [Hard] In how many ways can 8 people shake hands with each other exactly once?
A. 56 B. 28 C. 64 D. 36

Answer: B. Why: $8C2 = 8 \times 7/2 = 28$.

Q10. [Hard] How many diagonals does a hexagon (6 sides) have?
A. 9 B. 12 C. 15 D. 6

Answer: A. Why: Diagonals = $nC2 - n = 15 - 6 = 9$.

Q11. [Hard] [Trick] Find the number of derangements of 4 distinct objects (D(4)).
A. 6 B. 9 C. 12 D. 3

Answer: B. Why: $D(4) = 4![1 - 1/1! + 1/2! - 1/3! + 1/4!] = 24[1 - 1 + 0.5 - 0.1667 + 0.0417] = 24 \times 0.375 = 9$.

Q12. [Hard] How many 4-digit numbers (no repetition of digits) can be formed using digits 1-9 that are divisible by 5?
A. 224 B. 336 C. 448 D. 280

Answer: B. Why: No zero is available among digits 1-9, so the last digit must be 5 (the only multiple of 5 in the set). The remaining 3 positions are filled from the remaining 8 digits: $8P3 = 8 \times 7 \times 6 = 336$.

Q13. [Hard] How many ways can 3 boys and 2 girls be arranged in a row such that the 2 girls are always together?
A. 24 B. 48 C. 12 D. 120

Answer: B. Why: Treat 2 girls as 1 block + 3 boys = 4 units $\rightarrow 4! = 24$ arrangements of units $\times 2!$ arrangement within the block = $24 \times 2 = 48$.

A16. Quant Miscellaneous: Venn Diagrams, Age Problems & Partnership

A16.1 Venn Diagrams (Set Theory) [Level: Basic → Hard]

- This is the **quantitative/counting** use of Venn diagrams (contrast with the *logical-relationship* Venn diagrams used for classifying word-categories like "Diwali, Eid, Festival" — those test category containment/overlap conceptually, not counting).

$$\begin{aligned} \text{2-set: } |A \cup B| &= |A| + |B| - |A \cap B| \\ \text{3-set: } |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C| \end{aligned}$$

- TRAP: "Only A"** (exclusively A, not B) = $|A| - |A \cap B|$ — very commonly confused with just $|A|$ alone.
- TRICK for 3-region "only" problems: when you're told the sizes of pairwise-overlap regions (some of which may already EXCLUDE the triple-overlap, i.e., are "exactly two" regions, not "at least two"), always clarify whether a given overlap number is "exactly 2 categories" or "at least 2 (could include all 3)" — this distinction drives which subtraction to use.

Worked Example (Medium) — verified real exam question

In a survey: 10% are students+parents only, 10% are students+teachers+parents (all three), 15% are teachers+parents only, 35% are students+teachers only. Find % of only-teachers, only-parents, only-students (each group's total membership is 100%... i.e., each of the three named categories individually sums to 100% of ITS OWN circle when combined with its overlaps).

Only teachers = $100 - (35 + 10 + 15) = 40\%$. Only parents = $100 - (10 + 10 + 15) = 65\%$. Only students = $100 - (35 + 10 + 10) = 45\%$.

Worked Example (Hard) — verified real exam question

In a group, 6 speak Tamil, 15 speak Hindi, 6 speak Gujarati; no one speaks any other language. 2 people speak exactly two languages, 1 person speaks all three. Find total group size.

Only Tamil = $6 - 2 - 1 = 3$ (subtracting the 2-language and 3-language overlaps attributed here), Only Hindi = $15 - 2 - 1 = 12$, Only Gujarati = $6 - 1 = 5$ (assumed the two "2-language" speakers are Tamil+Hindi, not involving Gujarati, per the standard version of this question). Total = $(3 + 12 + 5)$ [only-one] + 2 [exactly-two] + 1 [all-three] = **23**.

A16.2 Age Problems [Level: Basic → Hard]

- Universal method: assign variables to PRESENT ages, translate every past/future condition into an equation, solve the linear system.
- TRAP (sign confusion): "**X years ago**" → $\text{present_age} - X$. "**After X years / X years hence**" → $\text{present_age} + X$. Getting this backwards is the single most common error in this topic — always re-read which direction the shift goes before writing the equation.
- Ratio-based ages: if "ratio of ages is a:b," let the ages be ax and bx (NOT a and b directly) — this preserves the ratio while giving you one variable to solve for.

Ways to Solve — Worked Examples (Basic → Medium, all hand-verified)

(1) Twice son's age + father's age = 70; twice father's age + son's age = 95. Find both.

$2S + F = 70$, $2F + S = 95$. Solving: **Father=40, Son=15**.

(2) Sum of father & son's ages = 90. 15 years ago, father's age was $3 \times$ son's age.

$F + S = 90$, $F - 15 = 3(S - 15)$. Solving: **Father=60, Son=30**.

(3) Sum of ages = 35, product of ages = 150.

$F + S = 35$, $F \times S = 150$ → quadratic $x^2 - 35x + 150 = 0$ → $x = 30$ or 5 . **Father=30, Son=5**.

(4) I am $3 \times$ my son's age. After 5 years I'll be $2.5 \times$ his age. Find both present ages.

$F = 3S$, $F + 5 = 2.5(S + 5)$. Solving: **Father=45, Son=15**.

Worked Examples (Hard) — ratio-based

(5) Ratio of A & B's ages 2 years ago was 5:3. After 2 years (from now), the ratio will be 7:5. Find present ages.

$(A-2):(B-2) = 5:3$ → $3A = 5B - 4$. $(A+2):(B+2) = 7:5$ → $5A = 7B + 4$. Solving simultaneously: **A=12, B=8**.

- TRAP:** Watch for ambiguous phrasing like "one year ago the ratio was X; **after 3 years** the ratio will be Y" — "after 3 years" can mean *3 years from now* OR *3 years from that past reference point*. A well-posed question should make this unambiguous (e.g., say "3 years from now" explicitly); if a question seems to give a non-integer/inconsistent answer under one reading, try the other reading before concluding an error.

A16.3 Partnership [Level: Basic → Hard]

$$\text{Profit-sharing ratio} = C_1 \times T_1 : C_2 \times T_2 : \dots$$

(Capital invested $\times$ Time invested, for EACH partner)

- TRAP:** If all partners invest for the SAME duration, ratio simplifies to just capital ratio — but the moment durations DIFFER, you **must** weight by time; splitting profit by capital alone (ignoring time) is the most common partnership error.
- Working partner:** may receive a fixed management fee/commission off the top BEFORE the remaining profit is split in the capital $\times$ time ratio among all partners — watch for this two-step split in "working vs sleeping partner" questions.
- Ways to solve capital-withdrawal/addition mid-year: split each partner's investment into time-weighted segments and sum them (e.g., invests 5000 for 4 months then withdraws to 3000 for remaining 8 months → effective $C \times T = 5000 \times 4 + 3000 \times 8$).

Worked Example (Medium)

A invests Rs. 6000 for the full year; B invests Rs. 9000 but only for 8 months. Find their profit-sharing ratio.

$C \times T$: A = $6000 \times 12 = 72,000$. B = $9000 \times 8 = 72,000$. Ratio = **1:1** (equal shares, despite B investing more capital, because A invested for longer).

Questions

(Easy → Hard)

Q1. [Easy] In a class, 60% play cricket, 50% play football, 30% play both. What % play at least one game?
A. 70% B. 80% C. 90% D. 100%

Answer: B. Why: $|CuF| = 60+50-30 = 80\%$.

Q2. [Easy] [Trap] In Q1's class, what % play ONLY cricket (not football)?

A. 60% B. 30% C. 20% D. 40%

Answer: B. Why: Only cricket = $60-30 = 30\%$ — not the full 60%.

Q3. [Medium] A father's age is 4 times his son's age now. 5 years ago, the father was 7 times as old as his son was then. Find their present ages.

A. Father 32, Son 8 B. Father 28, Son 7 C. Father 40, Son 10 D. Father 36, Son 9

Answer: C. Why: $F=4S$, $F-5=7(S-5) \rightarrow 4S-5=7S-35 \rightarrow 30=3S \rightarrow S=10$, $F=40$. Check: $F-5=35=7 \times (S-5)=7 \times 5=35 \checkmark$.

Q4. [Medium] Ram is 3 times as old as Shyam. After 15 years, Ram will be twice as old as Shyam. Find Shyam's present age.

A. 10 B. 12 C. 15 D. 20

Answer: C. Why: $R=3S$, $R+15=2(S+15) \rightarrow 3S+15=2S+30 \rightarrow S=15$.

Q5. [Hard] A invests Rs. 2000 for 6 months, B invests Rs. 3000 for 4 months, and C invests Rs. 6000 for 3 months. Find C's share of a total profit of Rs. 700.

A. Rs. 200 B. Rs. 250 C. Rs. 300 D. Rs. 350

Answer: C. Why: $C \times T$: $A=2000 \times 6=12000$, $B=3000 \times 4=12000$, $C=6000 \times 3=18000$. Ratio = $12:12:18 = 2:2:3$ (7 parts). C's share = $700 \times 3/7 = \text{Rs. } 300$.

Q6. [Hard] In a group, 10 speak Spanish, 20 speak German, and 8 speak Italian; no one speaks any other language. 3 people speak exactly Spanish+German (and no other), and 2 people speak all three languages. Find the total group size.

A. 28 B. 29 C. 31 D. 33

Answer: C. Why: Only Spanish = $10-3-2=5$. Only German = $20-3-2=15$. Only Italian = $8-2=6$ (no separate two-language-with-Italian group given). Only-one-language total = $5+15+6=26$. Total = $26 + 3$ (exactly-two) + 2 (all-three) = **31**.

A17. Sequence & Series, Statistics, Surds & Indices

A17.1 Arithmetic & Geometric Progressions [Level: Basic]

Arithmetic Progression (AP): common difference d is constant
 n th term = $a + (n-1)d$
 Sum of n terms = $n/2 [2a + (n-1)d] = n/2 \times (\text{first term} + \text{last term})$

Geometric Progression (GP): common ratio r is constant
 n th term = $a \cdot r^{(n-1)}$
 Sum of n terms = $a(r^n - 1)/(r - 1)$ [$r \neq 1$]

- TRICK: for an AP, Sum = (number of terms) $\times$ (average of first and last term) — faster than the full formula when you can spot first/last quickly (e.g., sum of 1 to 100 = $100 \times (1+100)/2 = 5050$, no need to invoke " $n/2[2a+(n-1)d]$ " explicitly).

A17.2 Statistics: Mean, Median, Mode [Level: Basic-Medium]

- **Mean** (raw data) = Sum of observations / Number of observations.
- **Median** = middle value when data is sorted (average of two middle values if count is even).
- **Mode** = most frequently occurring value.

Grouped data — Assumed Mean (Step-Deviation) shortcut:
 Mean = $A + h \times [\text{sum}(f_i \cdot u_i) / \text{sum}(f_i)]$
 where A = assumed mean (any convenient class mark), h = class width, $u_i = (x_i - A)/h$

Median (grouped) = $L + [(N/2 - CF) / f] \times h$
 L = lower boundary of median class, CF = cumulative frequency before median class,
 f = frequency of median class, h = class width

Mode (grouped) = $L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$
 f_1 = modal class frequency, f_0 = frequency of preceding class, f_2 = frequency of following class

- TRICK: use the step-deviation method (assumed mean + u_i substitution) whenever raw class-mark values are large — it converts big numbers into small integers, making hand calculation fast.

A17.3 Laws of Indices (Exponents) [Level: Basic — must be automatic]

$a^m \times a^n = a^{(m+n)}$
 $a^m / a^n = a^{(m-n)}$
 $(a^m)^n = a^{(mn)}$
 $a^0 = 1$ (for $a \neq 0$)
 $a^{(-n)} = 1/a^n$
 $a^{(1/n)} = n\text{-th root of } a$
 $(ab)^n = a^n \cdot b^n$

A17.4 Surds — Simplification & Rationalization [Level: Medium]

- A **surd** is an irrational root, e.g. $\sqrt{2}$, $\sqrt{3}$.
- **Rationalization**: remove a surd from the denominator by multiplying by its conjugate.

$$1/(\sqrt{a} + \sqrt{b}) \times (\sqrt{a} - \sqrt{b})/(\sqrt{a} - \sqrt{b}) = (\sqrt{a} - \sqrt{b})/(a - b)$$

- **Comparing surds of different orders** (e.g., is $\sqrt[3]{5}$ bigger or $\sqrt{3}$?): raise both to a common index (LCM of the root orders), then compare the resulting integer powers.
 Example: compare $\sqrt[3]{5}$ (cube root) and $\sqrt{3}$ (square root). LCM(2,3)=6. $\sqrt[3]{5} = 5^{(1/3)} = 5^{(2/6)} = (5^2)^{(1/6)} = 25^{(1/6)}$. $\sqrt{3} = 3^{(1/2)} = 3^{(3/6)} = (3^3)^{(1/6)} = 27^{(1/6)}$.
 Since $27 > 25$, $\sqrt{3} > \sqrt[3]{5}$.

A17.5 Comparing Exponent Towers [Level: Hard]

- TRICK: to compare a^m vs b^n , try to express both with a **common exponent** by rewriting the base as a power, then compare the resulting bases directly.
- Worked example: Which is bigger, 2^{300} or 3^{200} ?

$2^{300} = (2^3)^{100} = 8^{100}$
 $3^{200} = (3^2)^{100} = 9^{100}$
 Since $9 > 8$, $3^{200} > 2^{300}$.

A17.6 Basic Arithmetic Speed Tricks [Level: Medium]

- **Squaring a number ending in 5**: for $N = (10a+5)$, $N^2 = a(a+1)$ followed by "25". Example: $65^2 \rightarrow a=6$, $a(a+1)=6 \times 7=42 \rightarrow$ answer **4225**.
- **Multiplying near a round base**: for numbers close to 100 (or another power of 10), use $(100-x)(100-y) = 10000 - 100(x+y) + xy$ — faster than direct long multiplication.

Questions

Q1. [Easy] Find the sum of the first 20 natural numbers.
 A. 200 B. 210 C. 190 D. 220

Answer: B. **Why**: Sum = $n(n+1)/2 = 20 \times 21/2 = 210$.

Q2. [Easy] Simplify: $2^5 \times 2^{-2}$

A. 8 B. 2 C. 16 D. 4

Answer: A. Why: $2^5 \times 2^{-2} = 2^{(5-2)} = 2^3 = 8$.

Q3. [Medium] Rationalize: $1/(\sqrt{7} - \sqrt{5})$

A. $(\sqrt{7}+\sqrt{5})/2$ B. $(\sqrt{7}-\sqrt{5})/2$ C. $\sqrt{7}+\sqrt{5}$ D. $(\sqrt{7}+\sqrt{5})/12$

Answer: A. Why: Multiply by $(\sqrt{7}+\sqrt{5})/(\sqrt{7}+\sqrt{5})$: denominator becomes $7-5=2$, giving $(\sqrt{7}+\sqrt{5})/2$.

Q4. [Medium] Find 85^2 .

A. 7225 B. 7025 C. 7125 D. 7325

Answer: A. Why: $a=8$, $a(a+1)=8 \times 9=72$, append 25 → **7225**.

Q5. [Medium] [Trick] Which is greater: $\sqrt[3]{5}$ or $\sqrt{3}$?

A. $\sqrt[3]{5}$ B. $\sqrt{3}$ C. Equal D. Cannot be determined

Answer: B. Why: Raise both to power 6 (LCM of 3,2): $(\sqrt[3]{5})^6=5^2=25$, $(\sqrt{3})^6=3^3=27$. Since $27>25$, $\sqrt{3}$ is greater.

Q6. [Hard] [Trick] Which is greater: 2^{300} or 3^{200} ?

A. 2^{300} B. 3^{200} C. Equal D. Cannot be determined

Answer: B. Why: $2^{300}=(2^3)^{100}=8^{100}$; $3^{200}=(3^2)^{100}=9^{100}$. Since $9>8$, 3^{200} is greater.

Q7. [Hard] The mean of 10 observations is 24. If one observation of value 33 is removed, find the new mean.

A. 22 B. 23 C. 24 D. 25

Answer: B. Why: Original total = $24 \times 10 = 240$. New total = $240 - 33 = 207$, new count = 9. New mean = $207/9 = 23$.

Q8. [Hard] Find the sum: $3 + 6 + 9 + \dots + 300$.

A. 15150 B. 14850 C. 15000 D. 15300

Answer: A. Why: AP with $a=3, d=3$, last term = 300 → $n = (300-3)/3 + 1 = 100$. Sum = $n \times (\text{first} + \text{last})/2 = 100 \times 303/2 = 15150$.

A18. Mensuration & Geometry

A18.1 2D Mensuration Reference [Level: Basic]

Shape	Area	Perimeter
Square (side a)	a^2	$4a$
Rectangle (l,b)	$l \times b$	$2(l+b)$
Triangle (base b, height h)	$(1/2) \times b \times h$	sum of sides
Triangle (Heron's, sides a,b,c, $s=(a+b+c)/2$)	$\sqrt{s(s-a)(s-b)(s-c)}$	$a+b+c$
Equilateral triangle (side a)	$(\sqrt{3}/4) a^2$	$3a$
Circle (radius r)	πr^2	$2\pi r$
Parallelogram (base b, height h)	$b \times h$	$2(\text{sum of adjacent sides})$
Rhombus (diagonals d1,d2)	$(1/2) d_1 d_2$	$4 \times \text{side}$
Trapezium (parallel sides a,b, height h)	$(1/2)(a+b) h$	sum of sides

A18.2 3D Mensuration Reference [Level: Basic-Medium]

Solid	Volume	Curved/Lateral Surface Area	Total Surface Area
Cube (side a)	a^3	$4a^2$	$6a^2$
Cuboid (l,b,h)	lbh	$2h(l+b)$	$2(lb+bh+hl)$
Cylinder (r,h)	$\pi r^2 h$	$2\pi r h$	$2\pi r(h+r)$
Cone (r,h, slant $l=\sqrt{r^2+h^2}$)	$(1/3) \pi r^2 h$	$\pi r l$	$\pi r(l+r)$
Sphere (r)	$(4/3) \pi r^3$	—	$4\pi r^2$
Hemisphere (r)	$(2/3) \pi r^3$	$2\pi r^2$	$3\pi r^2$

- TRAP:** "Curved/Lateral Surface Area" excludes the flat base(s); "Total Surface Area" includes them. A question asking for the cost of painting only the curved surface of a cylindrical pillar needs the CSA, not TSA — misreading this is the #1 error in this chapter.

A18.3 Circle Theorems [Level: Medium]

- Angle in a semicircle = 90° (angle subtended by a diameter at any point on the circle).
- Angle subtended by an arc at the center = $2 \times$ angle subtended by the same arc at the circumference.

A18.4 Pythagorean Triples (Speed Trick) [Level: Medium]

- TRICK: memorize these common triples (and their multiples) to instantly solve right-triangle problems without computing square roots:

3-4-5 5-12-13 8-15-17 7-24-25
(and multiples: 6-8-10, 9-12-15, 10-24-26, etc.)

Questions

Q1. [Easy] Find the area of a rectangle with length 12 cm and breadth 7 cm.

A. 19 cm² B. 84 cm² C. 38 cm² D. 96 cm²

Answer: B. **Why:** Area = $l \times b = 12 \times 7 = 84$.

Q2. [Easy] [Trick] A right triangle has legs 9 cm and 12 cm. Find the hypotenuse.

A. 13 cm B. 15 cm C. 21 cm D. 18 cm

Answer: B. **Why:** 9-12-15 is 3x the 3-4-5 triple $\rightarrow$ hypotenuse = 15.

Q3. [Medium] Find the area of an equilateral triangle with side 8 cm.

A. $16\sqrt{3}$ cm² B. $32\sqrt{3}$ cm² C. $8\sqrt{3}$ cm² D. $24\sqrt{3}$ cm²

Answer: A. **Why:** Area = $(\sqrt{3}/4)a^2 = (\sqrt{3}/4)(64) = 16\sqrt{3}$.

Q4. [Medium] [Trap] Find the curved surface area of a cone with radius 7 cm and slant height 10 cm. (Use $\pi=22/7$)

A. 220 cm² B. 440 cm² C. 154 cm² D. 374 cm²

Answer: A. **Why:** CSA = $\pi r l = (22/7)(7)(10) = 220$ cm² (NOT total surface area, which would add $\pi r^2 = 154$ more).

Q5. [Medium] Find the volume of a cylinder with radius 7 cm and height 10 cm. (Use $\pi=22/7$)

A. 1540 cm³ B. 3080 cm³ C. 440 cm³ D. 770 cm³

Answer: A. **Why:** Volume = $\pi r^2 h = (22/7)(49)(10) = 1540$ cm³.

Q6. [Hard] Find the area of a triangle with sides 13, 14, 15 cm (use Heron's formula).

A. 84 cm² B. 91 cm² C. 75 cm² D. 98 cm²

Answer: A. **Why:** $s=(13+14+15)/2=21$. Area = $\sqrt{21(21-13)(21-14)(21-15)} = \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$.

Q7. [Hard] [Trap] A hemispherical bowl of radius 21 cm is to be painted only on the outer curved surface. Find the area to be painted. ($\pi=22/7$)

A. 1386 cm² B. 2772 cm² C. 4158 cm² D. 924 cm²

Answer: B. **Why:** Curved surface area of hemisphere = $2\pi r^2 = 2 \times (22/7) \times 441 = 2772$ cm² (total surface area $3\pi r^2 = 4158$ would incorrectly include the flat circular base).

Q8. [Hard] The diagonals of a rhombus are 16 cm and 12 cm. Find its area.

A. 96 cm² B. 192 cm² C. 48 cm² D. 144 cm²

Answer: A. **Why:** Area = $(1/2)(d_1)(d_2) = (1/2)(16)(12) = 96$.

Q9. [Hard] A ladder 25 m long is placed against a wall such that its foot is 7 m from the wall's base. How high up the wall does the ladder reach?
A. 24 m B. 18 m C. 20 m D. 22 m

Answer: A. **Why:** 7-24-25 is a Pythagorean triple $\rightarrow$ height = 24 m.

A19. Speed Mathematics (Fast Calculation Tricks)

A19.1 Square Root of a Perfect Square [Level: Basic]

- TRICK: (1) Pair digits from the right to estimate the digit-count of the root. (2) Use the unit digit of the square to narrow the possible unit digit of the root using this table:

Square ends in:	1	4	5	6	9	0
Root ends in:	1/9	2/8	5	4/6	3/7	0

(3) Bracket the leading digit(s) using nearby known squares (e.g., $20^2=400$, $30^2=900$) to pick between the two unit-digit candidates.

- Worked example: $\sqrt{7056}$. Pairs: "70" "56" → 2-digit root. Last digit of 7056 is 6 → root ends in 4 or 6. $80^2=6400$, $90^2=8100$ → root is between 80-90, closer to 84 (since 7056 is nearer 6400). Try 84: $84^2=7056$. $\sqrt{\text{Root}} = 84$.

A19.2 Square Root of an Imperfect Number (Approximation) [Level: Medium]

$$\sqrt{\text{N}} \sim R + (N - R^2) / (2R) \quad \text{where } R = \text{nearest lower perfect-square root}$$

- Worked example: $\sqrt{50}$. Nearest lower perfect square = 49 ($R=7$). $\sqrt{50} \approx 7 + (50-49)/(14) = 7 + 0.071 = \mathbf{7.071}$ (actual ≈ 7.0711 — very close).

A19.3 Fast Multiplication Near a Round Base [Level: Medium]

$$(\text{Base}+x)(\text{Base}+y) = \text{Base}^2 + \text{Base}(x+y) + xy$$

- Worked example: 104×107 (base=100, $x=4$, $y=7$): $100^2 + 100(11) + 28 = 10000 + 1100 + 28 = \mathbf{11128}$.
- Worked example (numbers below base): 97×96 (base=100, $x=-3$, $y=-4$): $10000 + 100(-7) + 12 = 10000 - 700 + 12 = \mathbf{9312}$.

A19.4 Clock Angle Formula [Level: Medium]

Angle between hour and minute hands = $|30H - 5.5M|$ degrees (H = hour 0-11, M = minutes)
If result > 180, subtract from 360 to get the smaller angle.

- Worked example: angle at 3:40. $|30(3) - 5.5(40)| = |90 - 220| = 130^\circ$.
- (See the dedicated Clocks chapter for the full theory behind this formula and additional clock problem types.)

A19.5 Counting Figures — General Approach [Level: Medium-Hard]

- TRICK: don't count triangles/squares by eye. Label every intersection point, systematically count how many ways base-lines/sub-triangles can combine using combinatorics (nCr -style counting per row/layer of the figure), and add contributions from each "size level" of shape separately.
- (See the dedicated Analytical Reasoning — Counting Figures chapter for the complete systematic method with worked examples.)

Questions

Q1. [Easy] Find $\sqrt{11025}$ using the digit-pattern method.

A. 105 B. 115 C. 95 D. 125

Answer: A. Why: Last digit 5 → root ends in 5. Pairs "1" "10" "25" → 3-digit root roughly, but 11025 is between $100^2=10000$ and $110^2=12100$, closer to $105^2=11025$. Check: $105^2=11025$ ✓.

Q2. [Easy] Multiply 98×97 using the base-100 method.

A. 9506 B. 9606 C. 9412 D. 9706

Answer: A. Why: Base=100, $x=-2$, $y=-3$: $10000 + 100(-5) + 6 = 10000 - 500 + 6 = 9506$.

Q3. [Medium] Approximate $\sqrt{65}$.

A. 7.8 B. 8.06 C. 8.5 D. 7.5

Answer: B. Why: Nearest lower perfect square = 64 ($R=8$). $\sqrt{65} \approx 8 + (65-64)/16 = 8 + 0.0625 = 8.06$.

Q4. [Medium] [Trick] Multiply 112×108 using the base method.

A. 12086 B. 12096 C. 12106 D. 12076

Answer: B. Why: Base=100, $x=12$, $y=8$: $10000 + 100(20) + 96 = 10000 + 2000 + 96 = 12096$.

Q5. [Medium] Find the angle between the hour and minute hands at 6:00.

A. 90° B. 180° C. 150° D. 120°

Answer: B. Why: $|30(6) - 5.5(0)| = |180 - 0| = 180^\circ$.

Q6. [Hard] Find the angle between the hour and minute hands at 7:20.

A. 90° B. 95° C. 100° D. 105°

Answer: C. Why: $|30(7) - 5.5(20)| = |210 - 110| = 100^\circ$.

Q7. [Hard] Find $\sqrt{9801}$ using the digit-pattern method.

A. 97 B. 99 C. 101 D. 93

Answer: B. Why: Last digit 1 → root ends in 1 or 9. 9801 is close to $100^2=10000$, so try 99: $99^2=9801$ ✓.

Q8. [Hard] Multiply 203×198 using the base-200 method.

A. 40114 B. 40214 C. 40194 D. 39994

Answer: C. Why: Base=200, $x=3$, $y=-2$: $200^2 + 200(1) + (3 \times -2) = 40000 + 200 - 6 = 40194$.

Part B — Engineering Mathematics

- B1. Linear Algebra
- B2a. Calculus — Limits, Continuity & Differentiability
- B2b. Calculus — Maxima-Minima & Integration
- B2c. Calculus — Numerical Methods
- B3. Trigonometry
- B4. Vectors

B1. Linear Algebra

B1.1 Matrix Basics & Special Matrix Types [Level: Basic]

- A **matrix** is a rectangular array of numbers arranged in rows and columns; an $m \times n$ matrix has m rows, n columns.
- Two matrices are **equal** only if they have the same order (dimensions) AND every corresponding entry matches.

Matrix Type	Defining Property
Row / Column matrix	Single row ($1 \times n$) / single column ($m \times 1$)
Square matrix	rows = columns ($n \times n$)
Diagonal matrix	Square, all off-diagonal entries = 0
Scalar matrix	Diagonal matrix with all diagonal entries equal
Identity matrix (I)	Scalar matrix with diagonal entries = 1
Zero / Null matrix	Every entry = 0
Symmetric matrix	$A^T = A$
Skew-symmetric matrix	$A^T = -A$ (diagonal entries must be 0)
Orthogonal matrix	$A^T A = I$ (so $A^{-1} = A^T$)
Idempotent matrix	$A^2 = A$
Involutory matrix	$A^2 = I$
Nilpotent matrix	$A^k = 0$ for some positive integer k

TRAP: Every diagonal element of a skew-symmetric matrix must be 0 (since $a_{ii} = -a_{ii}$ forces $a_{ii} = 0$) — a frequently tested one-liner.

B1.2 Matrix Addition, Multiplication & Transpose [Level: Basic-Medium]

Addition: $A + B$ defined only if A, B have the SAME order; add entry-wise.
 Multiplication: $(m \times n) \cdot (n \times p) = (m \times p)$ <- inner dimensions must match
 Matrix multiplication is NOT commutative in general: $AB \neq BA$

- TRAP:** AB may exist while BA doesn't (dimension mismatch), and even when both exist, $AB \neq BA$ in general.
- Transpose** A^T : rows and columns swapped. Key properties:

$(A^T)^T = A$
 $(A+B)^T = A^T + B^T$
 $(AB)^T = B^T \cdot A^T$ <- **TRAP:** order REVERSES, a classic trick question
 $(kA)^T = k \cdot A^T$

B1.3 Determinant [Level: Medium]

2x2: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$
 3x3 (expansion along row 1):

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

- Properties: $\det(AB) = \det(A) \cdot \det(B)$; $\det(A^T) = \det(A)$; $\det(kA) = k^n \cdot \det(A)$ for an $n \times n$ matrix (**TRAP:** the k scales by k^n , NOT just k); $\det = 0 \Leftrightarrow$ matrix is **singular** (non-invertible, rows/columns linearly dependent); swapping any two rows/columns flips the sign of the determinant; if any row/column is all zeros or two rows/columns are identical, $\det = 0$.
- Ways to solve for a triangular matrix:** $\det =$ product of diagonal entries (huge shortcut — never expand a triangular matrix the long way).

B1.4 Cofactor, Adjoint & Inverse [Level: Medium]

- Cofactor** $C_{ij} = (-1)^{i+j} \times$ (minor obtained by deleting row i , column j).
- Adjoint** $\text{adj}(A)$ = transpose of the cofactor matrix.
- Inverse:**

$A^{-1} = \text{adj}(A) / \det(A)$ <- exists ONLY if $\det(A) \neq 0$
 Key identity: $A \cdot \text{adj}(A) = \det(A) \cdot I$

- TRAP:** A matrix with $\det(A) = 0$ has NO inverse at all — don't attempt $\text{adj}(A)/0$.

B1.5 Trace [Level: Medium]

$\text{trace}(A)$ = sum of diagonal entries
 $\text{trace}(A+B) = \text{trace}(A) + \text{trace}(B)$
 $\text{trace}(AB) = \text{trace}(BA)$ <- TRUE even though $AB \neq BA$ in general (non-obvious, worth memorizing)

B1.6 Rank of a Matrix [Level: Medium-Hard]

- **Rank** = number of linearly independent rows (equivalently, columns) = number of non-zero rows after reducing to row-echelon form via elementary row operations (Gauss elimination).
- $\text{rank}(A) \leq \min(\text{rows}, \text{columns})$. A square $n \times n$ matrix has **full rank n** $\Leftrightarrow$ it is invertible $\Leftrightarrow \det(A) \neq 0$.
- **Ways to solve:** (1) Row-reduce to echelon form and count non-zero rows [general method]. (2) For a square matrix, quickly check $\det(A) \neq 0$ to confirm full rank without row-reducing at all [shortcut].

B1.7 System of Linear Equations [Level: Hard]

For a system written as $Ax = b$, compare $\text{rank}(A)$ [coefficient matrix] with $\text{rank}([A|b])$ [augmented matrix]:

$\text{rank}(A)$ vs $\text{rank}([A b])$	vs number of unknowns n	Result
$\text{rank}(A) = \text{rank}([A b]) = n$	—	Unique solution (consistent)
$\text{rank}(A) = \text{rank}([A b]) < n$	—	Infinite solutions (consistent)
$\text{rank}(A) \neq \text{rank}([A b])$	—	No solution (inconsistent)

- **TRAP:** A **homogeneous** system ($Ax = 0$) is NEVER inconsistent — it always has at least the trivial solution $x=0$. It has a **non-trivial** solution $\Leftrightarrow \det(A) = 0$ ($\text{rank}(A) < n$).
- **Ways to solve:** (1) Gauss elimination — reduce to echelon form, back-substitute. (2) Cramer's Rule — $x_i = \det(A_i) / \det(A)$, only valid for a square system with $\det(A) \neq 0$ (replace column i of A with b to get A_i). (3) Matrix method — $x = A^{-1}b$, again only if $\det(A) \neq 0$.

B1.8 Eigenvalues & Eigenvectors [Level: Hard]

Definition: $A \cdot v = \lambda v$ ($v \neq 0$ is the eigenvector, λ the eigenvalue)
Characteristic equation: $\det(A - \lambda I) = 0$

- **Shortcuts (avoid solving the full characteristic polynomial for verification):**

Sum of eigenvalues = $\text{trace}(A)$
Product of eigenvalues = $\det(A)$
Eigenvalues of a triangular (or diagonal) matrix = its diagonal entries, directly

- TRICK: given a 2×2 matrix, you can find both eigenvalues instantly from trace and determinant by solving $\lambda^2 - \text{trace} \cdot \lambda + \det = 0$, without writing out $\det(A - \lambda I)$ longhand.

Questions

Q1. [Easy] For matrix $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, find $\det(A)$.

A. 5 B. 8 C. 11 D. -5

Answer: A. **Why:** $\det = (2 \times 4) - (3 \times 1) = 8 - 3 = 5$.

Q2. [Easy] Which of these matrices is always invertible?

A. A zero matrix B. A matrix with $\det(A) = 0$ C. A matrix with $\det(A) \neq 0$ D. A singular matrix

Answer: C. **Why:** Invertibility requires a non-zero determinant by definition.

Q3. [Medium] If A is a 3×3 matrix with $\det(A) = 5$, find $\det(2A)$.

A. 10 B. 20 C. 40 D. 5

Answer: C. **Why:** $\det(kA) = k^n \det(A)$ for $n \times n$ matrix; here $k=2$, $n=3 \rightarrow 2^3 \times 5 = 40$.

Q4. [Medium] [Trap] For matrices A (2×3) and B (3×2), which of the following is generally FALSE?

A. AB exists and is 2×2 B. BA exists and is 3×3 C. $AB = BA$ D. Both AB and BA can be computed

Answer: C. **Why:** Even when both AB and BA exist, they generally have different orders here (2×2 vs 3×3) and are never equal in general — matrix multiplication is not commutative.

Q5. [Medium] A matrix A satisfies $A^2 = A$. What type of matrix is A ?

A. Nilpotent B. Idempotent C. Involutory D. Orthogonal

Answer: B. **Why:** $A^2 = A$ is the defining property of an idempotent matrix.

Q6. [Medium] [Trick] If A is a 4×4 upper triangular matrix with diagonal entries 2, 3, 5, 7, find $\det(A)$.

A. 17 B. 210 C. 35 D. 12

Answer: B. **Why:** For a triangular matrix, $\det = \text{product of diagonal entries} = 2 \times 3 \times 5 \times 7 = 210$ — no cofactor expansion needed.

Q7. [Hard] A system $Ax=b$ has 3 equations, 3 unknowns, with $\text{rank}(A) = 2$ and $\text{rank}([A|b]) = 3$. What is true of the system?

A. Unique solution B. Infinite solutions C. No solution D. Only the trivial solution

Answer: C. **Why:** $\text{rank}(A) \neq \text{rank}([A|b]) \rightarrow$ inconsistent $\rightarrow$ no solution.

Q8. [Hard] A 3×3 matrix A has $\text{trace}(A) = 9$ and $\det(A) = 24$. If two of its eigenvalues are 2 and 3, find the third.

A. 4 B. 3 C. 6 D. 9

Answer: A. **Why:** Sum of eigenvalues = $\text{trace} = 9 \rightarrow 2 + 3 + x = 9 \rightarrow x = 4$. (Check: product = $2 \times 3 \times 4 = 24 = \det(A)$ ✓, confirming consistency.)

Q9. [Hard] [Trap] For matrices A and B of compatible order, $(AB)^T$ equals:

A. $A^T B^T$ B. $B^T A^T$ C. $(BA)^T$ D. AB

Answer: B. **Why:** The transpose-of-a-product rule reverses order: $(AB)^T = B^T A^T$ — a very commonly reversed trap.

Q10. [Hard] A homogeneous system of 3 equations in 3 unknowns has a non-trivial solution. What must be true?

A. $\det(A) \neq 0$ B. $\det(A) = 0$ C. $\text{rank}(A) = 3$ D. The system is inconsistent

Answer: B. **Why:** A homogeneous system always has the trivial solution $x=0$; a non-trivial solution exists only when $\det(A)=0$ ($\text{rank}(A) < n$), making the coefficient matrix singular.

B2a. Calculus — Limits, Continuity & Differentiability

B2a.1 Limits — Basics & Indeterminate Forms [Level: Basic]

- A **limit** describes the value a function *approaches* as the input approaches some point — the function need not actually be defined (or attain that value) at the point itself.
- 7 Indeterminate Forms** (direct substitution gives a form with no fixed value — must be resolved before you can conclude anything):

$0/0$	infinity/infinity	$\text{infinity} - \text{infinity}$
$0 \times \text{infinity}$	1^infinity	0^infinity

- TRAP:** A limit that is NOT one of these 7 forms is already resolved by direct substitution — do not apply L'Hopital's or factorization tricks unnecessarily.

B2a.2 Ways to Solve Limits [Level: Basic-Medium]

Five distinct methods — pick based on the form of the expression:

- Direct Substitution** — plug in the value; use when the result is NOT indeterminate.
- Factorization Method** — cancel a common factor causing $0/0$ (typical for polynomial ratios), e.g. $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2) = \lim_{x \rightarrow 2} (x + 2) = 4$.
- Rationalization Method** — multiply by the conjugate to clear a surd causing $0/0$.
- Standard/Notable Special Limits** — memorize this reference block cold, it resolves most "trick" limits instantly without any algebra:

$\lim_{x \rightarrow 0} \sin(x)/x = 1$	$\lim_{x \rightarrow 0} \tan(x)/x = 1$
$\lim_{x \rightarrow 0} (1 - \cos x)/x = 0$	$\lim_{x \rightarrow 0} (e^x - 1)/x = 1$
$\lim_{x \rightarrow 0} \ln(1+x)/x = 1$	$\lim_{n \rightarrow \text{infinity}} (1 + 1/n)^n = e$
$\lim_{x \rightarrow 0} (a^x - 1)/x = \ln(a)$	

- L'Hopital's Rule** — for $0/0$ or infinity/infinity forms ONLY: differentiate numerator and denominator SEPARATELY (not as a quotient-rule derivative of the whole fraction), then re-evaluate; repeat if still indeterminate.
 - TRAP:** Applying L'Hopital's to a non-indeterminate expression gives a WRONG answer — always verify the form is $0/0$ or ∞/∞ immediately before each application, including after each repeated round.

B2a.3 Continuity [Level: Medium]

- f is **continuous at $x=a$** iff all three agree:

Left-Hand Limit ($x \rightarrow a^-$) = Right-Hand Limit ($x \rightarrow a^+$) = $f(a)$

- Sum, product, and composition of continuous functions are continuous.
- Types of Discontinuity:**

Type	Cause	Example pattern
Removable	Limit exists but $\neq f(a)$, or $f(a)$ undefined ("a hole")	$f(x) = (x^2 - 1)/(x - 1)$ at $x = 1$
Jump	LHL $\neq$ RHL	A step function
Infinite/Essential	The limit itself is infinite	$f(x) = 1/x$ at $x = 0$

GATE CS 2015 (Set 2, 1 Mark) — worked

$f(x) = x^{(-1/3)}$. A = area bounded by $f(x)$ and the x -axis for x in $[-1, 1]$. Which statements are TRUE? (1) f is continuous in $[-1, 1]$. (2) f is not bounded in $[-1, 1]$. (3) A is nonzero and finite.

A. 2 only B. 3 only C. 2 and 3 only D. 1, 2 and 3

Answer: C. f is undefined/blows up at $x=0$ (a change of 1 in x can produce an unbounded change in $f(x)$) → **not continuous**, so (1) is false. $f \rightarrow \pm\infty$ near $0 \rightarrow$ **unbounded**, (2) true. $A = 2 \int_0^1 x^{(-1/3)} dx = 2 \times [x^{(2/3)}/(2/3)]_0^1 = 2 \times (3/2) = 3$ (nonzero, finite) → (3) true.

GATE CS 2014 (Set 1, 2 Marks) — worked

f is continuous on $[0, 2]$, $f(0) = f(2) = -1$, $f(1) = 1$. Which must be TRUE?

A. $\exists y \in (0, 1)$: $f(y) = f(y+1)$ B. $\forall y \in (0, 1)$: $f(y) = f(2-y)$ C. max value on $(0, 2)$ is 1 D. $\exists y \in (0, 1)$: $f(y) = -f(2-y)$

Answer: A. Define $g(y) = f(y) - f(y+1)$ on $[0, 1]$. $g(0) = f(0) - f(1) = -1 - 1 = -2$ (negative). $g(1) = f(1) - f(2) = 1 - (-1) = 2$ (positive). g is continuous (f is continuous), so by the **Intermediate Value Theorem** there must exist $y \in (0, 1)$ with $g(y) = 0$, i.e. $f(y) = f(y+1)$. (B, C, D are not guaranteed by the given data alone.)

- TRICK: Whenever a GATE continuity question gives boundary values and asks "must be true," try constructing an auxiliary continuous function and applying the **Intermediate Value Theorem** — this is the standard technique behind this entire question family.

B2a.4 Differentiability [Level: Medium-Hard]

- Mathematical definition:** f is differentiable at $x=a$ if $\lim_{h \rightarrow 0} [f(a+h) - f(a)]/h$ exists (a finite value).
- Geometric definition:** a unique, well-defined tangent line exists at that point — no sharp corner, cusp, or vertical tangent.
- TRAP: Differentiability $\Rightarrow$ Continuity, but Continuity does NOT $\Rightarrow$ Differentiability.** Classic counter-example: $f(x) = |x|$ is continuous at $x=0$ but not differentiable there (sharp corner).
- Geometric red flags for non-differentiability: sharp corners/kinks, cusps, vertical tangents, and any point of discontinuity (which auto-disqualifies differentiability there).

GATE CS 2016 (Set 2, 1 Mark) — worked

$f(x)$ is a polynomial, $g(x) = f'(x)$. If degree of $(f(x) + f(-x))$ is 10, find degree of $(g(x) - g(-x))$.

Answer: 9. If f were odd, $f(x) + f(-x) = 0$ (degree undefined/zero) — contradicts "degree 10." So **f must be even:** $f(x) = f(-x) \Rightarrow f'(x) = -f'(-x)$ (derivative of an even function is odd). With $g = f'$: $g(x) - g(-x) = f'(x) - (-f'(x)) = 2f'(x)$. Since f has even degree 10, f' has degree 9.

GATE CS 2014 (Set 1, 2 Marks) — worked

$f(x) = x \sin(x)$ satisfies $f'(x) + f(x) + t \cos(x) = 0$. Find t .

A. -1 B. -2 C. 1 D. 2

Answer: B. $f(x) = x \cos x + \sin x$. $f'(x) = -x \sin x + \cos x + \cos x = -x \sin x + 2 \cos x$. Substituting: $(-x \sin x + 2 \cos x) + (x \sin x) + t \cos x = 0 \rightarrow 2 \cos x + t \cos x = 0 \rightarrow \cos x(t+2)=0 \rightarrow t = -2$.

B2a.5 Rolle's Theorem & Mean Value Theorems [Level: Hard]

Rolle's Theorem: if f continuous on $[a,b]$, differentiable on (a,b) , and $f(a)=f(b)$, then there EXISTS c in (a,b) with $f'(c) = 0$.

Lagrange's Mean Value Theorem (LMVT): if f continuous on $[a,b]$, differentiable on (a,b) , then there EXISTS c in (a,b) with $f'(c) = [f(b)-f(a)] / (b-a)$.
(Geometrically: some point has a tangent PARALLEL to the secant line through the endpoints.)

Cauchy's Extended MVT: for two functions f, g satisfying the LMVT conditions on $[a,b]$ ($g'(x) \neq 0$), there exists c in (a,b) with $[f(b)-f(a)] / [g(b)-g(a)] = f'(c) / g'(c)$.

- **TRAP:** Rolle's Theorem is just LMVT with $f(a)=f(b)$, forcing $f'(c)=0$ — recognize Rolle's as a **special case** of LMVT, not a separate unrelated result.
- Recognition pattern: any GATE question giving continuity+differentiability+boundary conditions and asking "must there exist a point where..." is almost always testing Rolle's/LMVT/IVT — construct the right auxiliary function.

Questions

(Easy → Hard; [Trick]/[Trap] tag shortcut/misconception-testing items; the 4 GATE PYQs above are reused here as Q9-Q12 with full verification)

Q1. [Easy] Evaluate $\lim_{x \rightarrow 0} \sin(3x)/x$.

A. 0 B. 1 C. 3 D. $1/3$

Answer: C. Why: Rewrite as $3 \times \sin(3x)/(3x)$; as $x \rightarrow 0$, $3x \rightarrow 0$ so $\sin(3x)/(3x) \rightarrow 1$, giving $3 \times 1 = 3$.

Q2. [Easy] Evaluate $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2)$.

A. 0 B. 2 C. 4 D. Does not exist

Answer: C. Why: Factor: $(x-2)(x+2)/(x-2) = x+2 \rightarrow$ at $x=2$, **4**.

Q3. [Medium] [Trick] Evaluate $\lim_{x \rightarrow 0} (e^{2x} - 1)/x$.

A. 0 B. 1 C. 2 D. e^2

Answer: C. Why: Using $\lim_{x \rightarrow 0} (e^{kx} - 1)/x = k$ (from the standard limit with substitution kx): here $k=2$, so answer = **2**.

Q4. [Medium] Evaluate $\lim_{x \rightarrow \infty} (3x^2 + 2x)/(x^2 + 5)$ using L'Hopital's Rule.

A. 0 B. 1 C. 3 D. ∞

Answer: C. Why: Form is ∞/∞ . Differentiate top & bottom: $(6x+2)/(2x) \rightarrow$ still ∞/∞ , differentiate again: $6/2 = 3$. (Faster: for equal-degree polynomials, the limit at ∞ is simply the ratio of leading coefficients, $3/1=3$.)

Q5. [Medium] [Trap] A student applies L'Hopital's Rule to $\lim_{x \rightarrow 0} (x+1)/(x+2)$. What is wrong?

A. Nothing, the answer is $1/2$ B. L'Hopital's cannot be applied since the form is NOT indeterminate (direct substitution gives $1/2$ immediately) C. The rule needs a trig function present D. L'Hopital's only works at $x \rightarrow \infty$

Answer: B. Why: Direct substitution gives $(0+1)/(0+2) = 1/2$ — a defined, non-indeterminate value; L'Hopital's Rule must never be invoked here (though coincidentally it would still give $1/2$, relying on it is conceptually wrong and fails on non-coincidental cases).

Q6. [Medium] Identify the type of discontinuity of $f(x) = (x^2 - 1)/(x - 1)$ at $x=1$ (with $f(1)$ separately defined as 5).

A. Jump B. Infinite C. Removable D. No discontinuity

Answer: C. Why: $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x+1) = 2$, which EXISTS but $\neq f(1)=5$ — a "hole" that could be removed by redefining $f(1)=2$. This is the textbook **removable discontinuity**.

Q7. [Hard] [Trap] Is $f(x)=|x|$ differentiable at $x=0$?

A. Yes, since it is continuous there B. No — continuity does not guarantee differentiability; $|x|$ has a sharp corner at 0 C. Yes, its derivative is 1 D. No, because it is discontinuous at 0

Answer: B. Why: f is continuous at 0 , but the left-hand derivative $(-1) \neq$ right-hand derivative $(+1)$ — a sharp corner, hence not differentiable, despite being continuous.

Q8. [Hard] A function f is continuous on $[1,5]$, differentiable on $(1,5)$, with $f(1)=f(5)=7$. By which theorem must some $c \in (1,5)$ have $f'(c)=0$?

A. Lagrange's MVT B. Rolle's Theorem C. Cauchy's Theorem D. L'Hopital's Rule

Answer: B. Why: Equal endpoint values ($f(1)=f(5)$) is exactly the extra condition that turns LMVT into **Rolle's Theorem**.

Q9. [Hard] $f(x) = x^{(-1/3)}$. A = area bounded by $f(x)$ and the x -axis for $x \in [-1,1]$. Which statements are TRUE? (1) f is continuous in $[-1,1]$. (2) f is not bounded in $[-1,1]$. (3) A is nonzero and finite. (GATE CS 2015, Set 2)

A. 2 only B. 3 only C. 2 and 3 only D. 1, 2 and 3

Answer: C. Why: f is undefined/unbounded at $x=0 \rightarrow$ not continuous (1 false), but unbounded (2 true); $A = 2 \int_0^1 x^{(-1/3)} dx = 2 \times (3/2) = 3$, finite and nonzero (3 true).

Q10. [Hard] f is continuous on $[0,2]$, $f(0)=f(2)=-1$, $f(1)=1$. Which must be TRUE? (GATE CS 2014, Set 1)

A. $\exists y \in (0,1): f(y)=f(y+1)$ B. $\forall y \in (0,1): f(y)=f(2-y)$ C. max value on $(0,2)$ is 1 D. $\exists y \in (0,1): f(y)=-f(2-y)$

Answer: A. Why: $g(y)=f(y)-f(y+1)$ is continuous, $g(0)=-2$, $g(1)=2 \rightarrow$ IVT guarantees a zero of g in $(0,1)$.

Q11. [Hard] f is a polynomial, $g=f$. If $\text{degree}(f(x)+f(-x))=10$, find $\text{degree}(g(x)-g(-x))$. (GATE CS 2016, Set 2)

A. 8 B. 9 C. 10 D. 11

Answer: B. Why: f must be even (else the sum would be 0), so $f'(x)=-f'(-x)$; $g(x)-g(-x)=2f'(x)$, degree = $10-1 = 9$.

Q12. [Hard] $f(x)=x \sin(x)$ satisfies $f''(x)+f(x)+t \cos(x)=0$. Find t . (GATE CS 2014, Set 1)

A. -1 B. -2 C. 1 D. 2

Answer: B. Why: $f'(x) = -x \sin x + 2 \cos x$; substituting and simplifying gives $x(t+2)=0$ for all x , so **$t=-2$** .

B2b. Calculus — Maxima-Minima & Integration

B2b.1 Stationary/Critical Points [Level: Basic]

- A **stationary (critical) point** is where $f'(x) = 0$ (or f' is undefined) — a candidate for a local maximum, minimum, or an inflection/saddle point.
- Monotonic functions:** f is strictly increasing on an interval if $f'(x) > 0$ throughout; strictly decreasing if $f'(x) < 0$ throughout.

B2b.2 Ways to Classify a Critical Point [Level: Medium]

Two methods — use the First if the sign pattern is easy to read, the Second if you can differentiate again quickly:

1. **First Derivative Test:** examine the sign of f' just before and after the critical point.

f' changes $+$ $\rightarrow$ $-$ at $x=c$ $\Rightarrow$ local MAXIMUM at c
 f' changes $-$ $\rightarrow$ $+$ at $x=c$ $\Rightarrow$ local MINIMUM at c
 f' does not change sign $\Rightarrow$ NOT an extremum (inflection-type point)

2. **Second Derivative Test:** evaluate $f''(c)$.

$f''(c) < 0$ $\Rightarrow$ local MAXIMUM
 $f''(c) > 0$ $\Rightarrow$ local MINIMUM
 $f''(c) = 0$ $\Rightarrow$ INCONCLUSIVE — must fall back to the First Derivative Test (or a higher-order test)

- TRAP:** $f'(c)=0$ does **not** mean "no extremum" — it means the second-derivative test simply fails to decide; the point could still be a max, min, or inflection point (e.g., $f(x)=x^4$ at $x=0$ is a minimum despite $f'(0)=0$).

Worked Example

Find and classify the critical points of $f(x) = x^3 - 3x$.

$$f'(x) = 3x^2 - 3 = 0 \rightarrow x = \pm 1.$$

$$f''(x) = 6x. \quad f''(1) = 6 > 0 \rightarrow \text{local minimum at } x=1 \text{ (value } f(1) = 1-3 = -2).$$

$$f''(-1) = -6 < 0 \rightarrow \text{local maximum at } x=-1 \text{ (value } f(-1) = -1+3 = 2).$$

B2b.3 Integration — Basics [Level: Basic]

- Indefinite Integral:** the antiderivative family, $\int f(x)dx = F(x) + C$ (arbitrary constant C , since differentiation of a constant is 0).
- Definite Integral:** $\int$ from a to b of $f(x)dx = F(b) - F(a)$, a single NUMBER, no "+C".

Standard forms:

Integral of $x^n dx$	$= x^{(n+1)}/(n+1) + C$	($n \neq -1$)
Integral of $1/x dx$	$= \ln x + C$	
Integral of $e^x dx$	$= e^x + C$	
Integral of $\sin(x) dx$	$= -\cos(x) + C$	
Integral of $\cos(x) dx$	$= \sin(x) + C$	
Integral of $\sec^2(x) dx$	$= \tan(x) + C$	

- Properties of definite integrals** (very exam-useful):

Integral(a to b) $f(x)dx$ = - Integral(b to a) $f(x)dx$
 Integral(a to a) $f(x)dx$ = 0
 Integral(θ to a) $f(x)dx$ = Integral(θ to a) $f(a-x)dx$ <- powerful symmetry trick

- TRICK:** The symmetry property above often converts a hard integral into one that, when ADDED to the original, cancels into something trivial — a classic GATE/JEE pattern: if $I = \int_0^a f(x)dx = \int_0^a f(a-x)dx$, then $2I = \int_0^a [f(x)+f(a-x)]dx$, which is frequently a constant.

B2b.4 Ways to Solve Integration Problems [Level: Medium-Hard]

Three principal techniques — choose based on the integrand's structure:

- Substitution Method (u-substitution):** for a composite function where one part's derivative appears elsewhere in the integrand. Set u = (inner function), rewrite entirely in terms of u , integrate, substitute back.
 - Example: $\int 2x \cos(x^2) dx \rightarrow$ let $u=x^2$, $du=2x dx \rightarrow \int \cos(u)du = \sin(u)+C = \sin(x^2)+C$.
- Partial Fractions Method:** for a rational function $P(x)/Q(x)$ — decompose the denominator's factors into simpler fractions with unknown constants, solve for those constants, then integrate each simple term (typically giving log or arctan forms).
 - Complication: a REPEATED linear factor $(x-a)^2$ needs TWO terms, $A/(x-a) + B/(x-a)^2$; an irreducible quadratic factor needs a linear numerator, $(Ax+B)/(x^2+bx+c)$ — forgetting these extra terms is the classic partial-fractions error.
- Integration by Parts:** $\int u \cdot dv = u \cdot v - \int v \cdot du$. Choose " u " (the part to differentiate) using the **LIATE** priority order — pick the first type present, top to bottom:

L - Logarithmic ($\ln x$, $\log x$)
 I - Inverse Trig ($\sin^{-1} x$, $\tan^{-1} x$)
 A - Algebraic (x , x^2 , polynomials)
 T - Trigonometric ($\sin x$, $\cos x$)
 E - Exponential (e^x , a^x)

- TRICK:** LIATE tells you which factor to differentiate (u) — the higher item in the list becomes u , the rest becomes dv . Example: $\int x \ln(x) dx \rightarrow$ "L" beats "A", so $u=\ln(x)$ (differentiate), $dv=x \cdot dx$ (integrate).
- TRAP:** Forgetting "+C" on an INDEFINITE integral is marked wrong in most exams; conversely, writing "+C" on a DEFINITE integral (which evaluates to a plain number) is also wrong — know which one you're doing.

Worked Example (Integration by Parts)

Evaluate $\int x \cdot e^x dx$.

LIATE: "A" (x) beats "E" (e^x) $\rightarrow u=x$ ($du=dx$), $dv=e^x dx$ ($v=e^x$).

$\int x \cdot e^x dx = x \cdot e^x - \int e^x dx = x \cdot e^x - e^x + C = e^x(x-1) + C$.

Questions

(Easy $\rightarrow$ Hard; [Trick]/[Trap] tag shortcut/misconception-testing items)

Q1. [Easy] Find the critical points of $f(x) = x^2 - 4x + 3$.

A. $x=1$ B. $x=2$ C. $x=3$ D. $x=4$

Answer: B. Why: $f'(x)=2x-4=0 \rightarrow x=2$.

Q2. [Easy] Classify the critical point of $f(x)=x^2-4x+3$ at $x=2$.

A. Maximum B. Minimum C. Inflection D. Cannot be determined

Answer: B. Why: $f''(x)=2 > 0$ (constant, always positive) $\rightarrow$ **minimum**.

Q3. [Medium] Evaluate $\int (3x^2 + 2x)dx$.

A. x^3+x^2+C B. $6x+2+C$ C. x^3+2x^2+C D. $3x^3+x^2+C$

Answer: A. Why: $\int 3x^2 dx = x^3$, $\int 2x dx = x^2 \rightarrow x^3 + x^2 + C$.

Q4. [Medium] [Trap] A student computes $\int_0^2 (3x^2)dx$ and writes the answer as " x^3+C evaluated" = $8+C$. What's wrong?

A. Nothing B. A definite integral yields a plain number — no " $+C$ " belongs in the final answer, correct value is 8 C. The formula for x^2 integral is wrong D. Limits should be swapped

Answer: B. Why: A definite integral's arbitrary constant always cancels: $[x^3]_0^2 = 8 - 0 = 8$, with no C at all.

Q5. [Medium] Classify the critical point at $x=0$ for $f(x) = x^4$.

A. Maximum (via 2nd derivative test) B. Minimum, but the 2nd derivative test is inconclusive here ($f''(0)=0$) so the 1st derivative test is needed C. Not a critical point D. Inflection point

Answer: B. Why: $f'(x)=4x^3=0$ at $x=0$; $f''(x)=12x^2$, so $f''(0)=0$ — inconclusive by 2nd derivative test. First derivative test: f' is negative for $x<0$, positive for $x>0 \rightarrow$ sign change ($-$ to $+$) confirms a **minimum**.

Q6. [Medium] [Trick] Using LIATE, which part of $\int x^2 \cdot \sin(x) dx$ should be "u" (differentiated)?

A. $\sin(x)$ B. x^2 C. Either, it doesn't matter D. Neither, use substitution instead

Answer: B. Why: LIATE ranks Algebraic (x^2) above Trigonometric ($\sin x$), so $u=x^2$ is differentiated, $dv=\sin(x)dx$ is integrated.

Q7. [Hard] Evaluate $\int 2x \cdot e^{(x^2)} dx$ using substitution.

A. $e^{(x^2)}+C$ B. $2e^{(x^2)}+C$ C. $x^2 e^{(x^2)}+C$ D. $e^{(x^2)}/2 + C$

Answer: A. Why: Let $u=x^2$, $du=2x dx \rightarrow \int e^u du = e^u + C = e^{(x^2)} + C$.

Q8. [Hard] Decompose $1/[(x-1)(x-2)]$ using partial fractions.

A. $1/(x-1) + 1/(x-2)$ B. $1/(x-2) - 1/(x-1)$ C. $1/(x-1) - 1/(x-2)$ D. $-1/(x-1) - 1/(x-2)$

Answer: B. Why: $1/[(x-1)(x-2)] = A/(x-1) + B/(x-2)$. Clearing denominators: $1=A(x-2)+B(x-1)$. At $x=1$: $1=A(-1) \rightarrow A=-1$. At $x=2$: $1=B(1) \rightarrow B=1$. So the decomposition = $-1/(x-1) + 1/(x-2) = 1/(x-2) - 1/(x-1)$.

Q9. [Hard] Evaluate $\int x \cdot \ln(x) dx$ using integration by parts.

A. $(x^2/2)\ln(x) - x^2/4 + C$ B. $x \cdot \ln(x) - x + C$ C. $(x^2/2)\ln(x) + C$ D. $\ln(x)^2/2 + C$

Answer: A. Why: LIATE: $u=\ln(x)$ ($du=dx/x$), $dv=x dx$ ($v=x^2/2$). $\int x \ln(x) dx = (x^2/2)\ln(x) - \int (x^2/2)(1/x) dx = (x^2/2)\ln(x) - \int (x/2) dx = (x^2/2)\ln(x) - x^2/4 + C$.

Q10. [Hard] [Trick] Using the symmetry property, evaluate $I = \int_0^{\pi} x \cdot \sin(x)/(1+\cos^2 x) dx$, given that the same substitution ($x \rightarrow \pi-x$) transforms $\sin(x)/(1+\cos^2 x)$ into itself unchanged.

A. 0 B. $\pi^2/4$ C. π D. 2π

Answer: B. Why: Let $f(x)=\sin(x)/(1+\cos^2 x)$; $f(\pi-x)=f(x)$ (given). Then $I = \int_0^{\pi} x \cdot f(x) dx$ and also $I = \int_0^{\pi} \pi \cdot f(x) dx$ (by the symmetry property). Adding: $2I = \pi \int_0^{\pi} f(x) dx$. Since $\int_0^{\pi} \sin(x)/(1+\cos^2 x) dx = [-\arctan(\cos x)]_0^{\pi} = \arctan(1) - \arctan(-1) = \pi/4 - (-\pi/4) = \pi/2$, we get $2I = \pi \cdot (\pi/2) \rightarrow I = \pi^2/4$.

B2c. Calculus — Numerical Methods

B2c.1 Foundations — Approximation, Error & Convergence [Level: Basic]

- Numerical methods give **approximate** answers — used when an exact solution is unavailable/difficult, only tabular/measured data exists, a computer must repeat calculations, or an answer is needed only to a stated accuracy (e.g., exact $\sqrt{2}$ vs numerical $\sqrt{2} \approx 1.4142$).
- Error sources: **Input** (data already approximate), **Round-off** (digits rounded during calculation), **Truncation** (an infinite process stopped after finite terms), **Discretisation** (continuous problem evaluated only at selected points), **Iteration** (process stopped before the true limiting value).

```
Let xt = true value, xa = approximate value.
True error:      Et = xt - xa
Absolute error:  |Et| = |xt - xa|
Relative error:  |xt - xa| / |xt|
Percentage error: Relative error x 100
When true value unknown, use consecutive iterates instead:
Approx. relative % error = |xn - xn-1| / |xn| x 100
```

- TRICK: Keep extra digits DURING calculation, round only the FINAL answer — early rounding is one of the most common sources of a "close but wrong" final answer in this whole chapter.
- Convergence & stopping criteria** (a safe algorithm uses an accuracy condition AND a max-iteration cap together): stop when $|x_n - x_{n-1}| < \text{tolerance}$, OR $|f(x_n)| < \text{tolerance}$, OR percentage error < permitted error, OR required decimal places stabilize, OR max iterations reached.

B2c.2 Root-Finding for Non-Linear Equations [Level: Basic-Medium]

- Linear equation: variable has power 1, no variables multiplied together (e.g., $2x+3=7$). Non-linear: power > 1 , or roots/exponentials/trig terms (e.g., $x^2-3x+1=0$, $e^x-x=0$).
- Bracketing (Intermediate Value) condition**: if f is continuous on $[a,b]$ and $f(a)f(b) < 0$, at least one root lies in (a,b) . **TRAP**: no sign change does NOT always mean no root exists — an even-multiplicity root can just touch the x -axis without crossing it.

Ways to solve $f(x)=0$ — four classic methods

1. Bisection Method (bracketing, most reliable, slowest)

```
Require f(a)f(b) < 0. Midpoint m=(a+b)/2.
If f(a)f(m)<0, new interval=[a,m]. If f(m)f(b)<0, new interval=[m,b]. If f(m)=0, m is exact root.
After n bisections: interval width = (b-a)/2^n ; max error of final midpoint = (b-a)/2^(n+1)
```

Use when reliability matters more than speed.

Worked example (GATE-style): $f(x)=x^3-x-2$, start $[1,2]$. Find the bracket after 4 iterations.

$f(1)=-2$, $f(2)=4$. $m_1=1.5000$, $f(m_1)<0 \rightarrow [1.5000, 2]$. $m_2=1.7500$, $f(m_2)>0 \rightarrow [1.5000, 1.7500]$. $m_3=1.6250$, $f(m_3)>0 \rightarrow [1.5000, 1.6250]$. $m_4=1.5625$, $f(m_4)>0 \rightarrow [1.5000, 1.5625]$ (approx root ≈ 1.53125).

2. Regula Falsi / False Position (bracketing, directed by function values instead of the plain midpoint)

```
Require f(a)f(b) < 0.
c = [a.f(b) - b.f(a)] / [f(b) - f(a)] (x-intercept of the line joining (a,f(a)),(b,f(b)))
Keep the half of [a,b] that still brackets the sign change.
```

Worked example: $x^2-2=0$, start $[1,2]$. $f(1)=-1$, $f(2)=2$. $c_1=[1(2)-2(1)]/[2-(-1)] = 4/3 = 1.3333$; $f(c_1) \approx -0.2222 \rightarrow$ new interval $[1.3333, 2]$. $c_2 \approx 1.4000$.

- Advantage over Bisection: converges faster on average since it's "directed," but can stagnate if one endpoint never updates.

3. Fixed-Point Iteration — rewrite $f(x)=0$ as $x=g(x)$; iterate $x_{n+1}=g(x_n)$ from a guess x_0 . Converges locally only if $|g'(x)| < 1$ near the root — a poor rearrangement of the SAME equation can diverge even though another rearrangement converges.

4. Newton-Raphson (open method, needs derivative, very fast near a simple root)

```
xn+1 = xn - f(xn)/f'(xn)
```

TRAP (failure cases): fails when $f'(x_n)=0$ (division by zero); a very small $f'(x_n)$ can cause a huge, unstable jump; a poor initial guess can converge to the WRONG root; oscillation/discontinuity can prevent convergence entirely.

GATE PYQ (Biotechnology 2025): $e^x-2=0$, $x_0=1$, find x_1 (round to 2 decimals). $f(x)=e^x-2$, $f'(x)=e^x$. $x_1 = 1-(e-2)/e = 2/e \approx 0.73576 \rightarrow 0.74$.

GATE PYQ (Chemical Engineering 2025): $f(x)=x^2-x-1=0$, $x^{(0)}=1$, find the SECOND iterate $x^{(2)}$. $f'(x)=2x-1$. $x^{(1)} = 1-(1-1-1)/(2-1) = 2$. $x^{(2)} = 2-(4-2-1)/(4-1) = 2-1/3 = 1.6667 \rightarrow 1.67$.

5. Secant Method (like Newton-Raphson, but the derivative is APPROXIMATED using two points — no derivative needed)

```
xn+1 = xn - [f(xn)(xn - xn-1)] / [f(xn) - f(xn-1)]
```

Worked example: $x^2-2=0$, $x_0=1$, $x_1=2$. $f(1)=-1$, $f(2)=2$. $x_2 = 2-[2(2-1)]/[2-(-1)] = 1.3333$. $f(x_2) \approx -0.2222$. $x_3 = 1.3333-[-0.2222(1.3333-2)]/[-0.2222-2] \approx 1.4000$.

- Faster than Bisection/Regula Falsi on average, but NOT guaranteed to converge for every starting pair.

Root-finding method comparison

Method	Needs derivative?	Bracket required?	Guaranteed convergence?	Speed
Bisection	No	Yes	Yes	Slow
Regula Falsi	No	Yes	Usually (can stagnate)	Medium
Fixed-Point	No	No	Only if $ g'(x) < 1$	Depends on g
Newton-Raphson	Yes	No	Only near a good guess	Fast (quadratic)
Secant	No	No (2 estimates)	Not guaranteed	Fast

- Selection rule: reliability → Bisection; interpolation bracket → Regula Falsi; derivative + good guess → Newton; no derivative + two estimates → Secant.

B2c.3 Systems of Linear Equations [Level: Medium]

- Matrix form: $\mathbf{Ax} = \mathbf{b}$. For a square system, a UNIQUE solution exists only when $\det(\mathbf{A}) \neq 0$.
- Augmented matrix $[\mathbf{A}|\mathbf{b}]$; valid row operations: swap two rows, scale a row by $k \neq 0$, replace a row with itself minus a multiple of another row — every operation must be applied to the WHOLE row.
- Pivot** = leading nonzero element used to eliminate a column. A zero pivot forces a row swap; a very small pivot magnifies round-off error (fixed by **Partial Pivoting**: swap in the row with the largest absolute value in that column before eliminating).

Ways to solve $\mathbf{Ax}=\mathbf{b}$ — direct methods

Gaussian Elimination — Forward elimination to upper-triangular form, then back-substitution (solve last equation first, substitute upward).

Worked example: $x+y+z=6$, $2x-y+z=3$, $x+2y-z=2$. Forward elimination → $[1 \ 1 \ 1|6]$; $[0 \ -3 \ -1|-9]$; $[0 \ 1 \ -2|-4]$ → swap R_2, R_3 → R_3+3R_2 → $[0 \ 0 \ -7|-21]$. Back-substitute: $\mathbf{z=3, y=2, x=1}$.

Gauss-Jordan — continue elimination until the coefficient matrix becomes the identity, $[\mathbf{A}|\mathbf{b}] \rightarrow [\mathbf{I}|\mathbf{x}]$; solution read directly, no back-substitution needed — more total operations than Gaussian Elimination, but directly gives RREF/matrix inverse.

LU Decomposition — $\mathbf{A} = \mathbf{LU}$ ($\mathbf{L}$ =lower-triangular, $\mathbf{U}$ =upper-triangular). Solve $\mathbf{Lz}=\mathbf{b}$ (forward substitution), then $\mathbf{Ux}=\mathbf{z}$ (back substitution). Most useful when the SAME $\mathbf{A}$ must be solved for MANY different $\mathbf{b}$ vectors (decompose once, reuse for every right-hand side).

Worked example: $2x+y=5$, $4x+3y=11$. $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} = \mathbf{LU}$ where $\mathbf{L} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $\mathbf{U} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$. $\mathbf{Lz}=\mathbf{b}$: $z_1=5$, $2(5)+z_2=11 \rightarrow z_2=1$. $\mathbf{Ux}=\mathbf{z}$: $y=1$, $2x+1=5 \rightarrow \mathbf{x=2, y=1}$.

- TRAP:** Gaussian Elimination/Gauss-Jordan/LU are **Direct methods** (exact after finite steps, best for small/medium dense systems). Jacobi/Gauss-Seidel below are **Iterative methods** (start from a guess, improve repeatedly — best for large, sparse systems).

Ways to solve $\mathbf{Ax}=\mathbf{b}$ — iterative methods

- Precondition: rearrange each equation to isolate one variable ($x=...$, $y=...$, $z=...$). **Diagonal Dominance** ($|a_{ii}| > \text{sum of other } |a_{ij}|$ in that row) is SUFFICIENT (not necessary) to guarantee convergence of both methods below.
- Jacobi Iteration:** every new value uses ONLY the PREVIOUS iteration's values (simultaneous update — parallelizable, but slower).
- Gauss-Seidel Iteration:** uses the NEWEST value immediately, as soon as it's computed (sequential update — usually faster, not parallelizable).
- TRAP:** "all old values" = Jacobi; "newest value immediately" = Gauss-Seidel — this recognition rule is the #1 way this pair is tested.

Shared worked example: $10x+y+z=12$; $2x+10y+z=13$; $2x+2y+10z=14$, all start at 0. Rearranged: $x=(12-y-z)/10$, $y=(13-2x-z)/10$, $z=(14-2x-2y)/10$.

Jacobi: Iteration 1 (all use old 0,0,0): $x^1=1.2$, $y^1=1.3$, $z^1=1.4$. Iteration 2 (all use iteration-1 values): $x^2=(12-1.3-1.4)/10=0.93$, $y^2=(13-2.4-1.4)/10=0.92$, $z^2=(14-2.4-2.6)/10=0.90$.

Gauss-Seidel: Iteration 1: $x^1=(12-0-0)/10=1.2$ (same start); y^1 uses the NEW x^1 immediately = $(13-2.4-0)/10=1.06$; z^1 uses NEW x^1 AND NEW y^1 = $(14-2.4-2.12)/10=0.948$. Gauss-Seidel's single first iteration (1.2, 1.06, 0.948) is already closer to the true (1,1,1) than Jacobi's first iteration (1.2,1.3,1.4) — illustrating its typically faster convergence.

- Residual** $\mathbf{r} = \mathbf{b} - \mathbf{A}\tilde{\mathbf{x}}$ (how well an approximate solution satisfies the equations) vs **Solution error** $\mathbf{e} = \mathbf{x} - \tilde{\mathbf{x}}$ (how far the approximate solution is from truth) — **TRAP:** a SMALL residual does NOT guarantee a small solution error, especially for **ill-conditioned** systems (small input changes cause large solution changes).

Linear-system method-selection guide

Need	Use
One system, direct/simple	Gaussian Elimination
Identity form / matrix inverse	Gauss-Jordan
Same A, many different b vectors	LU Decomposition
Large sparse system, simultaneous updates OK	Jacobi
Large sparse system, want faster practical convergence	Gauss-Seidel

B2c.4 Numerical Integration [Level: Medium]

$\mathbf{I} = \text{integral}(a \text{ to } b) \ f(x) \ dx$; Step size $\mathbf{h} = (b-a)/n$; Nodes $\mathbf{x_0=a, x_1=a+h, \dots, x_n=b}$; Ordinates $\mathbf{y_i=f(x_i)}$

- TRAP:** always verify equal spacing ($x_1-x_0 = x_2-x_1 = \dots$) before applying Simpson's rules — they require it; Trapezoidal can be applied interval-by-interval even with unequal widths.

Ways to solve — three composite rules

Trapezoidal Rule (straight-line/linear approximation per strip; ANY n works):

$\mathbf{I} \sim h/2 [y_0 + y_n + 2(y_1+y_2+\dots+y_{n-1})]$ weights: $\mathbf{1, 2, 2, \dots, 2, 1}$

GATE PYQ (Production & Industrial Engg 2024): $x=0,1,2,3,4$; $y=1,3,6,9,12$. $h=1$. $I \approx 1/2[1+12+2(3+6+9)] = 1/2[13+36] = \mathbf{24.5}$.

- Error order $\mathbf{O(h^2)}$; exact only for constant/linear functions; halving h cuts the leading error to $\sim 1/4$.

Simpson's One-Third Rule (quadratic/parabolic approximation; **n MUST be even**):

$$I \approx h/3 [y_0 + y_n + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})] \quad \text{weights: } 1, 4, 2, 4, 2, \dots, 4, 1$$

Worked example: $\int_0^2 (x^2+1)dx$, $n=2$, $h=1$. $y_0=1, y_1=2, y_2=5$. $I \approx 1/3[1+4(2)+5] = 14/3 \approx \mathbf{4.667}$ (this is EXACT here, since Simpson's 1/3 is exact up to degree-3 polynomials).

GATE PYQ (Chemical Engg 2023): $\int_{-1}^1 \sqrt{1-x^2}dx$, $h=0.5 \rightarrow$ nodes $-1, -0.5, 0, 0.5, 1$; $y_0=0, y_1=\sqrt{3}/2, y_2=1, y_3=\sqrt{3}/2, y_4=0$. $I \approx 1/6[4(\sqrt{3}/2 + \sqrt{3}/2) + 2] = 1/6(4\sqrt{3} + 2) = \mathbf{1/3 + 2/\sqrt{3}}$.

- Error order $O(h^4)$; exact for polynomials up to degree 3; halving h cuts leading error to $\sim 1/16$.

Simpson's Three-Eighth Rule (cubic approximation; **n MUST be divisible by 3**):

$$I \approx 3h/8 [y_0 + 3y_1 + 3y_2 + y_3] \quad (\text{basic form, } n=3) \quad \text{weights pattern: } 1, 3, 3, 2, 3, 3, \dots, 3, 1 \text{ (composite)}$$

Worked example: $\int_0^3 x^3 dx$, $n=3$, $h=1$. $y_0=0, y_1=1, y_2=8, y_3=27$. $I \approx 3/8[0+3(1)+3(8)+27] = 3/8(54) = \mathbf{20.25}$ (exact, matching $\int_0^3 x^3 dx = 81/4$).

- Error order also $O(h^4)$; exact up to degree 3.

Integration method comparison

Rule	Min. intervals	n condition	Weights	Exact up to degree	Error order
Trapezoidal	1	any	1, 2, ..., 1	1	$O(h^2)$
Simpson 1/3	2	even	1, 4, 2, 4, ..., 1	3	$O(h^4)$
Simpson 3/8	3	divisible by 3	1, 3, 3, 2, ..., 1	3	$O(h^4)$

- TRICK: if you're given an ODD number of intervals not divisible by 3 in a way Simpson 1/3 alone can't handle, split — e.g., 5 intervals = apply Simpson 1/3 over the first 2 and Simpson 3/8 over the remaining 3, then add the two partial results.
- TRAP (common mistakes): confusing NODES ($n+1$) with INTERVALS (n); applying Simpson 1/3 when n is odd; applying Simpson 3/8 when n isn't divisible by 3; rounding intermediate values too early.

B2c.5 Numerical Solution of Differential Equations (IVPs) [Level: Medium-Hard]

Standard form: $dy/dx = f(x, y)$, with initial condition $y(x_0)=y_0$.
Grid points: $x_n = x_0 + n \cdot h$; Slope at current point = $f(x_n, y_n)$

Ways to solve — single-step methods (all self-starting, need only y_0)

Euler's Method (simplest, uses only the STARTING slope of the interval):

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

Local error $O(h^2)$, **global error $O(h)$** (first-order method) — only the beginning-of-interval slope is used, so it's inaccurate if the slope changes a lot within the step.

Worked example: $dy/dx=x-y$, $y(0)=1$, $h=0.2$, find $y(0.4)$. Step 1: $f(0,1)=-1 \rightarrow y_1=1+0.2(-1)=0.8$. Step 2: $f(0.2,0.8)=-0.6 \rightarrow y_2=0.8+0.2(-0.6)=\mathbf{0.68}$.

GATE PYQ (Civil Engg Set 1, 2025): $dy/dx=y+2x-x^2$, $y(0)=1$, $h=0.5$, find $y(1.0)$. $f(0,1)=1 \rightarrow y_1=1.5$. $f(0.5,1.5)=2.25 \rightarrow y_2=1.5+0.5(2.25)=2.625 \rightarrow \mathbf{2.6}$.

Modified Euler / Heun's Method (predictor-corrector; averages the START and predicted END slope):

$$\begin{aligned} \text{Predictor: } y_{n+1}(p) &= y_n + h \cdot f(x_n, y_n) \\ \text{Corrector: } y_{n+1}(c) &= y_n + h/2 [f(x_n, y_n) + f(x_{n+1}, y_{n+1}(p))] \end{aligned}$$

Global error $O(h^2)$.

Worked example: $dy/dx=x+y$, $y(0)=1$, $h=0.2$, find $y(0.2)$. Predictor: $y_1^p=1+0.2(1)=1.2$. Predicted end-slope $f(0.2,1.2)=1.4$. Corrector: $y_1^c=1+0.1(1+1.4)=\mathbf{1.24}$.

RK2 (Midpoint Method) — uses the slope AT the midpoint of the interval:

$$k_1 = h \cdot f(x_n, y_n); \quad k_2 = h \cdot f(x_n + h/2, y_n + k_1/2); \quad y_{n+1} = y_n + k_2$$

Global error $O(h^2)$, 2 function evaluations.

Worked example: $dy/dx=x+y^2$, $y(0)=1$, $h=0.1$, find $y(0.1)$. $k_1=0.1(0+1)=0.1$. Midpoint: $x=0.05, y=1.05$. $k_2=0.1(0.05+1.05^2)=0.1(1.1525)=0.11525$. $y_1=1+0.11525=\mathbf{1.11525}$.

RK4 (Classical Fourth-Order Runge-Kutta) — combines 4 slope estimates (beginning, 2 midpoints, end) with weights **1:2:2:1**:

$$\begin{aligned} k_1 &= h \cdot f(x_n, y_n) \\ k_2 &= h \cdot f(x_n + h/2, y_n + k_1/2) \\ k_3 &= h \cdot f(x_n + h/2, y_n + k_2/2) \\ k_4 &= h \cdot f(x_n + h, y_n + k_3) \\ y_{n+1} &= y_n + (k_1 + 2 \cdot k_2 + 2 \cdot k_3 + k_4)/6 \end{aligned}$$

Global error $O(h^4)$, 4 function evaluations — the standard workhorse for high single-step accuracy. **TRAP:** h is already baked into every k_i — do NOT multiply by h again in the final update.

GATE PYQ (Agricultural Engg 2023): $dy/dx=2x+y$, $y(0)=1$, $h=0.2$, find $y(0.2)$. $k_1=0.2(0+1)=0.2$. $k_2=0.2 \cdot f(0.1, 1.1)=0.26$. $k_3=0.2 \cdot f(0.1, 1.13)=0.266$. $k_4=0.2 \cdot f(0.2, 1.266)=0.3332$. $y_1=1+(0.2+0.52+0.532+0.3332)/6=1.2642 \rightarrow \mathbf{1.264}$.

Single-step methods comparison

Method	Evaluations/step	Local error	Global error	If h halved, error becomes
Euler	1	$O(h^2)$	$O(h)$	$\sim 1/2$
Modified Euler / RK2	2	$O(h^3)$	$O(h^2)$	$\sim 1/4$
RK4	4	$O(h^5)$	$O(h^4)$	$\sim 1/16$

- A higher-order ODE (e.g., $y''+y=0$) is converted to a first-order SYSTEM before applying any of the above: let $u_1=y$, $u_2=y' \rightarrow u_1'=u_2$, $u_2'=-u_1$ (a general $d^2y/dx^2=F(x,y,y')$ becomes $u_1'=u_2$, $u_2'=F(x,u_1,u_2)$).

B2c.6 Multi-Step Methods [Level: Hard]

- Unlike single-step methods, multi-step methods use SEVERAL previous points ($y_n, f_n, f_{n-1}, f_{n-2}, \dots$) to predict the next value — they are generally **NOT self-starting** (need extra starting values, typically generated by running RK4 for the first few steps) and require equal step size throughout.

Adams-Bashforth Predictor (4-step, explicit):
 $y_{n+1}(p) = y_n + h/24 [55.f_n - 59.f_{n-1} + 37.f_{n-2} - 9.f_{n-3}]$
 Adams-Moulton Corrector (4th order, implicit — needs the predicted f_{n+1}):
 $y_{n+1}(c) = y_n + h/24 [9.f_{n+1}(p) + 19.f_n - 5.f_{n-1} + f_{n-2}]$
 Milne Predictor: $y_{n+1}(p) = y_{n-3} + 4h/3 [2f_n - f_{n-1} + 2f_{n-2}]$
 Simpson Corrector: $y_{n+1}(c) = y_{n-1} + h/3 [f_{n-1} + 4f_n + f_{n+1}]$

- Predictor-Corrector workflow:** P (predict) $\rightarrow$ E (evaluate f at the predicted point) $\rightarrow$ C (correct) $\rightarrow$ E (re-evaluate if another correction round is needed) — stop once two consecutive corrected values agree to the required accuracy.

Worked example ($dy/dx=y$, so $f=y$ everywhere; $x: 0.0, 0.1, 0.2, 0.3$, $y: 1.00000, 1.10517, 1.22140, 1.34986$ — these are e^x values): find $y(0.4)$ using 4-step Adams-Bashforth then correct with Adams-Moulton.

Predictor: $y_4^p = 1.34986 + (0.1/24)[55(1.34986) - 59(1.22140) + 37(1.10517) - 9(1)] = 1.34986 + 0.14196 = \mathbf{1.49182}$.

Corrector: $y_4^c = 1.34986 + (0.1/24)[9(1.49182) + 19(1.34986) - 5(1.22140) + 1.10517] \approx \mathbf{1.491826}$.

- TRAP:** "self-starting" is a single-step-method property — Euler/RK2/RK4 ARE self-starting; Adams-Bashforth/Adams-Moulton/Milne are NOT.
- Consistency** (numerical equation $\rightarrow$ differential equation as $h \rightarrow 0$), **Stability** (small disturbances don't grow uncontrollably), and **Convergence** (numerical solution $\rightarrow$ exact solution as $h \rightarrow 0$) are three DIFFERENT properties — for linear multi-step methods, Consistency + Zero-stability TOGETHER imply Convergence. **TRAP:** a high-order method is not automatically useful — an unstable method can still magnify tiny errors without bound.

B2c.7 Complete Selection Map [Level: Reference]

Category	Use this when...
Non-linear root	Bisection (reliability); Regula Falsi (bracketed, directed); Fixed-Point ($x=g(x)$ form); Newton (derivative + good guess); Secant (no derivative, two estimates)
Linear system	Gaussian (one direct solve); Gauss-Jordan (need identity/inverse); LU (same A, many b's); Jacobi (parallel, all-old updates OK); Gauss-Seidel (sequential, want faster convergence)
Integration	Trapezoidal (general/simple); Simpson 1/3 (n even); Simpson 3/8 (n divisible by 3)
ODE	Euler (basic estimate); Modified Euler/RK2 (2-slope improvement); RK4 (high-accuracy single step); Multi-step (many equally-spaced points already available); Predictor-Corrector (refine a prediction)

Questions

(Easy $\rightarrow$ Hard; several are real official GATE PYQs or the source's own final mixed test, all independently verified)

Q1. [Easy] Bisection starts with an interval of width 3. What is the interval width after 5 bisections?

A. $3/8$ B. $3/16$ C. $3/32$ D. $3/64$

Answer: C. **Why:** Width after n bisections = original/ $2^n = 3/2^5 = 3/32$.

Q2. [Easy] Solve using Gaussian Elimination-style substitution: $2x+y=5$, $x-y=1$.

A. $x=1, y=3$ B. $x=2, y=1$ C. $x=3, y=-1$ D. $x=1, y=2$

Answer: B. **Why:** Adding both equations: $3x=6 \rightarrow x=2$, then $y=x-1=1$.

Q3. [Medium] Using Simpson's One-Third Rule with $h=1$, evaluate $\int_0^2 x^2 dx$.

A. 2 B. $7/3$ C. $8/3$ D. 3

Answer: C. **Why:** $y_0=0, y_1=1, y_2=4$. $I \approx 1/3[0+4(1)+4]=8/3$ (exact, matching the true integral).

Q4. [Medium] Using one Euler step for $dy/dx=x+y$, $y(0)=1$, $h=0.1$, find $y(0.1)$.

A. 1.00 B. 1.05 C. 1.10 D. 1.11

Answer: C. **Why:** $y_1 = 1 + 0.1 \cdot f(0,1) = 1 + 0.1(0+1) = 1.10$.

Q5. [Medium] [Trap] What is the global (accumulated) error order of the classical RK4 method?

A. $O(h)$ B. $O(h^2)$ C. $O(h^3)$ D. $O(h^4)$

Answer: D. **Why:** RK4's 4 weighted slope evaluations give 4th-order global accuracy — much better than Euler's $O(h)$ or RK2's $O(h^2)$.

Q6. [Medium] [Trap] Which of these methods is self-starting (needs only the initial condition, not extra prior points)?

A. Four-step Adams-Bashforth B. Milne Predictor C. RK4 D. Adams-Moulton Corrector

Answer: C. **Why:** RK4 is a single-step method needing only (x_0, y_0) ; the other three are multi-step methods requiring several previously computed points.

Q7. [Hard] [GATE PYQ] Using Newton-Raphson on $f(x)=e^x-2=0$ with $x_0=1$, find x_1 (round to 2 decimals).

A. 0.63 B. 0.69 C. 0.74 D. 0.82

Answer: C. **Why:** $x_1 = x_0 - f(x_0)/f'(x_0) = 1 - (e-2)/e = 2/e \approx 0.73576 \rightarrow 0.74$.

Q8. [Hard] [GATE PYQ] For $f(x)=x^2-x-1=0$ with $x^{(0)}=1$, find the SECOND Newton-Raphson iterate $x^{(2)}$ (round to 2 decimals).

A. 1.50 B. 1.60 C. 1.67 D. 1.75

Answer: C. **Why:** $x^{(1)} = 1 - (-1-1)/(2-1) = 2$. $x^{(2)} = 2 - (4-2-1)/(4-1) = 2 - 1/3 = 1.6667 \rightarrow 1.67$.

Q9. [Hard] [Trap] A system's coefficient matrix satisfies strict diagonal dominance. What does this guarantee?

A. The system has no solution B. Both Jacobi and Gauss-Seidel iterations are guaranteed to converge C. The determinant is zero D. Gaussian Elimination will fail

Answer: B. **Why:** Strict diagonal dominance is a SUFFICIENT (not necessary) condition for convergence of both Jacobi and Gauss-Seidel.

Q10. [Hard] [GATE PYQ] Using the Trapezoidal Rule on the table $x=0,1,2,3,4$ and $y=1,3,6,9,12$, estimate $\int_0^4 y(x)dx$.

A. 22.0 B. 23.5 C. 24.5 D. 26.0

Answer: C. Why: $h=1$. $I \approx 1/2[y_0+y_4+2(y_1+y_2+y_3)] = 1/2[1+12+2(18)] = 1/2(49) = 24.5$.

Q11. [Hard] [GATE PYQ] Using RK4 for $dy/dx=2x+y$, $y(0)=1$, $h=0.2$, find $y(0.2)$ (round to 3 decimals).

A. 1.240 B. 1.264 C. 1.280 D. 1.300

Answer: B. Why: $k_1=0.2$, $k_2=0.26$, $k_3=0.266$, $k_4=0.3332$. $y_1=1+(0.2+0.52+0.532+0.3332)/6=1.264$.

Q12. [Hard] [Trick] A 4-step Adams-Bashforth predictor gives $y_4^p=1.49182$ for $dy/dx=y$ with $y_3=1.34986$. Using the Adams-Moulton corrector

$y_4^c=y_3+h/24[9f_4^p+19f_3-5f_2+f_1]$ with $h=0.1$, $f_2=1.22140$, $f_1=1.10517$, find y_4^c .

A. 1.481826 B. 1.491826 C. 1.501826 D. 1.511826

Answer: B. Why: $y_4^c = 1.34986 + (0.1/24)[9(1.49182)+19(1.34986)-5(1.22140)+1.10517] \approx 1.491826$ — the corrector barely adjusts the predictor here since the function is smooth.

B3. Trigonometry

B3.1 Basic Ratios & Sign Rules [Level: Basic]

$\sin$, $\cos$, $\tan$ and their reciprocals $\operatorname{cosec} = 1/\sin$, $\sec = 1/\cos$, $\cot = 1/\tan$
 $\tan = \sin/\cos$, $\cot = \cos/\sin$

- **ASTC rule (sign by quadrant)** — mnemonic "**All Silver Tea Cups**":

Quadrant I (0–90): All functions positive
 Quadrant II (90–180): only $\sin$ (and cosec) positive
 Quadrant III (180–270): only $\tan$ (and $\cot$) positive
 Quadrant IV (270–360): only $\cos$ (and $\sec$) positive

- **TRICK:** to find any trig ratio of an angle outside 0–90°, first pin down its quadrant (for the sign) then reduce it to the equivalent acute-angle ratio using the allied-angle rules below.

B3.2 Standard Angles & Allied Angles [Level: Basic-Medium]

Angle	0°	30°	45°	60°	90°
$\sin$	0	1/2	1/√2	√3/2	1
$\cos$	1	√3/2	1/√2	1/2	0
$\tan$	0	1/√3	1	√3	undefined

- **Allied angle rule:** for angles of the form $(n \cdot 90^\circ \pm \theta)$:

If n is ODD (90 +/- theta, 270 +/- theta): function NAME CHANGES ($\sin \leftrightarrow \cos$, $\tan \leftrightarrow \cot$, $\sec \leftrightarrow \operatorname{cosec}$)
 If n is EVEN (180 +/- theta, 360 +/- theta): function NAME STAYS THE SAME
 In both cases: the SIGN is whatever that function's sign would be in the resulting quadrant (use ASTC).

e.g., $\sin(90+\theta) = \cos\theta$ (name changes, and $90+\theta$ lands in QII where $\sin$ is positive); $\cos(180+\theta) = -\cos\theta$ (name same, QIII, $\cos$ is negative there).

B3.3 Compound Angle & Double Angle Formulas [Level: Medium]

$\sin(A+B) = \sin A \cos B + \cos A \sin B$ $\cos(A+B) = \cos A \cos B - \sin A \sin B$
 $\sin(A-B) = \sin A \cos B - \cos A \sin B$ $\cos(A-B) = \cos A \cos B + \sin A \sin B$
 $\tan(A+B) = (\tan A + \tan B) / (1 - \tan A \tan B)$

Double angle (set $B=A$):

$\sin(2A) = 2 \sin A \cos A$
 $\cos(2A) = 1 - 2 \sin^2 A = 2 \cos^2 A - 1 = \cos^2 A - \sin^2 A$

B3.4 Core Identities [Level: Medium]

$\sin^2(\theta) + \cos^2(\theta) = 1$
 $1 + \tan^2(\theta) = \sec^2(\theta)$
 $1 + \cot^2(\theta) = \operatorname{cosec}^2(\theta)$

- **TRAP:** these three identities are just Pythagoras in disguise (divide $\sin^2 + \cos^2 = 1$ by $\cos^2$ or $\sin^2$ respectively to get the other two) — don't memorize them as unrelated facts.

B3.5 Properties of Triangles [Level: Hard]

Sine Rule: $a/\sin A = b/\sin B = c/\sin C = 2R$ (R = circumradius)
 Cosine Rule: $a^2 = b^2 + c^2 - 2bc \cos A$
 Projection Formula: $a = b \cos C + c \cos B$
 Area of triangle = $(1/2) \cdot a \cdot b \cdot \sin C$
 Area = $r \cdot s$ (r = inradius, s = semi-perimeter)
 Area = $(abc)/(4R)$ (R = circumradius)

- **TRICK:** given all 3 sides (no angles), find Area via Heron's formula first, then get $r = \text{Area}/s$ and $R = abc/(4 \times \text{Area})$ directly — no need to find any angle first.

B3.6 Heights & Distances [Level: Medium-Hard]

- **Angle of elevation:** angle from the horizontal UP to an object above eye level. **Angle of depression:** angle from the horizontal DOWN to an object below. Both are measured from the **horizontal**, not the vertical — the single most common student error here.

Core relation: $\tan(\text{angle}) = (\text{vertical height}) / (\text{horizontal distance})$

- **Ways to solve multi-triangle problems:** set up one right triangle per vantage point, express the unknown height/distance in terms of $\tan(\text{angle})$ for each, then equate or combine.

Worked Examples (verified)

A 20 m tall pole casts a shadow 10 m long. Find the angle of elevation of the sun.
 $\tan(\theta) = 20/10 = 2 \rightarrow \theta = \arctan(2) \approx \mathbf{63.43^\circ}$.

From a point 40 m from a tower's base, the angle of elevation to its top is 30° . Find the tower's height.

height = $40 \times \tan(30^\circ) = 40/\sqrt{3} = 40\sqrt{3}/3 \approx 23.09$ m.

Two towers of height 50 m and 80 m stand 100 m apart. Find the point on the line joining their bases from which the angles of elevation to both tops are equal.

Let the point be x m from the 50 m tower (so $100-x$ from the 80 m tower). Equal elevation angle $\Rightarrow$ equal height/distance ratio: $50/x = 80/(100-x) \rightarrow 50(100-x) = 80x \rightarrow 5000 = 130x \rightarrow x = 500/13 \approx 38.46$ m from the shorter tower.

B3.7 Inverse Trigonometric Functions (ITF) [Level: Hard]

Function	Principal value range
$\sin^{-1}x$	$[-\pi/2, \pi/2]$
$\cos^{-1}x$	$[0, \pi]$
$\tan^{-1}x$	$(-\pi/2, \pi/2)$
$\cot^{-1}x$	$(0, \pi)$

- **TRAP:** $\sin^{-1}$ and $\cos^{-1}$ do NOT share the same principal-value range — a very commonly tested mix-up.

B3.8 Trigonometric Equations — General Solutions [Level: Hard]

```

sin(theta) = sin(alpha) => theta = n.pi + (-1)^n . alpha      (n an integer)
cos(theta) = cos(alpha) => theta = 2n.pi +/- alpha
tan(theta) = tan(alpha) => theta = n.pi + alpha

```

Questions

Q1. [Easy] In which quadrant is $\tan(\theta)$ positive but $\sin(\theta)$ negative?

A. I B. II C. III D. IV

Answer: C. **Why:** By ASTC, Quadrant III has only Tan (and cot) positive; sin is negative there.

Q2. [Easy] Find the value of $\sin(30^\circ) + \cos(60^\circ)$.

A. 0.5 B. 1 C. 1.5 D. 2

Answer: B. **Why:** $\sin 30^\circ = 1/2$, $\cos 60^\circ = 1/2$, sum = 1.

Q3. [Medium] Evaluate $\sin(120^\circ)$.

A. $1/2$ B. $-1/2$ C. $\sqrt{3}/2$ D. $-\sqrt{3}/2$

Answer: C. **Why:** $120^\circ = 180^\circ - 60^\circ$; name stays "sin" (even multiple of 90 rule, since $180 = 2 \times 90$), and QII keeps sin positive $\rightarrow \sin(120^\circ) = \sin(60^\circ) = \sqrt{3}/2$.

Q4. [Medium] [Trap] Evaluate $\cos(90^\circ + \theta)$.

A. $\cos \theta$ B. $-\cos \theta$ C. $\sin \theta$ D. $-\sin \theta$

Answer: D. **Why:** 90 is an ODD multiple of $90^\circ \rightarrow$ function name CHANGES ($\cos \rightarrow \sin$); $90^\circ + \theta$ lies in QII where cos is negative $\rightarrow$ result is $-\sin \theta$.

Q5. [Medium] A ladder 10 m long rests against a wall making a 60° angle of elevation with the ground. How high up the wall does it reach?

A. 5 m B. $5\sqrt{3}$ m C. 8.66 m D. Both B and C

Answer: D. **Why:** height = $10 \times \sin(60^\circ) = 10 \times (\sqrt{3}/2) = 5\sqrt{3} \approx 8.66$ m — B and C are the same value in different forms.

Q6. [Hard] From a point 40 m from a tower's base, the angle of elevation to its top is 30° . Find the tower's height.

A. 20 m B. 23.09 m C. 40 m D. 69.28 m

Answer: B. **Why:** height = $40 \times \tan(30^\circ) = 40/\sqrt{3} \approx 23.09$ m.

Q7. [Hard] In a triangle, $a=7$, $b=8$, $c=9$. Using the cosine rule, find $\cos A$.

A. 0.5 B. 0.6 C. 0.7 D. 0.4

Answer: B. **Why:** $\cos A = (b^2 + c^2 - a^2)/(2bc) = (64 + 81 - 49)/(2 \times 8 \times 9) = 96/144 = 0.6$.

Q8. [Hard] [Trap] What is the principal value range of $\cos^{-1}x$?

A. $[-\pi/2, \pi/2]$ B. $[0, \pi]$ C. $(-\pi/2, \pi/2)$ D. $[-\pi, \pi]$

Answer: B. **Why:** $\cos^{-1}x$ has range $[0, \pi]$, distinct from $\sin^{-1}x$'s $[-\pi/2, \pi/2]$ — a frequently swapped pair.

Q9. [Hard] Solve for the general solution: $\sin(\theta) = 1/2$.

A. $\theta = n\pi + (-1)^n(\pi/6)$ B. $\theta = 2n\pi \pm (\pi/6)$ C. $\theta = n\pi + (\pi/6)$ D. $\theta = n\pi/6$

Answer: A. **Why:** For $\sin \theta = \sin \alpha$ ($\alpha = \pi/6$ here since $\sin(\pi/6) = 1/2$), the general solution form is $\theta = n\pi + (-1)^n \alpha$.

Q10. [Hard] Two towers of height 50 m and 80 m stand 100 m apart on level ground. Find the distance from the 50 m tower to the point between them where the angles of elevation to both tops are equal.

A. 30.77 m B. 38.46 m C. 50 m D. 61.54 m

Answer: B. **Why:** Equal-ratio condition $50/x = 80/(100-x)$ gives $130x = 5000 \rightarrow x = 500/13 \approx 38.46$ m.

B4. Vectors

B4.1 Basic Concepts & Types [Level: Basic]

- **Vector**: has both magnitude and direction (e.g., displacement, velocity, force). **Scalar**: magnitude only (e.g., mass, temperature, time).
- Component form: $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, magnitude $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$.
- Types: **Position vector** (from origin to a point), **Unit vector** (magnitude 1, direction-only), **Zero vector** (magnitude 0, undefined direction), **Coinitial vectors** (same starting point), **Collinear vectors** (parallel/along the same line), **Coplanar vectors** (lie in the same plane), **Free vector** (position doesn't matter, only magnitude+direction).

$$\text{Unit vector along } \mathbf{a} = \mathbf{a} / |\mathbf{a}|$$

B4.2 Vector Addition — Triangle & Parallelogram Law [Level: Basic]

- **Triangle Law**: if two vectors are the first two sides of a triangle (taken in order, tip-to-tail), their sum is the third side (tail of first to tip of second, closing the triangle).
- **Parallelogram Law**: if two vectors are adjacent sides of a parallelogram from a common point, their sum is the diagonal through that point.
- **TRAP**: both laws describe the SAME resultant — Triangle Law is just the Parallelogram Law's diagonal viewed as two triangles; don't treat them as giving different answers.

B4.3 Section Formula & Direction Cosines/Ratios [Level: Medium]

$$\text{Position vector of point dividing AB in ratio } m:n \text{ (internally)} = (\mathbf{m} \cdot \mathbf{B} + \mathbf{n} \cdot \mathbf{A}) / (m+n)$$

- **Direction Cosines (DCs)**: $l, m, n = \cos(\text{angles the vector makes with } x, y, z \text{ axes})$; always satisfy $l^2 + m^2 + n^2 = 1$.
- **Direction Ratios (DRs)**: any numbers proportional to the DCs (don't need to satisfy the sum-of-squares=1 condition themselves) — e.g., the components a_1, a_2, a_3 of a vector are one valid set of DRs.
- **Collinear/parallel vectors test**: $\mathbf{a}$ and $\mathbf{b}$ are parallel $\Leftrightarrow \mathbf{a} = \lambda \mathbf{b}$ for some scalar λ (Condition 1) $\Leftrightarrow$ their components are proportional: $a_1/b_1 = a_2/b_2 = a_3/b_3$ (Condition 2 — same test, ratio form). If $\lambda > 0$, same direction; if $\lambda < 0$, opposite direction.

B4.4 Dot (Scalar) Product [Level: Medium]

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cdot \cos(\theta) = a_1 \cdot b_1 + a_2 \cdot b_2 + a_3 \cdot b_3 \quad (\text{result is a SCALAR})$$

- **Ways to use it**: (1) Find the **angle** between two vectors: $\cos(\theta) = (\mathbf{a} \cdot \mathbf{b}) / (|\mathbf{a}| |\mathbf{b}|)$. (2) Test **perpendicularity**: $\mathbf{a} \cdot \mathbf{b} = 0 \Leftrightarrow \mathbf{a} \perp \mathbf{b}$. (3) **Scalar projection** of $\mathbf{a}$ onto $\mathbf{b} = (\mathbf{a} \cdot \mathbf{b}) / |\mathbf{b}|$.
- **TRAP**: Dot product of a vector with itself gives its magnitude SQUARED: $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.

B4.5 Cross (Vector) Product [Level: Medium-Hard]

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cdot \sin(\theta) \cdot \mathbf{n}_{\text{hat}} \quad (\text{result is a VECTOR, perpendicular to BOTH } \mathbf{a} \text{ and } \mathbf{b})$$

- **TRAP**: Dot product $\rightarrow$ scalar (uses cos); Cross product $\rightarrow$ vector (uses sin) — the single most tested distinction in this chapter. Also $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$ (anti-commutative — order matters, unlike the dot product where $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$).
- **Ways to use it**: (1) Test **parallel vectors**: $\mathbf{a} \times \mathbf{b} = 0 \Leftrightarrow \mathbf{a} \parallel \mathbf{b}$. (2) **Area of parallelogram** formed by $\mathbf{a}, \mathbf{b} = |\mathbf{a} \times \mathbf{b}|$. (3) **Area of triangle** formed by $\mathbf{a}, \mathbf{b} = \frac{1}{2} |\mathbf{a} \times \mathbf{b}|$.

Questions

Q1. [Easy] Find the unit vector along $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$.

A. $(3\mathbf{i}+4\mathbf{j})/5$ B. $(3\mathbf{i}+4\mathbf{j})/7$ C. $3\mathbf{i}+4\mathbf{j}$ D. $(4\mathbf{i}+3\mathbf{j})/5$

Answer: A. Why: $|\mathbf{a}| = \sqrt{9+16} = 5$, unit vector $= \mathbf{a}/|\mathbf{a}| = (3\mathbf{i}+4\mathbf{j})/5$.

Q2. [Easy] What is the result type of a dot product of two vectors?

A. Vector B. Scalar C. Matrix D. Unit vector

Answer: B. Why: Dot product always yields a scalar (magnitude only, no direction).

Q3. [Medium] Given $\mathbf{a} = 2\mathbf{i}+4\mathbf{j}+6\mathbf{k}$ and $\mathbf{b} = \mathbf{i}+2\mathbf{j}+3\mathbf{k}$, are they collinear?

A. Yes, since components are proportional (ratio 2:1 throughout) B. No C. Only if magnitudes are equal D. Cannot be determined

Answer: A. Why: $a_1/b_1=2, a_2/b_2=2, a_3/b_3=2$ — all ratios equal $\Rightarrow \mathbf{a} = 2\mathbf{b}$, so they're parallel/collinear.

Q4. [Medium] If $\mathbf{a} \cdot \mathbf{b} = 0$ for two non-zero vectors, what can you conclude?

A. They are parallel B. They are perpendicular C. They are equal D. One of them is a zero vector

Answer: B. Why: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos(\theta) = 0$ with both magnitudes non-zero forces $\cos(\theta)=0$, i.e., $\theta=90^\circ$.

Q5. [Medium] [Trap] If $\mathbf{a} \times \mathbf{b} = 0$ for two non-zero vectors, what can you conclude?

A. They are perpendicular B. They are parallel C. They have equal magnitude D. Their dot product is also 0

Answer: B. Why: $\mathbf{a} \times \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \sin(\theta) \hat{n} = 0$ with non-zero magnitudes forces $\sin(\theta)=0$, i.e., $\theta=0^\circ$ or 180° — parallel vectors. (Contrast directly with Q4's dot-product case — the two products test opposite conditions.)

Q6. [Hard] Find the area of the triangle formed by vectors $\mathbf{a} = 2\mathbf{i}+3\mathbf{j}$ and $\mathbf{b} = \mathbf{i}+4\mathbf{j}$ (treating them as two sides from a common vertex).

A. 2.5 B. 5 C. 10 D. 11

Answer: A. Why: $\mathbf{a} \times \mathbf{b}$ (in 2D, the k-component) $= (2 \times 4 - 3 \times 1) = 8 - 3 = 5$, so $|\mathbf{a} \times \mathbf{b}| = 5$. Area of triangle $= \frac{1}{2} \times 5 = 2.5$.

Q7. [Hard] Find the angle between vectors $\mathbf{a} = \mathbf{i}+\mathbf{j}$ and $\mathbf{b} = \mathbf{i}-\mathbf{j}$.

A. 0° B. 45° C. 90° D. 180°

Answer: C. Why: $\mathbf{a} \cdot \mathbf{b} = (1)(1) + (1)(-1) = 0 \rightarrow$ perpendicular $\rightarrow 90^\circ$.

Q8. [Hard] [Trick] Point P divides segment AB in ratio 2:3 (internally), where $\mathbf{A}=(1,2)$ and $\mathbf{B}=(6,7)$ (as position vectors). Find P.

A. (3,4) B. (2,4) C. (4,5) D. (3,5)

Answer: A. Why: $\mathbf{P} = (\mathbf{m} \cdot \mathbf{B} + \mathbf{n} \cdot \mathbf{A}) / (\mathbf{m} + \mathbf{n}) = (2(6,7) + 3(1,2)) / 5 = ((12+3), (14+6)) / 5 = (15, 20) / 5 = (3, 4)$.

Part C — Verbal & Logical Reasoning

- C1. Analogy & Number/Letter Series
- C2. Puzzle Solving
- C3. Syllogisms
- C4. Statement-Based Reasoning (Conclusion, Argument, Assumption, Course of Action)
- C5. Calendar
- C6. Seating Arrangements
- C7. Coding-Decoding
- C8. Blood Relations & Directions
- C9. Clocks
- C10. Ranking, Order & Sequence; Logical Sequence of Words
- C11. Inequality & Data Sufficiency

C1. Analogy & Number/Letter Series

C1.1 Analogy — Basic Concept [Level: Basic]

- **Analogy** = a pair of items sharing a specific relationship; given a first pair, find the missing term in a second pair that shares the SAME relationship.
- TRICK: The master skill is to **name the relationship explicitly** before searching for the answer — once named, applying it is mechanical.

Common Relationship Types (reference list)

Relationship	Example pair
Synonym	Big : Large
Antonym	Hot : Cold
Part-Whole	Wheel : Car
Worker-Tool	Carpenter : Saw
Worker-Product	Baker : Bread
Object-Function	Knife : Cut
Cause-Effect	Fire : Burn
Degree/Intensity	Warm : Hot
Class-Member	Fruit : Mango
Male-Female	Lion : Lioness

- Letter Analogy: apply a fixed **positional shift** in the alphabet (e.g., B:D :: F:H, each letter +2).
- Number Analogy: apply a fixed **arithmetic rule** (e.g., 4:16 :: 5:25, rule = square; or 3:7 :: 5:11, rule = double+1).

C1.2 Number Series — Classification Checklist [Level: Basic-Medium]

TRICK: Before hunting for a pattern, run this checklist **in order** — this classification IS the shortcut:

1. Check constant DIFFERENCE (Arithmetic series)
2. Check constant RATIO (Geometric series)
3. Check difference-of-differences (Two-stage series)
4. Check odd/even position split (Alternating/combination series)
5. Check squares/cubes with an offset (Square/Cube-based series)
6. Check prime numbers
7. Check a mixed operation pattern (e.g., $x2+1$, $x2+2$, $x2+3\dots$)

Series Types with Examples

Type	Rule	Example
Arithmetic	constant d	3,7,11,15,19 (d=4)
Geometric	constant r	2,6,18,54 (r=3)
Two-stage	differences form a series	2,3,5,8,12 (diffs: 1,2,3,4)
Alternating/combination	two interleaved rules	1,10,3,8,5,6 (odd pos +2, even pos -2)
Square-based	$n^2 \pm \text{offset}$	3,6,11,18,27 ($n^2+2: 1+2,4+2,9+2\dots$)
Mixed operation	changing operation per step	2,5,11,23,47 ($\times 2+1$ each time)

- **TRAP:** A **two-stage series** (differences forming their own series) is very commonly mistaken for a simple arithmetic series if only the first two gaps are checked — always verify at least 3-4 consecutive gaps before concluding the rule is constant.

C1.3 Letter / Alphabet Series [Level: Medium]

- Map letters to positions: A=1, B=2, ... Z=26 (and the reverse mapping Z=1...A=26 for "reverse alphabet" questions — always check which convention a question uses).
- Apply the same classification checklist (C1.2) to the position numbers instead of raw letters.
- Repeating letter-group patterns (e.g., AB, CD, EF, GH... — groups shift together) — treat each group as one "term" and find the group-level rule.

C1.4 Brute-Force Trick for Tough Series [Level: Hard]

TRICK: When no pattern is obvious quickly, compute **all simple differences AND all simple ratios** between consecutive terms simultaneously in a small table, and scan both rows for ANY consistency (constant value, or a second-level pattern) — this is faster than guessing randomly and catches two-stage/mixed-operation series quickly.

Questions

(Easy → Hard; [Trick] rewards the classification checklist, [Trap] tests common series-type misidentification)

Q1. [Easy] Doctor : Hospital :: Teacher : ?

A. Student B. School C. Book D. Chalk

Answer: B. **Why:** Relationship = workplace of the worker; a doctor works at a hospital, a teacher works at a school.

Q2. [Easy] Find the next term: 5, 10, 15, 20, ?

A. 22 B. 24 C. 25 D. 30

Answer: C. **Why:** Arithmetic series, constant difference +5.

Q3. [Easy] BD : FH :: JL : ?

A. NP B. MO C. NO D. OP

Answer: A. Why: Each letter shifts +4 positions (B→F→J→N, D→H→L→P): B,D → F,H → J,L → N,P.

Q4. [Medium] [Trap] Find the next term: 2, 6, 12, 20, 30, ?

A. 36 B. 40 C. 42 D. 44

Answer: C. Why: This LOOKS arithmetic-ish but differences are 4,6,8,10 (increasing by 2 each time — a two-stage series, not simple arithmetic). Next difference = 12, so next term = $30+12 = 42$. (Pattern is actually $n(n+1)$: $1\times 2, 2\times 3, 3\times 4, 4\times 5, 5\times 6, 6\times 7=42$.)

Q5. [Medium] Find the next term: 3, 9, 27, 81, ?

A. 162 B. 243 C. 324 D. 729

Answer: B. Why: Geometric series, constant ratio $\times 3$. $81\times 3 = 243$.

Q6. [Medium] [Trick] Find the next term: 1, 4, 9, 18, 35, ?

A. 60 B. 64 C. 68 D. 70

Answer: C. Why: Differences: 3,5,9,17. Differences-of-differences: 2,4,8 — doubling each time. Next difference-of-difference = 16, so next difference = $17+16 = 33$, giving $35+33 = 68$. The brute-force diff table (checklist step 3) catches this in seconds.

Q7. [Medium] Find the missing letter series term: Z, W, T, Q, ?

A. N B. M C. O D. P

Answer: A. Why: Reverse alphabet positions 26,23,20,17 (constant difference -3) → next = $14 = N$.

Q8. [Hard] Find the next term: 4, 5, 9, 14, 23, 37, ?

A. 56 B. 60 C. 61 D. 65

Answer: B. Why: Fibonacci-style: each term = sum of previous two ($4+5=9$, $5+9=14$, $9+14=23$, $14+23=37$, $23+37=60$).

Q9. [Hard] [Trap] Find the odd one out: 2, 3, 5, 7, 9, 11

A. 2 B. 9 C. 11 D. 3

Answer: B. Why: All others are prime numbers; $9 = 3^2$ is composite — a prime-number-series check (checklist step 6) reveals the outlier.

Q10. [Hard] Complete the analogy: $3 : 28 :: 5 : ?$

A. 124 B. 126 C. 130 D. 150

Answer: B. Why: Rule: $n^3 + 1$. $3^3+1=28$ ✓. So $5^3+1 = 125+1 = 126$.

Q11. [Hard] Find the next term in the alternating series: 1, 20, 4, 17, 9, 14, 16, ?

A. 9 B. 11 C. 13 D. 15

Answer: B. Why: Odd positions (1st,3rd,5th,7th): 1,4,9,16 = squares ($1^2, 2^2, 3^2, 4^2$) → next odd-position term(9th) not asked. Even positions(2nd,4th,6th,8th): 20,17,14,? decreasing by 3 → next = 11.

Q12. [Hard] [Trick] Find the next term: 100, 96, 89, 79, 66, ?

A. 50 B. 52 C. 54 D. 56

Answer: A. Why: Differences: 4,7,10,13 (increasing by 3 — two-stage series). Next difference = 16, so $66-16 = 50$.

C2. Puzzle Solving

C2.1 The Master Strategy [Level: Basic]

Puzzles (seating/floor/ranking/box arrangements with multiple clues) are solved by **systematic deduction**, not by guessing. Follow this sequence every time:

1. Read ALL clues once before writing anything down.
2. Fill in DIRECT/absolute clues first (e.g., "X is on floor 3").
3. Use ELIMINATION for relative/comparative clues (e.g., "X is above Y").
4. Build a GRID (rows = entities, columns = attributes) – track possibilities on paper/in a table, never purely in your head.
5. After each new deduction, RE-SCAN all remaining clues – a fact you just fixed often unlocks a clue that seemed unusable before.

- TRICK: Always process the MOST restrictive/direct clue first (one that fixes an exact value), since it shrinks the possibility space fastest for every other clue.

C2.2 Common Puzzle Types [Level: Basic-Medium]

Type	Typical clue style
Linear/Floor arrangement	People/objects placed in a line or building floors
Circular/Rectangular seating	People seated around a table, facing in/out
Comparison/Ranking	"more than," "taller than," "just before/after"
Box/Stacking	Boxes stacked in an order, with colour/weight/size clues
Blood-relation-embedded	Puzzle clues also encode family relationships
Day/Date-embedded	Clues reference days of the week or dates

- **TRAP:** For **floor-based** puzzles, always re-read whether "floor 1" is stated as the ground/bottom floor or a specific reference floor — this convention is stated explicitly in the question and reverses the entire ordering if misread.
- **TRAP:** For **comparison clues**, "X is more than Y" and "X is less than Y" are opposite directions — misreading the direction is the single most common puzzle error. Underline the comparison word every time.

C2.3 Worked Example — Grid Method [Level: Medium]

Four friends P, Q, R, S live on 4 different floors (1 lowest to 4 highest) of a building.

Clues: (1) Q lives immediately above P. (2) S does not live on floor 1 or floor 4. (3) R lives on floor 4.

Step 1 (direct clue): R = floor 4 (fixed).

Step 2 (elimination): $S \neq 1$, $S \neq 4$ (R already has 4) $\rightarrow S = 2$ or 3.

Step 3 (relative clue): Q is immediately above P $\rightarrow (P,Q)$ is one of (1,2) or (2,3) or (3,4). Since R=4, (3,4) is impossible for Q. So $(P,Q) = (1,2)$ or $(2,3)$.

Step 4 (combine): If $(P,Q)=(1,2)$, remaining floor 3 = S $\rightarrow$ but Step 2 said $S=2$ or 3; $S=3$ is fine here.

If $(P,Q)=(2,3)$, remaining floor 1 = S $\rightarrow$ but Step 2 forbids $S=1$. Contradiction — reject this branch.

Conclusion: $P=1$, $Q=2$, $S=3$, $R=4$.

C2.4 Higher-Difficulty Puzzles [Level: Hard]

- Multi-attribute puzzles (e.g., floor AND colour AND profession simultaneously) — extend the grid to one column per attribute; solve attributes in the order that has the most direct clues first.
- Puzzles embedding blood relations or day/date facts require solving the "embedded" sub-puzzle (see Blood Relations / Calendar chapters) before it constrains the main arrangement.
- When two branches both seem consistent after direct clues, defer them explicitly (label as Case 1 / Case 2) and continue applying remaining clues to each branch separately until one is eliminated by a contradiction — don't try to force a single answer prematurely.

Questions

(Easy $\rightarrow$ Hard; [Trap] tests misreading a comparative/direction clue)

Q1. [Easy] Five boxes A, B, C, D, E are stacked with A at the top and E at the bottom in that fixed order. Which box is directly below C?

A. A B. B C. D D. E

Answer: C. Why: Order top $\rightarrow$ bottom is A,B,C,D,E, so D is directly below C.

Q2. [Easy] In a row of 5 people facing north, Ravi is 2nd from the left and 4th from the right. How many people are in the row?

A. 4 B. 5 C. 6 D. 7

Answer: B. Why: Position from left + position from right – 1 = total $\rightarrow 2+4-1 = 5$.

Q3. [Medium] [Trap] A is taller than B. C is shorter than B. Who is the shortest?

A. A B. B C. C D. Cannot be determined

Answer: C. Why: $A > B > C$, so C is shortest. (Trap: misreading "C is shorter than B" as "C is taller" reverses the whole order.)

Q4. [Hard] Four boxes W, X, Y, Z are stacked one on another (position 1 = topmost, 4 = bottommost). Z is at the bottom. W is directly above X. Y is not at the top. Find the order from top to bottom.

A. W, X, Y, Z B. X, W, Y, Z C. W, Y, X, Z D. Y, W, X, Z

Answer: A. Why: $Z=4$ (given). W directly above X means W's position = X's position–1; the only fit among remaining positions {1,2,3} is $W=1$, $X=2$ (the alternative $W=2$, $X=3$ would force $Y=1$, violating "Y is not at the top"). So $Y=3$. Order top $\rightarrow$ bottom: W, X, Y, Z.

Q5. [Hard] Among five friends, A is taller than B but shorter than C. D is taller than C. E is shorter than B. Who is the tallest and who is the shortest?

A. C tallest, B shortest B. D tallest, E shortest C. D tallest, B shortest D. A tallest, E shortest

Answer: B. Why: Chain the clues: $D > C > A > B > E$. Tallest = D, shortest = E.

C3. Syllogisms

C3.1 The Four Statement Types [Level: Basic]

Code	Name	Form	Example
A	Universal Affirmative	All A are B	All cats are animals
E	Universal Negative	No A is B	No cat is a dog
I	Particular Affirmative	Some A are B	Some cats are black
O	Particular Negative	Some A are not B	Some cats are not black

- A syllogism gives 2+ statements (premises) and asks which of several conclusions **logically follow** — using **ONLY** the given statements, disregarding real-world/commonly-known facts.

C3.2 The Venn Diagram Method — Core Principle [Level: Basic-Medium]

1. Draw the diagram(s) consistent with the given statements, using **MINIMUM** overlap first.
2. A conclusion is **TRUE** only if it holds in **EVERY** possible diagram consistent with the statements.
3. A conclusion is **FALSE** if even **ONE** valid diagram can be drawn where it fails.

- TRICK: This "must hold in ALL possible diagrams" rule is the single most important principle in this chapter — most wrong answers come from checking only the first (most obvious) diagram and stopping.

C3.3 Key Inference Rules [Level: Medium]

- Valid chain:** "All A are B" + "All B are C" $\Rightarrow$ "All A are C" (transitive, valid).
- TRAP:** "All A are B" + "All C are B" gives **NO relation** between A and C — two statements funnelling into the *same* B do not connect A and C. (e.g., "All cats are animals" + "All dogs are animals" tells you nothing about cats vs dogs.)
- "Some A are B" only guarantees its **conversion**: "Some B are A" — nothing stronger.
- TRAP:** "Some A are B" does **not** make "Some A are not B" false — both can be simultaneously true depending on the diagram (e.g., some cats are black AND some cats are not black, both true together).
- "No A is B" converts validly to "No B is A" (universal negatives convert freely).
- "All A are B" does NOT convert to "All B are A" — only to "Some B are A" (a universal affirmative only converts to a particular).

C3.4 Possibility-Type Questions [Level: Hard]

- Three distinct answer categories: **Definitely True**, **Definitely False**, **Possibly True** (true in at least one valid diagram, even if not all).
- TRICK: If a conclusion is not "Definitely True" by the all-diagrams rule, check whether it can be **Possibly True** before marking it false outright — many higher-difficulty questions specifically test this middle category.

Questions

(Easy $\rightarrow$ Hard; [Trap] targets the classic same-B fallacy and the Some-A-are-B scope trap)

Q1. [Easy] Statements: All pens are books. All books are tables.

Conclusions: (I) All pens are tables. (II) Some tables are pens.

A. Only I follows B. Only II follows C. Both I and II follow D. Neither follows

Answer: C. Why: All-All chain gives All pens are tables (I, valid transitive chain) which in turn implies Some tables are pens (II, valid conversion of I).

Q2. [Easy] Statements: No apple is a mango. All mangoes are fruits.

Conclusion: Some fruits are not apples.

A. Follows B. Does not follow C. Possibly true only D. Cannot be determined

Answer: A. Why: Since no apple is a mango, and all mangoes are fruits, the mango-fruits (which exist) are guaranteed to be non-apple fruits — so some fruits (the mangoes) are definitely not apples.

Q3. [Medium] [Trap] Statements: All doctors are engineers. All lawyers are engineers.

Conclusion: Some doctors are lawyers.

A. Follows B. Does not follow C. Possibly true D. Definitely false

Answer: B. Why: Both groups funnel into the SAME class (engineers) via separate "All...are engineers" statements — this gives NO relation between doctors and lawyers. Classic same-B fallacy.

Q4. [Medium] Statements: Some students are athletes. All athletes are disciplined.

Conclusion: Some students are disciplined.

A. Follows B. Does not follow C. Possibly true only D. Cannot be determined

Answer: A. Why: The athletes who are students (guaranteed to exist by "Some students are athletes") are also disciplined (by "All athletes are disciplined") — so some students are definitely disciplined.

Q5. [Medium] [Trap] Statement: Some flowers are red.

Conclusion: Some flowers are not red.

A. Follows B. Does not follow C. Both can be simultaneously true, but conclusion isn't forced D. Definitely false

Answer: C. Why: "Some A are B" does not rule out "Some A are not B" — both could hold, but the given statement alone does NOT force the conclusion to be true; it's merely not contradicted.

Q6. [Hard] Statements: All squares are rectangles. No rectangle is a triangle. Some triangles are polygons.

Conclusion: (I) No square is a triangle. (II) Some polygons are not squares.

A. Only I follows B. Only II follows C. Both follow D. Neither follows

Answer: C. Why: All squares are rectangles + No rectangle is a triangle $\Rightarrow$ No square is a triangle (I, valid — every square, being a rectangle, is excluded from triangles). This is equivalent to "No triangle is a square" (E-type statements convert freely). Combined with "Some triangles are polygons," those triangle-polygons are guaranteed non-square (since no triangle is a square) — so some polygons are definitely not squares (II, also valid).

Q7. [Hard] Statements: Some books are pens. Some pens are pencils. All pencils are erasers.

Conclusion: Some erasers are books.

A. Definitely follows B. Possibly true only C. Definitely false D. Cannot be determined

Answer: B. Why: A chain of two "Some" statements (books-pens, pens-pencils) does NOT guarantee the SAME pens are involved in both — the pens that are books might be entirely different from the pens that are pencils. So the eraser-book link is only possible, not definite.

Q8. [Hard] Statement: All coins are metals.

Which of these is a valid conclusion?

A. All metals are coins B. Some metals are coins C. No metal is a coin D. Some coins are not metals

Answer: B. Why: A universal affirmative (All A are B) converts validly only to a particular (Some B are A) — never to another universal (All B are A), which would be an unwarranted reversal.

Q9. [Hard] Statements: No teacher is lazy. Some lazy people are students.

Conclusion: Some students are not teachers.

A. Follows B. Does not follow C. Possibly true only D. Definitely false

Answer: A. Why: The lazy people who are students (guaranteed by "Some lazy people are students") cannot be teachers (since "No teacher is lazy" excludes all teachers from being lazy) — so those particular students are definitely not teachers.

C4. Statement-Based Reasoning (Conclusion, Argument, Assumption, Course of Action)

(One of the highest-yield logical reasoning topics — asked in almost every IT-placement and competitive exam. The four sub-types below are frequently mixed up; the table in C4.1 is the single most important thing to internalize.)

C4.1 The Four Types at a Glance [Level: Basic]

Type	What it asks	Judged by
Conclusion	What logically follows FROM the statement?	Strict derivability — no outside knowledge
Argument	Is a given "Yes/No, because..." reason strong or weak?	Relevance + practicality
Assumption	What is silently taken for granted BEHIND the statement?	Implicit necessity — statement fails without it
Course of Action	What practical step should follow?	Feasibility + direct remedy

- **TRAP:** **Conclusion** is derived *from* the statement (looks forward, what the statement implies); **Assumption** is presupposed *behind* the statement (looks backward, what the statement silently relies on to make sense). Students constantly swap these two directions.

C4.2 Statement & Conclusion [Level: Basic-Medium]

- A conclusion must follow **logically and directly** from the statement alone — never bring in outside/common-sense world knowledge, and never require an assumption to bridge the gap.
- **Five standard tips (from official test-prep convention):**
 1. **Do NOT assume** an unstated cause/reason. *Statement:* "Teacher scolded Ram in front of the class." → "The teacher disliked Ram" does **not** follow (many other causes are equally possible — late to class, incomplete homework, etc.).
 2. **Advice or a clearly-implied result is always true** if it is directly related to the statement. *Statement:* "Petrol may become rare due to rising consumption." → "Petroleum products should be used efficiently" **follows** (a direct, related piece of advice).
 3. **Avoid conclusions built on a pre-assumption.** *Statement:* "Many hospital admissions occurred; junk food is assumed poisonous." → "Junk food is ALWAYS poisonous" does **not** follow — it converts a hedge ("assumed") into an absolute, pre-assumed fact.
 4. **Ignore morality/ethics — judge only logical fit.** *Statement:* "Many medicines contain fish oil." → "Some vegetarians consume such medicines to cure diseases" **follows**, even though it may seem to contradict vegetarian ethics — logical relevance beats moral comfort.
 5. Watch for **absolute-scope words** (ALL, NO, EVERY, ONLY) introduced in a conclusion that the statement never licensed.

Worked Examples (from official practice sets)

- *Statement:* "Shyam is one of the probable students for securing 1st rank." *Conclusions:* (1) Shyam will secure 1st rank (2) Shyam will not secure 1st rank. → **Either 1 or 2 follows** — the statement restricts the outcome to exactly two mutually exclusive possibilities, so one of the two conclusions *must* be true even though neither is individually certain.
- *Statement:* "50% off on all Electronic Goods up till 31st January." *Conclusions:* (1) After 31 Jan, no discount will be provided (2) No sale of electronic goods after 31 Jan. → **Only conclusion 1 follows.** The offer's end-date implies the discount stops, but nothing says sales themselves stop — conclusion 2 overreaches.
- *Statement:* "Due to recession, Company X fired 200 employees." *Conclusions:* (1) Company X is well known for firing people (2) All competitors are also affected by the recession. → **Neither follows** — (1) wrongly generalizes a single event into a reputation/history claim; (2) wrongly extends the statement to unrelated companies.
- **TRAP:** If a statement is about ONE specific entity (a country, a company, a person), a conclusion generalizing to "all others" or "no others" almost never follows — this single pattern accounts for a large share of "neither follows" answers.

C4.3 Statement & Argument [Level: Medium]

- Judge whether each argument is **Strong** (directly relevant, practical, addresses the real crux of the issue) or **Weak** (irrelevant, superficial, an appeal to an unrelated example/country, or purely emotional with no practical backing).
- **Checklist for a strong argument:** (a) relevant to the exact issue raised, (b) based on a practical/factual consequence, not a vague generality, (c) not simply "it happens elsewhere" (what another country/company does is rarely, by itself, a strong reason for domestic policy).
- NOTE: Argument-strength questions are the most subjective sub-type in this topic — official answer keys sometimes differ between books. Apply the checklist consistently rather than chasing a single "always correct" rule.

Worked Examples

- *Should military training be compulsory?* Arg 1: "Yes, every child should go through it" (asserts the conclusion again in absolute terms — **weak**, no supporting reason given). Arg 2: "No, it is not compulsory in USA" (irrelevant — a foreign country's policy is not a reason for domestic policy — **weak**). → **Neither is strong.**
- *Should the Disarmament Policy be adopted?* Arg 1: "Yes, it promotes peace and encourages other nations to follow" (practical, addresses real consequence — **strong**). Arg 2: "No, it poses a security threat" (practical, addresses the core risk — **strong**). → **Both are strong** — this is the standard pattern when both sides cite a direct, practical consequence rather than a generality.
- *Should more big industries be started in urban areas?* Arg 1: "Yes, creates job opportunities" (**strong**, practical). Arg 2: "No, increases population and pollution" (**strong**, practical). → **Both strong.**

C4.4 Statement & Assumption [Level: Medium-Hard]

- An assumption is **implicit** — not stated, but *necessary* for the statement to make sense. Test: "If I deny this assumption, does the statement's logic collapse?" If yes, it is a valid (implicit) assumption.
- **Five official rules for spotting a FALSE assumption:**
 1. Words like **EVERY, EACH, NONE, ALL, ONLY** in the assumption → almost always **FALSE** (too absolute to be a safe hidden premise).
 2. An assumption **not tied to** the statement's actual content → **FALSE**.
 3. An assumption about the **past or future** (when the statement is about the present) → **FALSE**.
 4. An assumption that **contradicts** the statement → **FALSE**.
 5. If the statement is **general** but the assumption is **specific** (or vice versa) → **FALSE**.

Worked Examples (rules applied)

- *Statement*: "Liquid soap should be used instead of soap bars to wash hands." *Assumption*: "Soap bars are harsher to the skin than liquid soap." → **Valid/implicit** — this is exactly the unstated reason that makes the recommendation sensible; none of the 5 false-rules apply.
- *Statement*: "Electric Vehicles produce less air pollutants." *Assumption*: "EVs will be the ONLY means of transport in future." → **FALSE** (Rule 1: contains "ONLY"; also Rule 3: talks about the future).
- *Statement (warning)*: "Cellphone use is prohibited beyond this point." *Assumption*: "This warning is not necessary." → **FALSE** (Rule 4: directly contradicts the statement — a warning being posted presupposes it IS necessary).
- *Statement (warning)*: "Paper will get wet because of water." *Assumption*: "Water will NOT make the paper wet." → **FALSE** (Rule 4: a direct, explicit contradiction of the statement).
- *Statement*: "Seats allotted only for physically disabled people." *Assumption*: "ONLY females who are physically disabled can occupy such seats." → **FALSE** (Rule 1: contains "ONLY"; also Rule 5: the statement is gender-general, the assumption narrows it to a specific gender with no basis).
- **TRICK**: Scan first for ALL/EVERY/NONE/ONLY — a huge fraction of "false assumption" answers can be eliminated on this word-spotting check alone, before doing any deeper logical analysis.

C4.5 Statement & Course of Action [Level: Hard]

- A course of action is valid only if it is a **practical, feasible, and administratively reasonable step that directly follows** as a solution/response to the problem in the statement — not merely a related idea, and not something requiring resources/authority far beyond what's implied.
- **TRAP**: A course of action that is *morally desirable* but **impractical or too extreme/costly** for the stated problem is usually rejected — the standard is feasibility, not idealism.
- *Worked example* — *Statement*: "Absenteeism among factory workers has risen sharply in the last quarter." *Suggested actions*: (1) The factory should shut down immediately. (2) Management should investigate the cause of absenteeism and consider incentive schemes. → Action (1) is disproportionate/impractical (**does not follow**); Action (2) is a direct, feasible, practical response (**follows**).

Questions

(ordered Easy → Hard; [Trap] flags a classic misconception)

Q1. [Easy] *Statement*: "Ravi got the highest marks in his class this term." *Conclusion*: "Ravi is the most intelligent student in the class." Does the conclusion follow?

A. Yes, it follows directly B. No — marks reflect this term's performance only, not a general intelligence claim C. Cannot be determined D. It's a strong argument, not a conclusion question

Answer: B. Why: The statement is about one term's marks; "most intelligent" is a broader, unstated generalization — classic overreach (Tip 1: do not assume).

Q2. [Easy] *Statement*: "This medicine should be taken only after meals." Which is the implicit assumption?

A. The medicine is safe on an empty stomach B. Taking it on an empty stomach could cause harm/reduced effect C. All medicines should be taken after meals D. The medicine has no side effects

Answer: B. Why: The instruction only makes sense if there's a reason to avoid an empty stomach; option C uses "all" (Rule 1 — false), option A contradicts the statement's intent (Rule 4 — false).

Q3. [Medium] *Statement*: "Company Z's advertisement says, 'If you love adventure, buy our shoes.'" *Conclusion*: "People who buy the shoes love adventure." Does it follow?

A. Yes B. No — the statement is a conditional aimed at adventure-lovers, it does not claim only they will buy the shoes C. Cannot say D. It's an assumption, not a conclusion

Answer: B. Why: "If you love adventure, buy our shoes" does not logically imply the converse (that all buyers love adventure) — a classic if-then vs converse trap.

Q4. [Medium] [Trap] *Statement*: "The company has decided to promote employees based on performance, not seniority." *Assumption*: "Seniority alone does not guarantee good performance." Is this a valid assumption?

A. Valid — it's exactly the unstated premise behind the policy change B. Invalid — it uses the word "alone" so it's absolute C. Invalid — it talks about the past D. Cannot be determined

Answer: A. Why: The policy shift only makes sense if seniority isn't a reliable performance proxy — none of the 5 false-rules apply (no ALL/EVERY/ONLY, not past/future, not contradicting, not a scope mismatch).

Q5. [Medium] *Statement*: "Should smoking be banned in public places?" *Argument*: "Yes, because it protects non-smokers from passive smoking-related health risks." Is this argument strong or weak?

A. Strong — direct, practical, addresses the core issue B. Weak — too general C. Weak — it's an appeal to emotion D. Cannot be determined

Answer: A. Why: It cites a specific, practical, directly relevant consequence (health risk to non-smokers) — meets the strong-argument checklist fully.

Q6. [Medium] [Trap] *Statement*: "Should India increase its defence budget?" *Argument*: "No, because our neighboring country X has a much smaller defence budget." Is this strong or weak?

A. Strong B. Weak — comparison to another country's budget size is not, by itself, a relevant domestic reason C. Strong, since it's factual D. Cannot say

Answer: B. Why: Being factual isn't enough — the argument must be *relevant* to whether India's own security/practical needs justify the increase; a neighbor's smaller budget doesn't address that.

Q7. [Hard] *Statement*: "All employees earning above ₹10 lakh/year must pay a special surcharge." *Conclusion*: "No employee earning ₹10 lakh or below pays this surcharge." Does it follow?

A. Yes, it follows directly by contrast B. No — the statement doesn't state what happens below the threshold, only what happens above it C. Cannot be determined either way D. It's a course of action, not a conclusion

Answer: A. Why: This is a valid logical contrapositive-style conclusion: the statement fixes the surcharge condition to "above ₹10 lakh" as the entire triggering criterion, so anyone not meeting it is excluded — this is one of the few cases where the "opposite" DOES follow, because the original statement itself draws an exact, exclusive threshold (unlike vaguer statements in Q1/Q3 where such inferences overreach).

Q8. [Hard] *Statement*: "Road accidents have increased by 40% in the city over the past year." *Suggested courses of action*: (1) Immediately ban all private vehicles in the city. (2) Increase traffic police patrolling and install speed cameras at accident-prone junctions. Which follows?

A. Only 1 B. Only 2 — practical, feasible, and directly targets the stated problem C. Both D. Neither

Answer: B. Why: Action 1 is disproportionate and infeasible; Action 2 is a direct, practical, standard administrative response to rising accidents.

Q9. [Hard] [Trap] *Statement*: "The exam result of School A has been the best in the district for the third consecutive year." *Conclusion*: "School A will also top the district next year." Does it follow?

A. Yes — the trend guarantees it B. No — past performance across 3 years does not logically guarantee a future (4th year) outcome C. Cannot be determined — insufficient data on other schools D. Both B and C capture the flaw, but B is the core reasoning error

Answer: D. Why: This combines two errors: (1) any claim about the FUTURE from a purely past-tense statement is invalid (Rule 3 territory from the Assumption rules, applicable to conclusions too), and (2) the statement is not even a conclusion, it's a prediction. B correctly identifies why it fails, C is also technically true — D acknowledges both but flags B as the primary reasoning error tested.

Q10. [Hard] *Statement*: "Government warning on cigarette packets: 'Smoking is injurious to health.'" *Assumption*: "People read what is printed on cigarette packets." Is this a valid assumption?

A. Valid — the warning's entire purpose depends on people reading it B. Invalid — contains no absolute word C. Invalid — it's about the future D. Cannot be determined

Answer: A. Why: The warning is pointless unless we assume it will be read — a textbook example of a valid implicit assumption (none of the 5 false-rules trigger: no ALL/EVERY/ONLY, not past/future-only, doesn't contradict, matches scope).

C5. Calendar

(A pure-formula chapter — once the odd-days method is internalized, every question type below becomes mechanical. Source slides for this topic were image-only; all content below is standard, verified calendar arithmetic.)

C5.1 Odd Days — The Foundational Idea [Level: Basic]

- A week has 7 days. **Odd days** = days left over after removing complete weeks from a span.
- Ordinary year = 365 days = 52 weeks + **1 odd day**. Leap year = 366 days = 52 weeks + **2 odd days**.

Days in each month (ordinary year):
 Jan 31 Feb 28 Mar 31 Apr 30 May 31 Jun 30
 Jul 31 Aug 31 Sep 30 Oct 31 Nov 30 Dec 31
 (Leap year: Feb = 29)

- Odd days per month (ordinary year, mod 7): Jan=3, Feb=0, Mar=3, Apr=2, May=3, Jun=2, Jul=3, Aug=3, Sep=2, Oct=3, Nov=2, Dec=3.

C5.2 Leap Year Rule [Level: Basic]

A year is a leap year IF divisible by 4,
 EXCEPT a century year (divisible by 100),
 which is a leap year ONLY IF also divisible by 400.

- TRAP:** 1900 and 2100 are **NOT** leap years (divisible by 100, not by 400); 2000 and 2400 **ARE** leap years (divisible by 400). This century-exception is the single most tested leap-year fact.

C5.3 Odd Days Over Long Spans — Breaking a Year into Pieces [Level: Basic-Medium]

Odd days in 100 years = 5
 Odd days in 200 years = 3
 Odd days in 300 years = 1
 Odd days in 400 years = 0 (and repeats: 800, 1200... also = 0)

- TRICK: Because 400 years always contribute exactly 0 odd days, **the calendar exactly repeats every 400 years** — this single fact is *why* the century rule (divisible by 400) exists at all: it keeps the 400-year block's leap-day count constant.
- "Leap year calendar repeats after 28 years"** — true ONLY when no century year falls inside that 28-year span (a 28-year gap normally contains exactly 7 leap years including itself $\rightarrow 7 \times 2 + 21 \times 1 = 35$ odd days = exactly 0 mod 7). **TRAP:** if a non-leap century year (1900, 2100...) falls inside the 28-year window, this shortcut breaks and a longer/adjusted gap is needed instead.

C5.4 Finding the Day of the Week for Any Date [Level: Medium]

Weekday code: 0=Sunday 1=Monday 2=Tuesday 3=Wednesday 4=Thursday 5=Friday 6=Saturday
 Total odd days = (odd days from all COMPLETE years before the given year)
 + (odd days from all COMPLETE months before the given month, in the given year)
 + (the date number itself)
 Answer weekday = (Total odd days) mod 7 --> look up in the code table above

- Ways to compute "odd days from complete years":**
 - Direct/reference method:** count forward or backward from a known anchor date (e.g., "1 Jan 2000 = Saturday") using the per-century/leap counting rules above.
 - Formula method:** for Y complete years, odd days = $[Y + \text{floor}(Y/4) - \text{floor}(Y/100) + \text{floor}(Y/400)] \bmod 7$ (this single formula automatically encodes the century exception).

Worked Example (verified)

Find the day of the week on **15 August 1947**.
 Complete years before 1947 = 1946 = 1900 + 46.
 • 1900 years $\rightarrow$ split as 4×400 (0 odd days each) + 300 (1 odd day) = **1 odd day**.
 • Remaining 46 years (1901–1946): leap years among them = 1904, 1908, ..., 1944 = **11** leap years. Odd days = $46 + 11 = 57 \rightarrow 57 \bmod 7 = 1$.
 • Total for 1946 complete years = $1 + 1 = 2$ odd days.
 Complete months before August in 1947 (not a leap year): Jan(3)+Feb(0)+Mar(3)+Apr(2)+May(3)+Jun(2)+Jul(3) = 16 $\rightarrow 16 \bmod 7 = 2$.
 Add the date itself: **15** $\rightarrow 15 \bmod 7 = 1$.
 Total = $2 + 2 + 1 = 5 \rightarrow$ weekday code 5 = **Friday**. ✓ (Matches the well-known historical fact that India's independence fell on a Friday.)

C5.5 Date-to-Date Relationship Problems [Level: Medium-Hard]

A useful way to classify these problems by rising difficulty:

- Same year, no century/leap complication** — just count elapsed days between the two dates mod 7.
- Spanning a Feb 29** — **TRAP:** if the range crosses a leap year's Feb 29, add **one extra odd day** for that crossing; forgetting this is the most common error in this category.
- Spanning one or more centuries** — must invoke the 100/200/300/400-year odd-day table (C5.3) rather than just counting leap years by inspection.
- Multi-century + partial-year combination** — combine (1)+(3): break the total span into whole-century blocks plus a remainder of years plus a remainder of months/days, and sum all three odd-day contributions mod 7.

C5.6 Which Months Share the Same Calendar? [Level: Medium]

In a **non-leap** year, these month-pairs/groups always start on the same weekday (verified: elapsed days between each pair's 1st is a multiple of 7):

January <-> October (273 days apart = 39 weeks exactly)
 February <-> March <-> November
 April <-> July (91 days apart = 13 weeks exactly)
 September<-> December (91 days apart = 13 weeks exactly)

- **TRAP:** The **February↔March↔November** group holds only in a **non-leap** year — a leap year's extra day in February shifts March (and everything after) by one day, breaking this specific triple (April↔July and September↔December are unaffected either way, since they don't span Feb 29).

C5.7 Same Calendar in Different Years [Level: Hard]

Two years have an **identical full calendar** (every date falls on the same weekday) only if **both** conditions hold:

1. Total odd days between the two years is a multiple of 7, AND
2. Both years have the SAME leap-year status (both leap, or both non-leap)

- **TRAP:** Condition 2 is the one students forget — matching odd-day count alone is not sufficient; if one year is leap and the other isn't, the Feb 29 shift means the calendars cannot fully coincide even if the odd-day arithmetic superficially lines up.
- Non-leap years typically repeat every **6 or 11 years** later (varies with the specific leap years skipped in between); the **full 400-year cycle** is the only guarantee that works unconditionally for every date pairing.

Questions

(ordered Easy → Hard; all answers independently verified by odd-day computation)

Q1. [Easy] Which of these is a leap year?

A. 1900 B. 2000 C. 2100 D. 1800

Answer: B. **Why:** 2000 is divisible by 400 → leap. 1900, 2100, 1800 are divisible by 100 but not 400 → not leap.

Q2. [Easy] How many odd days are there in 400 years?

A. 5 B. 3 C. 1 D. 0

Answer: D. **Why:** The 400-year block always contains exactly 0 odd days — this is why the calendar repeats every 400 years.

Q3. [Medium] If 1 January of a certain non-leap year is a Monday, what day is 1 October of the same year?

A. Sunday B. Monday C. Tuesday D. Wednesday

Answer: B. **Why:** January and October are always the same weekday in a non-leap year (273 days = exactly 39 weeks apart).

Q4. [Medium] [Trick] If 1 February of a non-leap year is a Thursday, what day is 1 March of the same year?

A. Wednesday B. Thursday C. Friday D. Saturday

Answer: B. **Why:** February has 28 days in a non-leap year = exactly 4 weeks (0 odd days), so March 1 falls on the identical weekday as February 1.

Q5. [Medium] [Trap] A person says "the calendar of 2001 is identical to the calendar of 2002 since consecutive years usually differ by just 1 odd day, and I've heard some consecutive years match." Is this reasoning valid?

A. Yes B. No — 2001 has 1 odd day of its own, so 2002 starts one weekday later; they cannot be identical C. Yes, if both are non-leap D. Cannot be determined

Answer: B. **Why:** Any single non-leap year always contributes exactly 1 odd day, meaning the following year's calendar shifts by exactly one weekday — consecutive years can never have an identical calendar (a minimum 2-year gap, specifically 5, 6, or 11 years depending on leap placement, is required).

Q6. [Hard] What was the day of the week on 1 January 1901?

A. Monday B. Tuesday C. Wednesday D. Thursday

Answer: B. **Why:** Odd days for 1900 complete years = 1 (from C5.3: $1900 = 4 \times 400 + 300 \rightarrow 0 + 1 = 1$). 1 Jan 1901 is the day right after 31 Dec 1900, so total odd days = 1 (no months/dates elapsed yet in the new count since we want 1 Jan itself, i.e., 1900 complete years + 1 day) $\rightarrow 1 + 1 = 2 \rightarrow$ weekday code 2 = Tuesday.

Q7. [Hard] Find the day of the week on 26 January 1950 (Republic Day of India — a well-known historical anchor).

A. Wednesday B. Thursday C. Friday D. Saturday

Answer: B. **Why:** Complete years before 1950 = $1949 = 1900(1 \text{ odd day}) + 49 \text{ years}(1901-1949)$. Leap years in that 49-year span: $1904 \dots 1948 = 12$ leap years. Odd days = $49 + 12 = 61 \rightarrow 61 \bmod 7 = 5$. Total for 1949 years = $1 + 5 = 6$. Complete months before January = 0. Add date 26 $\rightarrow 26 \bmod 7 = 5$. Total = $6 + 0 + 5 = 11 \rightarrow 11 \bmod 7 = 4 \rightarrow$ weekday code 4 = Thursday.

Q8. [Hard] [Trap] Two dates are exactly 5 years apart (e.g., from 1 Jan Year X to 1 Jan Year X+5), and there are exactly 1 leap year in between. What is the total odd-day shift between the two 1-Januaries?

A. 5 B. 6 C. 7 (i.e., same weekday) D. 4

Answer: B. **Why:** 5 years contribute a base of 5 odd days, plus 1 extra for the single leap year crossed = 6 total odd days shift — NOT a full week, so the two dates fall on different weekdays ($6 \bmod 7 = 6$, not 0). This illustrates why "same calendar" requires a very specific odd-day-multiple-of-7 AND matching leap status (C5.7), not just "roughly similar" year gaps.

C6. Seating Arrangements

C6.1 Basic Concept [Level: Basic]

- A block of raw clues (positional + relative statements) must be converted into one definite arrangement of people/objects.
- Four standard formats: **Single Row**, **Double Row** (two rows facing each other), **Circular Table**, **Polygonal/Square Table** (mixed corner + edge seating).

C6.2 The Left-Right Facing Trap [Level: Basic — the single most important idea in this chapter]

Person facing NORTH: their Left/Right = same as an outside observer's Left/Right (as drawn on a map).
 Person facing SOUTH: their Left/Right = MIRRORED vs an outside observer (observer's left = person's right).

- TRAP:** When two people in a row face **opposite directions**, "immediate left" for one is "immediate right" for the other — always fix an orientation (usually North, matching map convention) and re-derive left/right for anyone facing the opposite way; never assume everyone's left/right agree.
- Circular table:** facing **inward** (toward center) → your right hand points **clockwise** (as drawn from above). Facing **outward** (away from center) → your right hand points **counter-clockwise** — the exact reverse of facing inward. Mixed inward/outward arrangements require checking each person's facing individually before applying "left/right."

C6.3 Master Method [Level: Basic-Medium]

Step 1: Count total people; draw a position template (numbered seats) as a "Seating Frame."
 Step 2: Place every ABSOLUTE/direct clue first (e.g., "X sits at the left end," "Y sits 2nd from right").
 Step 3: Use RELATIVE clues (immediate left/right, "between," "not adjacent to") to eliminate remaining slots.
 Step 4: Re-scan all clues after every placement — a new fact often unlocks another clue.
 Step 5: If 2 arrangements still satisfy every clue, check whether the question asks something true in BOTH (then it's still answerable) or need

C6.4 Fully Worked Example — Double Row [Level: Hard]

(Real puzzle, independently solved and verified below.)

Five boys (Ashwin, Dipesh, Eshan, Chetan, Bipin) and five girls (Parul, Komal, Radha, Savita, Vimla) sit in two rows facing each other — boys in one row, girls in the other. Clues: Eshan is immediate right of Bipin, opposite Parul, not at any end. Radha is immediate right of Komal, opposite Chetan, at one of the ends. Ashwin is opposite Komal, who is 3rd to the right of Savita. Dipesh and Vimla are NOT opposite each other.

Solving (positions 1–5 left→right, column i of one row directly opposite column i of the other):

```
Radha = immediate right of Komal AND at an end -> Radha must be at position 5 (rightmost);
  (position 1 is impossible: nothing sits "left of" position 1 for Komal to occupy)
=> Girls: pos5=Radha, pos4=Komal
Radha opposite Chetan => Boys: pos5=Chetan
Ashwin opposite Komal => Boys: pos4=Ashwin
Komal is 3rd right of Savita => Savita=pos1 (Girls)
Remaining boys {Bipin,Eshan,Dipesh} fill positions {1,2,3}; Eshan=Bipin+1, Eshan not at an end.
Try Bipin=1,Eshan=2,Dipesh=3 -> Parul(opposite Eshan)=pos2, Vimla=pos3
  -> Dipesh(pos3) opposite Vimla(pos3): VIOLATES "not opposite" clue. Reject.
Try Bipin=2,Eshan=3,Dipesh=1 -> Parul(opposite Eshan)=pos3, Vimla=pos2
  -> Dipesh(pos1) opposite Savita(pos1): no violation. ACCEPTED.

Final: Boys: 1-Dipesh 2-Bipin 3-Eshan 4-Ashwin 5-Chetan
        Girls: 1-Savita 2-Vimla 3-Parul 4-Komal 5-Radha
```

- Q: Who is in the middle of the boys' row? → position 3 = **Eshan** (a "None of these" answer if Eshan isn't listed as an option — classic trap where the obvious-looking named options are all wrong).
- Q: Who is immediately right of Parul (pos 3)? → pos 4 = **Komal**.
- Q: Who is opposite Dipesh (pos 1)? → pos 1 girls = **Savita**.
- Q: How many boys between Ashwin (pos 4) and Bipin (pos 2)? → only position 3 (Eshan) → **1**.

C6.5 Fully Worked Example — Circular, Facing Outward [Level: Hard]

Six friends A,B,C,D,E,F sit around a circular table **facing outward**. E is between B and C. B and F are exactly opposite each other. A is exactly to the right of B.

Solving (positions 1–6 clockwise; facing outward ⇒ "right" = counter-clockwise ⇒ right-neighbor of seat k is seat k–1):

```
Let B = seat 1. Opposite (3 apart on a 6-seat table) => F = seat 4.
A = right of B = counter-clockwise neighbor of seat1 = seat 6.
Remaining seats {2,3,5} for {C,D,E}. E is between B and C => E adjacent to B(seat1) => E=seat2.
E's other neighbor (seat3) must be C (E is "between B and C") => C = seat3. Remaining D = seat5.

Final (clockwise): 1-B 2-E 3-C 4-F 5-D 6-A
```

- Q: Who is between D(5) and B(1)? → going the short way 5→6→1 passes seat 6 = **A**.
- Q: Who is exactly to the right of D(5)? → right = counter-clockwise neighbor = seat 4 = **F**.
- Q: Neighbors of A(6)? → seats 5 and 1 = **D and B**.

C6.6 Double-Row "Opposite" Convention & Polygonal Tables [Level: Hard]

- **TRAP:** In a double-row facing-each-other setup, two people in the SAME numbered column (counted left-to-right by an outside observer for both rows) are "opposite" each other — do not re-flip left/right per row; only re-flip left/right when a question asks about a specific PERSON's own left/right, not about column position.
- **Polygonal/square tables** (people at corners facing center + people at edge-midpoints facing outward, or similar mixed setup): treat corner-facing-center and edge-facing-outward people with their OWN separate left/right rule (from C6.2) — the two groups' "right" directions can point opposite ways even though they share one table, so re-derive per person, never assume uniformly.

Questions

(All fully self-contained text puzzles, Easy → Hard; [Trap] marks the facing-direction reversal trap.)

Q1. [Easy] Five friends sit in a row facing North. If P is immediately to the left of Q, and Q is immediately to the left of R, what is P's position relative to R?

A. Immediately right of R B. Two places to the left of R C. Immediately left of R D. Cannot be determined

Answer: B. **Why:** P—Q—R left to right, so P is 2 places left of R (both facing North, so map-left = their own left; no reversal needed here).

Q2. [Medium] [Trap] Six people sit in a row. A faces North, B faces South. If A says "C is to my right," and B is sitting exactly where A is looking from the opposite side, is C also to B's right?

A. Yes, always B. No — since B faces the opposite direction, C would be to B's LEFT C. Cannot be determined without more information D. Only if C sits between A and B

Answer: B. **Why:** A and B face opposite directions, so their left/right senses are mirrored; whichever side is "right" for A is "left" for B (C6.2's core trap).

Q3. [Medium] Six friends P, Q, R, S, T, U sit in a row facing North. R is immediately right of Q. Q is 2nd from the left end. T is immediately left of Q. S is at the rightmost (6th) end. Who is at the leftmost end?

A. T B. P C. U D. Cannot be determined

Answer: A. **Why:** Q is fixed at position 2. T is immediately left of Q ⇒ T = position 1 = the leftmost end.

Q4. [Medium] Six friends sit around a circular table facing the CENTER. E is between B and D. A is to the immediate right of B. If facing outward instead (all else identical), would A still be to the immediate right of B?

A. Yes, right stays the same regardless of facing B. No — facing outward reverses the clockwise/counter-clockwise sense of "right," so A's seat relative to B would need to be re-derived C. Only if E also changes seats D. Cannot be determined

Answer: B. **Why:** "Right" flips between clockwise and counter-clockwise depending on whether the group faces inward or outward — this is C6.2's central rule.

Q5. [Hard] (Using the double-row puzzle solved in C6.4) Who is the immediate neighbor of Chetan on the boys' side?

A. Ashwin only B. Dipesh only C. Bipin and Ashwin D. Eshan and Ashwin

Answer: A. **Why:** Boys row = Dipesh(1)-Bipin(2)-Eshan(3)-Ashwin(4)-Chetan(5); Chetan (pos 5, an end) has only ONE neighbor: Ashwin (pos 4).

Q6. [Hard] (Using the circular facing-outward puzzle in C6.5) If B and D swap seats, who becomes the immediate right neighbor of F?

A. E B. C C. The new occupant of seat 5 D. A

Answer: B. **Why:** Original: 1-B 2-E 3-C 4-F 5-D 6-A. Swapping B(seat1)↔D(seat5) only changes seats 1 and 5: new layout is 1-D 2-E 3-C 4-F 5-B 6-A. Right-neighbor of F(seat4) = seat(4−1) = seat 3 = **C**, untouched by the swap.

Q7. [Hard] [Trap] In a double row facing each other, X (row 1) is exactly opposite Y (row 2). X says Y is sitting to X's left-front. Is Y necessarily sitting toward what an outside observer would call the "left" side of the table?

A. Yes, always B. Not necessarily — "left" from X's own facing direction may be the observer's right if X faces a direction where left/right is mirrored C. Only if both rows face North D. Cannot be determined ever

Answer: B. **Why:** As in C6.2/C6.6, a person's own stated left/right depends on their facing direction, not a fixed map direction — always convert to the observer's frame before matching against absolute clues like "the left end of the hall."

Q8. [Hard] Eight people sit at a square table: four at corners (facing center) and four at edge-midpoints (facing outward). P sits at a corner. Q sits at an edge-midpoint immediately clockwise (as drawn) of P. From Q's own perspective, is P to Q's left or right?

A. Left, because Q faces outward and P is clockwise of Q, which is Q's left B. Right, always C. Depends only on P's facing, not Q's D. Cannot be determined

Answer: A. **Why:** Q faces outward, so Q's "right" = counter-clockwise (per C6.2); since P is clockwise of Q, P must be on Q's LEFT — a direct application of the outward-facing reversal rule to a mixed polygonal table (C6.6).

Q9. [Hard] Using the double-row puzzle in C6.4, if the girls' row is now described from THEIR own left (i.e., mirrored versus the observer used in the solution), who is at "girls' own rightmost end"?

A. Savita B. Radha C. Komal D. Vimla

Answer: A. **Why:** The girls face the boys (opposite direction to the observer's implicit "left-to-right" numbering used for boys); the girls' own right end is the observer's LEFT end of the girls' row = position 1 = Savita — another direct application of the facing-reversal trap, now on a solved arrangement.

C7. Coding-Decoding

C7.1 Basic Concept [Level: Basic]

- A word/number/phrase is transformed by a **consistent hidden rule**. You are given one or more example transformations and must decode the rule, then apply it to a new input.
- Master method for every type below: **compare position-by-position** between the given input and its code; do not guess the rule from the whole word at once.

C7.2 Type 1 — Letter/Word-Based Coding [Level: Basic-Medium]

Alphabet position: A=1, B=2, C=3, ... Z=26
 Reverse position: position + reverse-position = 27 (so Z=1st-from-end, A=26th-from-end, etc.)

- Fixed shift:** every letter moves forward/back by the SAME amount (e.g., CAT → FDW is a +3 shift on each letter).
- Variable shift:** the shift amount changes per letter position (e.g., +1, +2, +3... — check the shift for letter 1, then letter 2, etc., before assuming it's fixed).
- Reversal coding:** letters reversed in order, and/or each letter replaced by its "opposite" via the 27-minus rule ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$...).
- TRAP:** Never assume a fixed shift after checking only the first letter — always verify against at least 2 letters before locking in the rule (a variable shift looks identical to a fixed one on a single letter).

C7.3 Type 2 — Number-Based Coding [Level: Medium]

- Letters/words mapped to numbers, either via straightforward alphabet position, or via an **arbitrary assigned code** that must be cross-referenced across 2+ given example sentences to isolate which number belongs to which word (treat it like solving simultaneous equations: find a word common to two examples, its number must be common too — the differing words then map to the differing numbers).

C7.4 Type 3 — Mixed Alphabet/Word/Number Coding [Level: Medium-Hard]

- Combines Types 1 and 2 in the same question. Solve by isolating ONE part of the rule at a time (e.g., first determine how numbers are assigned, holding the letter-rule aside, then determine the letter-rule using the now-known number pattern).

C7.5 Type 4 — Symbol-Based Coding [Level: Medium]

- Letters/words replaced by arbitrary symbols. This is a pure substitution-cipher lookup once a mapping table is built from the given examples — no arithmetic trick, just careful bookkeeping of which symbol equals which letter/word.

C7.6 Type 5 — Language/"Chinese Coding" Substitution [Level: Hard]

- A fictional-language sentence maps to an English sentence; you're given 2+ paired example sentences and must decode a new sentence (or a specific word).
- Method (substitution-elimination, like solving simultaneous equations):**

Step 1: Find a fictional word that appears in TWO given example sentences.
 Step 2: Find the English word common to both corresponding English sentences.
 => That fictional word means that common English word.
 Step 3: Eliminate the now-known word from each sentence; the remaining fictional word(s) must map to the remaining English word(s) in that sentence.
 Step 4: Repeat until every word is mapped, or until you've mapped the specific word(s) the question asks about.

- Worked example: "sim tam kol" = "roses are red" and "sim del pic" = "roses smell nice". Common fictional word = "sim"; common English word = "roses" ⇒ **sim = roses**. Then "tam kol" = "are red" and "del pic" = "smell nice" (no further overlap needed if the question only asks for "roses").

Questions

(Easy → Hard; [Trick] = reward for knowing a shortcut; [Trap] = common misconception)

Q1. [Easy] If CAT is coded as DBU, what is the coding rule?

A. Each letter -1 B. Each letter +1 C. Reverse the letters D. Replace with opposite letter

Answer: B. Why: $C \rightarrow D(+1)$, $A \rightarrow B(+1)$, $T \rightarrow U(+1)$ — consistent +1 shift on every letter.

Q2. [Easy] Using the same +1 rule from Q1, how is DOG coded?

A. EPH B. CNF C. EOG D. EPI

Answer: A. Why: $D \rightarrow E$, $O \rightarrow P$, $G \rightarrow H \Rightarrow$ EPH.

Q3. [Medium] [Trap] If BAT is coded as YZG (using $A \leftrightarrow Z$, $B \leftrightarrow Y$, ... 27-minus-position rule), what is CAT coded as?

A. XZG B. XZH C. ZYG D. YZH

Answer: A. Why: Position(B)=2→27-2=25=Y ✓; A=1→26=Z ✓; T=20→7=G ✓ (confirms the rule). For CAT: C=3→24=X; A=1→26=Z; T=20→7=G ⇒ **XZG**.

Q4. [Medium] In a code, ROSE is written as 6821 and CHAIR is written as 73456. What is the code for SEARCH?

A. 214673 B. 246173 C. 214763 D. 146273

Answer: A. Why: From ROSE: R=6, O=8, S=2, E=1. From CHAIR (R=6 consistent): C=7, H=3, A=4, I=5. SEARCH = S-E-A-R-C-H = 2-1-4-6-7-3 = **214673**.

Q5. [Medium] [Trick] If MONEY is coded as NQQID, what shift pattern is being used?

A. Fixed shift of +1 B. Fixed shift of +2 C. Variable shift, increasing by 1 for each successive letter D. Reversal coding

Answer: C. Why: $M(+1)=N$, $O(+2)=Q$, $N(+3)=Q$, $E(+4)=I$, $Y(+5)=D$ (wrapping past Z: $25+5=30$, $30-26=4=D$) — the shift grows by 1 each letter, exactly the C7.2 trap of assuming a single fixed shift after checking only the first letter.

Q6. [Hard] In a symbol code, "@" = A, "#" = B, "\$" = C, and words are spelled out symbol-by-symbol in order. If "BAG" is coded as "#@%", what does "%" represent?

A. A B. B C. G D. Cannot be determined from other examples

Answer: C. Why: B→#, A→@ (both already known), so by elimination the remaining symbol % must represent the remaining letter G.

Q7. [Hard] Type 5: "pil mit sog" = "flowers are beautiful" and "rin mit kot" = "roses are fragrant". What does "mit" mean?

A. flowers B. are C. beautiful D. fragrant

Answer: B. Why: "mit" is common to both fictional sentences; "are" is the only word common to both English sentences ⇒ mit = are.

Q8. [Hard] Using Q7's data, what does "pil" most likely mean, given "pil" appears only in the first sentence alongside "sog"?

A. flowers B. are C. beautiful D. Cannot be determined from these two sentences alone

Answer: D. **Why:** With only 2 sentences and "pil"/"sog" both unique to sentence 1 (mapping to "flowers"/"beautiful" in some order), there is no third overlapping sentence to disambiguate which of "pil" or "sog" is "flowers" vs "beautiful" — a genuine coding-decoding question always needs enough overlapping examples to fully pin down every word; this is the classic trap of assuming a unique answer exists when the given data is actually insufficient.

Q9. [Hard] [Trap] A code assigns EACH letter of "STRONG" a number equal to its position in the alphabet MINUS its position in the word (1st letter, 2nd letter, etc.). What number codes the letter 'O' (4th letter of STRONG)?

A. 11 B. 15 C. 19 D. 9

Answer: A. **Why:** Position of O in alphabet = 15. Position in word STRONG: S(1)T(2)R(3)O(4)N(5)G(6) $\Rightarrow$ O is 4th. $15 - 4 = 11$.

C8. Blood Relations & Directions

C8.1 Blood Relations — Basic Concept [Level: Basic]

- Deduce a relationship between two people from a chain of stated relations (often through an unnamed intermediate person).
- Core reference list** (memorize — this list IS the shortcut):

Given relation	Resolves to
Mother's/Father's son	Brother (or self, if speaker is male — always check for self-reference)
Mother's/Father's daughter	Sister (or self, if speaker is female)
Father's son's son	Nephew (brother's son) — or own son, if "son" refers to the speaker
Mother's father	Grandfather
Father's sister	Paternal aunt
Mother's brother	Maternal uncle (mama)
Son's wife	Daughter-in-law
Husband's/Wife's sister	Sister-in-law
Husband's/Wife's brother	Brother-in-law
Sister's husband	Brother-in-law

C8.2 Method & the Self-Reference Trap [Level: Medium]

Step 1: Work from the OUTERMOST relation inward (start decoding from the last-stated link closest to the person named in the question), OR build a small generation-level family tree in text (Grandparents -> Parents -> Speaker's generation -> Children).

Step 2: Track gender as "unknown" until a pronoun (he/she) or explicit word (brother/sister/son/daughter) fixes it.

Step 3: ALWAYS check whether a long relation chain loops back to the ORIGINAL speaker — e.g., "my father's son" is NOT always "my brother"; if the speaker has no other brother, "my father's son" IS the speaker himself.

- TRAP:** This self-reference loop is the single most common trap in the entire chapter — always ask "could this relation just mean the speaker?" before naming a new person.

C8.3 Coded Blood Relations [Level: Medium-Hard]

- Relations replaced by symbols/operators, e.g., " $A + B$ " = A is B's mother, " $A \times B$ " = A is B's brother, " $A - B$ " = A is B's father, " $A \div B$ " = A is B's sister (the exact symbol-meaning mapping is always GIVEN in the question — never assumed).
- Method: substitute the given meanings and trace the chain exactly as in C8.2, one link at a time, writing out the plain-language relation as you go rather than trying to hold the whole symbolic chain in your head.

C8.4 Directions — Basic Concept [Level: Basic]

Right-turn sequence (clockwise): N -> E -> S -> W -> N
 Left-turn sequence (counter-clockwise): N -> W -> S -> E -> N

- 4 main directions (N,E,S,W) + 4 intermediates (NE,SE,SW,NW) at 45° between them.

C8.5 Coordinate-Tracking Method [Level: Medium — the master technique]

Treat North/South as the Y-axis (+ for North, - for South).
 Treat East/West as the X-axis (+ for East, - for West).
 Add up every leg of the journey on its own axis; do NOT try to visualize the whole path — just accumulate (x, y) separately.
 Final straight-line/shortest distance from start = $\sqrt{x^2 + y^2}$ [Pythagoras]

- This coordinate method is far more reliable than mental path-tracing once a journey has 3+ turns.

C8.6 Shadow Direction [Level: Basic — a standalone fact]

- At **sunrise**, the sun is in the East, so a shadow falls toward the **West**.
- At **sunset**, the sun is in the West, so a shadow falls toward the **East**.
- TRAP:** Students frequently reverse these — anchor it by remembering "the shadow always falls AWAY from the sun."

Questions

(Easy → Hard; [Trap] tags the two classic error categories: self-referential blood relations, and sunrise/sunset shadow confusion.)

Q1. [Easy] Pointing to a man, a woman says, "His mother is my mother's only daughter." How is the man related to the woman?

A. Brother B. Son C. Father D. Uncle

Answer: B. **Why:** "My mother's only daughter" = the speaker herself (she has no sisters). So "his mother" = the speaker $\Rightarrow$ the man is her **son**.

Q2. [Easy] A is B's father. B is C's brother. How is A related to C?

A. Father B. Grandfather C. Uncle D. Brother

Answer: A. Why: B and C are siblings, and A is B's father $\Rightarrow$ A is also C's **father** (same parent for both siblings, absent any statement to the contrary).

Q3. [Medium] Pointing to a photograph, a man says, "She is the daughter of my grandfather's only son." How is the woman in the photo related to the man?

A. Sister B. Wife C. Daughter D. Cannot be determined

Answer: A (or the speaker himself, if female — but since the speaker refers to "she," treat as distinct). **Why:** "My grandfather's only son" = the speaker's father (assuming the speaker's father is that only son) $\Rightarrow$ "daughter of my father" = the speaker's **sister**. [Trap check: could "my grandfather's only son" be the speaker himself? Only if the speaker is male and has no father alive in the frame — the standard reading takes it as the speaker's father since a woman is being described as a separate person "she".]

Q4. [Medium] [Trap] A man said to a woman, "Your mother's husband's sister is my aunt." How is the woman related to the man?

A. Sister B. Cousin C. Niece D. Mother

Answer: B. Why: "Your mother's husband" = the woman's father. "Father's sister" = the woman's paternal aunt. That same aunt is also the man's aunt $\Rightarrow$ the man and the woman share a grandparent through that aunt's sibling (the man's parent and the woman's father are siblings) $\Rightarrow$ they are **cousins**.

Q5. [Medium] In a family, P is the mother of Q. Q is the brother of R. R is the son of S. How is S related to P?

A. S is P's husband B. S is P's brother C. S is P's daughter D. Cannot be determined

Answer: A. Why: Q and R are siblings (same parents). P is their mother, so R's father must be P's husband. Since R is "the son of S," S is R's father $\Rightarrow$ **S is P's husband**.

Q6. [Hard] Coded relations: "A + B" means A is the mother of B; "A $\times$ B" means A is the brother of B; "A – B" means A is the father of B. If "P + Q $\times$ R – S" holds, how is S related to P?

A. Grandson B. Granddaughter C. Grandfather D. Son

Answer: A. Why: P+Q $\Rightarrow$ P is Q's mother. Q $\times$ R $\Rightarrow$ Q is R's brother (so Q and R are siblings, same mother P). R–S $\Rightarrow$ R is S's father. So S is R's child, R is P's child $\Rightarrow$ S is P's **grandson** (assuming S's gender is male per "grandson"; if ungendered, "grandchild" — question presumes male by the options given).

Q7. [Hard] Starting from point X, a man walks 5 km North, then turns right and walks 3 km, then turns right again and walks 5 km, then turns left and walks 4 km. How far is he from X, and in which direction?

A. 7 km, East B. 7 km, North C. 9 km, East D. 3 km, North

Answer: A. Why (coordinate method): Start (0,0). North 5 $\Rightarrow$ (0,5). Turn right (facing North $\rightarrow$ right = East), walk 3 $\Rightarrow$ (3,5). Turn right again (facing East $\rightarrow$ right = South), walk 5 $\Rightarrow$ (3,0). Turn left (facing South $\rightarrow$ left = East), walk 4 $\Rightarrow$ (7,0). Final position (7,0) $\Rightarrow$ **7 km East** of X.

Q8. [Hard] A man walks 8 km East, then 6 km South, then 8 km West. How far and in which direction is he from the start, and what is the straight-line distance?

A. 6 km South B. 6 km, direction South of start C. 10 km, South-West D. 14 km South

Answer: B. Why (coordinate method): (0,0) $\rightarrow$ East 8 $\rightarrow$ (8,0) $\rightarrow$ South 6 $\rightarrow$ (8,-6) $\rightarrow$ West 8 $\rightarrow$ (0,-6). Net displacement (0,-6) $\Rightarrow$ exactly **6 km due South** — the East and West legs (both 8 km) cancel out completely, leaving only the South component.

Q9. [Hard] [Trap] One morning, a man's shadow fell exactly to his LEFT as he stood still. Which direction was he facing?

A. North B. South C. East D. West

Answer: A. Why: Morning sun is in the East, so shadows fall toward the West (C8.6). Facing North, West is on your left (standard map orientation) — so the shadow-on-the-left condition places him **facing North**.

Q10. [Hard] [Trap] A tower's shadow is longer than the tower's height. Is it more likely to be near sunrise/sunset or near noon?

A. Near noon, when the sun is directly overhead B. Near sunrise or sunset, when the sun is low on the horizon, casting a long shadow C. Shadow length is unrelated to time of day D. Cannot be determined

Answer: B. Why: The lower the sun's angle above the horizon (as at sunrise/sunset), the longer the shadow cast by a fixed-height object; near noon the sun is high, giving the shortest shadow.

C9. Clocks

C9.1 Basic Mechanics [Level: Basic]

- Clock dial = circle = 360° , divided into 60 minute-spaces ($60 \times 6^\circ = 360^\circ$).
- Hour hand ("short hand") and Minute hand ("long hand") speeds:

Minute hand: $360 \text{ deg} / 60 \text{ min} = 6 \text{ deg/minute}$
 Hour hand: $360 \text{ deg} / (12 \times 60) \text{ min} = 0.5 \text{ deg/minute}$
 Relative speed (minute hand gains on hour hand) = $6 - 0.5 = 5.5 \text{ deg/minute}$

- TRICK: The minute hand is exactly **12 times faster** than the hour hand ($6 / 0.5 = 12$) — this single ratio underlies every clock formula below.

C9.2 Angle Between Hands [Level: Basic-Medium]

Angle = $|30H - 5.5M|$ degrees
 (H = hour shown, 0-11 form; M = minutes past that hour)
 If the result exceeds 180 degrees, subtract it from 360 to get the actual (reflex-avoided) angle.

- Derivation: Hour hand position from 12 = $30H + 0.5M$ degrees. Minute hand position from 12 = $6M$ degrees. Difference = $30H + 0.5M - 6M = 30H - 5.5M$.

Ways to solve "angle at exact o'clock" (M=0)

1. **Formula method:** $|30H - 0| = 30H$ (or $360 - 30H$ if > 180).
2. **Counting method:** each hour-mark = 30° apart; count the number of hour-marks between the hands and multiply by 30.
 - 5 o'clock: 5 marks $\times 30^\circ = 150^\circ$. 8 o'clock: 4 marks (shorter way round) $\times 30^\circ = 120^\circ$. 2 o'clock: 2 $\times 30^\circ = 60^\circ$.

Worked Examples (increasing difficulty)

At 5:30 $\rightarrow |30(5) - 5.5(30)| = |150 - 165| = 15^\circ$.
 At 8:40 $\rightarrow |30(8) - 5.5(40)| = |240 - 220| = 20^\circ$.
 At 2:13 $\rightarrow |30(2) - 5.5(13)| = |60 - 71.5| = 11.5^\circ$.

C9.3 Hands Coincide / Opposite / Perpendicular [Level: Medium — the highest-yield facts in this chapter]

Hands coincide (angle = 0°): occurs every $720/11 = 65 + 5/11$ minutes
 Hands opposite (angle = 180°): occurs every $720/11$ minutes (offset by half a cycle)
 Hands perpendicular (angle = 90°): occurs twice as often as coincidence

- **TRAP:** In any 12-hour span, hands coincide **11 times, not 12** (naive students assume once per hour = 12). Reason: between 11:00 and 1:00, they coincide only ONCE — at 12:00 — not twice. Same undercounting applies to "opposite" (11 times per 12h, 22 times/day) and "perpendicular" (22 times per 12h, 44 times/day).
- TRICK: List of coincidence times starting from 12:00 — 12:00, 1:05(5/11), 2:10(10/11), 3:16(4/11)... each **65+5/11 min** after the previous.
- To find WHEN hands coincide between H and H+1 o'clock: solve $30H - 5.5M = 0 \rightarrow M = 60H/11$.

Between 6 and 7: $M = 360/11 = 32 + 8/11$ minutes, i.e. 6:32 $\frac{8}{11}$.

C9.4 Mirror Image & Water Image of Clock Time [Level: Medium]

Mirror image time (reflect left-right, vertical mirror) = $12:00 - \text{given time}$ (borrow 12:00 if needed)
 Water image time (reflect top-bottom, horizontal mirror) = $6:00 - \text{given time}$ (borrow 12:00 if needed)

- **TRAP:** These use **different** reference points (12:00 for mirror, 6:00 for water) — do not swap them.

Mirror image of 8:40 $\rightarrow 12:00 - 8:40 = 3:20$.
 Mirror image of 4:12 $\rightarrow 12:00 - 4:12 = 7:48$.
 Water image of 10:10 $\rightarrow 6:00 - 10:10 \rightarrow \text{borrow } 12\text{h} \rightarrow 18:00 - 10:10 = 7:50$.

C9.5 Distance Traveled by Hands [Level: Medium]

- Each hand traces a circle of circumference $2\pi r$ (r = hand length) every time it completes a revolution. Minute hand: 1 revolution/hour. Hour hand: 1 revolution/12 hours.

A clock's long hand (minute hand) = 8 cm, short hand (hour hand) = 7 cm. Runs for 4 days (=96 hours). Total distance covered by both hands?
 Minute hand: 96 revolutions $\rightarrow 96 \times 2\pi(8) = 1536\pi$ cm.
 Hour hand: $96/12 = 8$ revolutions $\rightarrow 8 \times 2\pi(7) = 112\pi$ cm.
Total = $1536\pi + 112\pi = 1648\pi$ cm.

C9.6 Gaining / Losing (Faulty) Clocks [Level: Hard]

- Core idea: a faulty clock's **shown/displayed time** and **real elapsed time** run at different rates. Always convert via a proportion: (faulty minutes shown) : (real minutes elapsed).
- **TRAP:** The common WRONG shortcut is to just multiply "total hours elapsed" by the per-hour gain/loss and add/subtract directly — this silently assumes the elapsed-hours count is measured in *real* time, but if the elapsed hours are counted *on the faulty clock's face*, that approach is invalid. Always set up the full proportion instead.

Worked Example (classic multi-part problem)

A clock is set right at Monday 10:00 AM. It **loses 6 minutes every hour** (i.e., in 60 real minutes, the faulty clock advances only 54 minutes). Find the actual time when the faulty clock shows **Friday 3:00 PM**.
 Step 1 — elapsed time *shown on the faulty clock*: Monday 10 AM $\rightarrow$ Friday 3 PM = 4 days + 5 hours = **101 hours** = 6060 faulty-minutes.

Step 2 — convert to real minutes using the ratio 54 (faulty) : 60 (real): Real minutes = $6060 \times (60/54) = \mathbf{6733.33 \text{ min} = 112 \text{ h } 13 \text{ min } 20 \text{ s}}$.

Step 3 — add to the start time: Monday 10:00 AM + 112h13m20s = (112h = 4 days 16h) → Friday 10 AM + 16h = **Saturday 2:13:20 AM**.

TRAP: A popular wrong shortcut computes $101 \times 6 = 606 \text{ minutes}$ and simply adds this to Friday 3 PM, giving Saturday 1:06 AM — this is **WRONG** because it treats the 101 hours as real elapsed time rather than faulty-clock-displayed time; always use the Step-2 proportion.

More Worked Examples

A clock loses 1% of time in week 1, then gains 2% in week 2 (1 week = 168 hours). Set right at Sunday 12 noon. What does it show 14 days later?

Week 1: loses 1% of 168h = 1.68h. Week 2: gains 2% of 168h = 3.36h. Net after 2 weeks = $-1.68 + 3.36 = \mathbf{+1.68 \text{ h ahead}}$.

Real time 14 days later = Sunday 12 noon again. Clock shows 1.68h = 1h 40min 48s ahead → **1:40:48 PM**.

A watch is 2 minutes slow at noon Monday, and 4 min 48 sec fast at 2 PM the following Monday (uniform gain throughout). When was it exactly correct?

Total swing = from -2 min to $+4 \text{ min } 48 \text{ s}$ = 6.8 minutes gained, over (7 days 2 hours =) 170 hours → rate = $6.8/170 = 0.04 \text{ min/hour}$.

Watch is correct once it has gained back its initial 2-minute deficit: time needed = $2/0.04 = \mathbf{50 \text{ hours}}$ after noon Monday = 2 days 2 hours → **2:00 PM Wednesday**.

Questions

(Several of these are real, previously-asked exam questions — verified by hand.)

Q1. [Easy] Find the angle between hour and minute hand at 7 o'clock.

A. 150° B. 180° C. 210° D. 120°

Answer: A. Why: $[30 \times 7 - 5.5 \times 0] = 210^\circ$. Since the clock-angle convention always reports the smaller of the two angles formed ($\leq 180^\circ$), the answer is $360 - 210 = \mathbf{150^\circ}$.

(Counting method check: 7 o'clock is 5 hour-marks apart the short way round → $5 \times 30^\circ = 150^\circ$.)

Q2. [Easy] What is the angle between the hands at 5:30?

A. 45° B. 30° C. 15° D. 60°

Answer: C. Why: $[30(5) - 5.5(30)] = [150 - 165] = \mathbf{15^\circ}$.

Q3. [Medium] Find the acute angle between the hour and minute hand at 2:13 PM.

A. 16.5° B. 18° C. 13.5° D. 11.5°

Answer: D. Why: $[30(2) - 5.5(13)] = [60 - 71.5] = \mathbf{11.5^\circ}$.

Q4. [Medium] [Trick] At what time between 6 and 7 o'clock do the hands of a clock coincide?

A. 6:13 B. 6:36 C. 6:32 + 8/11 min D. 6:35 + 8/11 min

Answer: C. Why: Solve $30(6) - 5.5M = 0 \rightarrow M = 180/5.5 = 32 + 8/11 \text{ minutes}$.

Q5. [Medium] [Trap] How many times do the hands of a clock coincide in 5 hours?

A. 4 B. 3 C. 2 D. 6

Answer: A. Why: Coincidences occur every $65 + 5/11 \approx 65.45 \text{ minutes}$ = 300 minutes available/ $65.45 \approx 4.58 \rightarrow \mathbf{4 \text{ complete coincidences}}$ in 5 hours (300 minutes).

Q6. [Medium] [Trap] Between 9 AM and 9 PM of the same day, how many times are the minute and hour hands together?

A. 9 B. 10 C. 11 D. 12

Answer: C. Why: They coincide once per hour on average, but between 11 and 1 o'clock the coincidence at 12:00 is shared/counted once — over any 12-hour span the count is **11**, not 12.

Q7. [Medium] [Trap] How many times in a day are the hands of a clock in a straight line but pointing in opposite directions?

A. 20 B. 22 C. 24 D. 48

Answer: B. Why: Opposite-direction alignment happens 11 times per 12 hours (between 5 and 7 o'clock it happens only once, at 6:00) → 22 times in a full day.

Q8. [Medium] [Trick] Find the mirror image of the time 8:40.

A. 3:20 B. 2:30 C. 8:20 D. 9:20

Answer: A. Why: Mirror image = $12:00 - 8:40 = \mathbf{3:20}$.

Q9. [Medium] If a clock shows 4 hours 12 minutes, what time does its mirror image show?

A. 8h 48m B. 9h 48m C. 10h 48m D. 7h 48m

Answer: D. Why: $12:00 - 4:12 = \mathbf{7:48}$.

Q10. [Hard] A clock's long hand (minute hand) is 8 cm and short hand (hour hand) is 7 cm. If the clock runs for 4 days, find the total distance covered by both hands.

A. $1824\pi \text{ cm}$ B. $1724\pi \text{ cm}$ C. $1648\pi \text{ cm}$ D. $2028\pi \text{ cm}$

Answer: C. Why: 4 days = 96 h. Minute hand: $96 \text{ rev} \times 2\pi(8) = 1536\pi$. Hour hand: $96/12 = 8 \text{ rev} \times 2\pi(7) = 112\pi$. Total = **$1648\pi \text{ cm}$** .

Q11. [Hard] [Trap] A church clock is set right at Monday 10:00 AM. It loses 6 minutes every hour. What is the actual time when the faulty clock shows 3:00 PM on Friday?

A. Saturday 1:06 AM B. Friday 4:54 AM C. Saturday 2:13:20 AM D. Friday 3:00 AM

Answer: C. Why: Faulty-clock-elapsed = 101 h = 6060 faulty-min. Real minutes = $6060 \times (60/54) = 6733.33 \text{ min} = 112 \text{ h } 13 \text{ m } 20 \text{ s}$. Monday 10 AM + 112h13m20s =

Saturday 2:13:20 AM. (Option A results from the common WRONG shortcut of directly multiplying 101×6 minutes and adding — see the TRAP note in C9.6.)

Q12. [Hard] A clock loses 1% of time during the first week, then gains 2% during the next week. If set right at 12 noon Sunday, what time will it show exactly 14 days later?

A. 1:40:48 B. 1:36:48 C. 1:41:24 D. 10:19:12

Answer: A. Why: Net gain over 2 weeks = $-1.68 \text{ h} + 3.36 \text{ h} = +1.68 \text{ h} = 1 \text{ h } 40 \text{ min } 48 \text{ s}$ ahead of the real 12 noon → **1:40:48 PM**.

Q13. [Hard] A watch is 2 minutes slow at noon on Monday and 4 min 48 sec fast at 2 PM the following Monday, gaining uniformly throughout. When was it exactly correct?

A. 2 PM Tuesday B. 2 PM Wednesday C. 3 PM Thursday D. 1 PM Friday

Answer: B. Why: Total gain of 6.8 minutes over 170 hours → rate 0.04 min/h . Time to gain back the initial 2-min deficit = $2/0.04 = 50 \text{ hours} = 2 \text{ days } 2 \text{ hours}$ after noon Monday = **2 PM Wednesday**.

Q14. [Hard] [Trick] How many "palindrome" times (digit sequence reads the same forwards and backwards, e.g. 5:45 → "545") occur on a 12-hour clock face from 12:00 noon to the next midnight?

A. 43 B. 51 C. 58 D. 57

Answer: D. Why: For single-digit hours 1-9: format H:M1M2 is a palindrome whenever $M2 = H$ ($M1$ free, 0-5) → 6 times each $\times 9 \text{ hours} = 54$. For two-digit hours 10, 11, 12: exactly one palindrome each (10:01, 11:11, 12:21) = 3 more. Total = $54 + 3 = \mathbf{57}$.

C10. Ranking, Order & Sequence; Logical Sequence of Words

C10.1 Ranking — Core Formula [Level: Basic]

$$\text{Total people} = (\text{Rank from the front}) + (\text{Rank from the back}) - 1$$

- TRICK: this single formula solves the vast majority of ranking questions — memorize it as the anchor and derive whichever unknown is asked (total, or one of the two ranks) by rearranging it.
- Worked example: Rohan is 7th from the top and 12th from the bottom in a class. Total students = $7+12-1 = 18$.

C10.2 Change of Position (Someone Moves in a Row/Rank) [Level: Medium]

- If a person's rank from the front changes from a to b (moves back in rank, i.e., more people now ahead), the number of people who "overtaken" them = $b - a$. If asked how many are between two ranks, subtract and adjust by 1 depending on whether endpoints are included.
- TRAP:** "number of people between rank x and rank y " = $|x - y| - 1$ (exclude both endpoints), NOT $|x-y|$ — a very commonly missed off-by-one error.

C10.3 Ranking in Both Directions Simultaneously [Level: Medium]

- When ranks from left AND right (or top AND bottom) are both given for the SAME row, use the total-count formula (C10.1) directly.
- When comparing two DIFFERENT people's positions in the same row, set up each person's rank-from-front and rank-from-back as separate equations sharing the same total, then solve simultaneously.

C10.4 Distance-Direction-Ranking Combined Puzzles [Level: Hard]

- Some questions combine a physical arrangement (people standing in a line facing a direction) with ranking — draw the row as a simple left-to-right diagram with position numbers, mark which way each person faces, then answer rank/distance/direction sub-questions directly off the diagram rather than trying to reason abstractly.

C10.5 Logical Sequence of Words [Level: Medium]

- Given a set of words, arrange them in a logically meaningful order — common patterns:
 - Size/hierarchy order** (e.g., Village → Town → City → Metropolis).
 - Process/chronological order** (e.g., seed → sapling → tree → fruit, or the sequential steps of an activity like Application → Interview → Offer → Joining).
 - Dictionary/alphabetical order** — used when the question explicitly wants alphabetical arrangement of otherwise unrelated words.
 - Age/time order** (e.g., Infant → Child → Adolescent → Adult → Old age).
- TRICK: identify the CATEGORY of logical relationship first (size, time, process, hierarchy) before attempting to order the words — trying to order without first spotting the underlying dimension leads to guessing.
- TRAP:** some words may fit more than one plausible order (e.g., could be ordered by size OR by a different real-world criterion) — always pick the ordering criterion that uses ALL the given words consistently, discarding an ordering that leaves an odd one out unless the question explicitly allows it.

Questions

Q1. [Easy] In a row of students, Meena is 9th from the left and 15th from the right. How many students are in the row?

A. 24 B. 23 C. 25 D. 22

Answer: B. Why: Total = $9+15-1 = 23$.

Q2. [Easy] If there are 30 people in a queue and a person is 12th from the front, what is their rank from the back?

A. 18 B. 19 C. 20 D. 17

Answer: B. Why: Total = front + back - 1 → $30 = 12 + \text{back} - 1 \rightarrow \text{back} = 19$.

Q3. [Medium] [Trap] In a line of 20 people, how many people are standing between the 5th person from the left and the 15th person from the left?

A. 10 B. 9 C. 11 D. 8

Answer: B. Why: Between (exclusive of both endpoints) = $|15-5| - 1 = 10 - 1 = 9$.

Q4. [Medium] Arjun is ranked 10th from the top in a class. If there are 35 students in total, what is his rank from the bottom?

A. 25 B. 26 C. 24 D. 27

Answer: B. Why: Total = front+back-1 → $35 = 10+\text{back}-1 \rightarrow \text{back} = 26$.

Q5. [Medium] Arrange in logical (chronological/process) order: (1) Seed (2) Fruit (3) Flower (4) Plant

A. 1-4-3-2 B. 1-3-4-2 C. 4-1-3-2 D. 1-2-3-4

Answer: A. Why: Natural growth sequence: Seed → Plant → Flower → Fruit.

Q6. [Hard] Arrange in a logical size order: (1) Village (2) Country (3) Town (4) City (5) District

A. 1-3-4-5-2 B. 1-3-5-4-2 C. 3-1-4-5-2 D. 1-4-3-5-2

Answer: B. Why: Increasing size/administrative hierarchy: Village → Town → District → City → Country. (District is administratively above a City in this common ordering convention, sitting between Town and Country.)

Q7. [Hard] In a row of children facing North, Rakesh is 7th from the left and 4th from the right. A new child joins the row at the right end. What is the new total number of children?

A. 10 B. 11 C. 9 D. 12

Answer: B. Why: Original total = $7+4-1 = 10$; adding one more child at the end makes the new total 11.

Q8. [Hard] [Trap] In a class of 40, Priya's rank from the top improves from 15th to 10th after a re-test. How many students did she overtake?

A. 5 B. 10 C. 15 D. 25

Answer: A. Why: Rank improved from 15 to 10 → she moved past exactly $(15-10)=5$ students who were previously ahead of her.

C11. Inequality & Data Sufficiency

C11.1 Decoding Inequality Symbols [Level: Basic]

- Standard symbols: $>$ (greater than), $<$ (less than), $\geq$ (at least/greater-or-equal), $\leq$ (at most/less-or-equal), $=$ (equal).
- Many reasoning questions replace these with arbitrary CODED symbols and give you a key, e.g.:

```
A @ B means A > B
A # B means A < B
A $ B means A = B
A & B means A ≥ B
A * B means A ≤ B
```

- TRICK: Always decode the ENTIRE statement chain into plain $>$ $<$ $=$ $\geq$ $\leq$ symbols FIRST, before doing any reasoning. Trying to reason directly in coded symbols is the #1 cause of careless errors.

C11.2 Chaining Inequalities [Level: Medium]

- A chain like $A > B > C = D \geq E$ lets you relate any two elements by combining the links between them.
- TRAP:** Only an unbroken chain of **strict** relations ($>$, $<$, $=$) gives a **strict** conclusion between two endpoints. The moment a $\geq$ or $\leq$ appears anywhere in the path between two elements, the *derived* relation between those two endpoints is also only "at least" / "at most" — **not** strict — even if every other link in that path is strict.
- TRICK: To test whether a given conclusion "X (relation) Y" *definitely* follows, trace the unbroken path of links from X to Y in the decoded chain and combine them:

```
All links strict (>, <, =) along the path -> conclusion is STRICT
Any single link in the path is ≥ or ≤ -> conclusion is only "at least" / "at most"
No unbroken path connects X and Y at all -> NO definite relation can be concluded
```

Worked Example

Decoded chain: $A > B > C = D \geq E$.

Conclusion I: $A > E$ — Path $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ has links $>, >, =, \geq$. The last link is $\geq$, so the combined relation is only $A \geq E$, not strict. **Conclusion I does NOT definitely follow.**

Conclusion II: $A > D$ — Path $A \rightarrow B \rightarrow C \rightarrow D$ has links $>, >, =$, all strict. So $A > D$ **definitely follows.**

A Second Common Trap Pattern: Two Elements Both Related to a Common Third

Decoded: $A > B, C > B$ (i.e., both A and C are greater than the same B), and $C \leq E$.

Conclusion: $A > C$? — A and C are both compared only to B, never to each other directly, and no chain connects A to C. **No definite relation between A and C can be concluded** — this is the single most common inequality trap: two elements sharing a common "anchor" element are NOT thereby related to each other.

Conclusion: $C \leq E$? — Given directly. **Definitely true.**

C11.3 Data Sufficiency — The Framework [Level: Basic]

- Data Sufficiency does **not** ask you to solve the question — only to judge whether the given statement(s) provide **enough** information to answer it with a single, unique result.
- Standard 5-option framework (memorize this cold, it's identical across every exam that uses this format):

Option	Meaning
(A)	Statement I alone is sufficient, but Statement II alone is not
(B)	Statement II alone is sufficient, but Statement I alone is not
(C)	BOTH together are sufficient, but neither alone is
(D)	EACH statement alone is sufficient (either one works on its own)
(E)	Even both statements together are NOT sufficient

C11.4 The Solving Method — Order Matters [Level: Medium]

```
Step 1: Read the question. Identify exactly what needs a UNIQUE value/answer.
Step 2: Judge Statement I COMPLETELY ALONE. Pretend Statement II does not exist.
Step 3: Judge Statement II COMPLETELY ALONE. Pretend Statement I does not exist (and forget any partial progress from Step 2).
Step 4: ONLY IF both are insufficient alone, combine them and judge together.
```

- TRAP:** Never let information from Statement II "leak" into your judgment of Statement I in isolation (or vice versa) — this is the most common process error, and it silently produces the wrong option even when the underlying math is done correctly.
- TRICK: "Sufficient" means the statement pins down **exactly one** possible value/answer. If a statement is consistent with **two or more** different possible answers, it is **not** sufficient — even if it narrows the possibilities down a lot.

C11.5 The Classic Quadratic Trap [Level: Hard]

- TRAP:** A statement that looks like it "gives you x" via an equation may actually yield **two** valid roots (e.g., from a quadratic), making it **insufficient** even though it appears to pin down a value.

Example: "What is x ?" Statement: $x^2 = 9$
 This does NOT give a unique x — it gives $x = 3$ OR $x = -3$.
 $\Rightarrow$ Statement alone is INSUFFICIENT, despite looking like a complete equation.

Questions

(ordered Easy $\rightarrow$ Hard; [Trick] tests a shortcut, [Trap] tests a classic misconception)

Q1. [Easy] Given: $A > B$, $B > C$. Which is definitely true?

A. $A > C$ B. $A = C$ C. $A < C$ D. Cannot be determined

Answer: A. Why: Both links are strict $>$, so they chain directly: $A > B > C$ implies $A > C$.

Q2. [Easy] Decode: " $P \& Q$ " means $P \geq Q$, and " $P \# Q$ " means $P < Q$. Given $P \& Q$ and $Q \# R$, what is the relation between P and R ?

A. $P > R$ B. $P \geq R$ C. P could be less than, equal to, or greater than R D. $P < R$

Answer: C. Why: $P \geq Q$ and $Q < R$. Chaining "at least" with "less than" doesn't fix P relative to R : e.g., $P=Q=5, R=6$ gives PR . No definite relation exists.

Q3. [Medium] [Trap] Given: $A > B > C = D \geq E$. Does " $A > E$ " definitely follow?

A. Yes, always B. No — only $A \geq E$ is guaranteed C. No relation can be determined D. $A < E$ is guaranteed instead

Answer: B. Why: The path $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ has links $(>, >, =, \geq)$. Since the final link is non-strict $(\geq)$, the combined result is only $A \geq E$, not strict $A > E$.

Q4. [Medium] Given: $X > Y$, $Z > Y$. Which conclusion definitely follows?

A. $X > Z$ B. $Z > X$ C. $X = Z$ D. No definite relation between X and Z

Answer: D. Why: X and Z are each compared only to the common element Y , never to each other — a classic "shared anchor" trap. Both being greater than Y says nothing about their relative order.

Q5. [Medium] "What is the value of x ?" Statement I: $x + 5 = 12$. Statement II: $x^2 = 49$.

A. I alone sufficient, II not B. II alone sufficient, I not C. Both together needed D. Neither alone nor together sufficient

Answer: A. Why: Statement I: $x = 7$, unique — sufficient. Statement II: $x = 7$ or $x = -7$, two possible values — NOT sufficient (the classic quadratic trap).

Q6. [Medium] "Is n divisible by 3?" Statement I: n is divisible by 9. Statement II: n is divisible by 4.

A. I alone sufficient, II not B. II alone sufficient, I not C. Both together needed D. Each alone is sufficient

Answer: A. Why: Divisible by 9 ($=3 \times 3$) guarantees divisible by 3 — definite YES, sufficient alone. Divisible by 4 says nothing about divisibility by 3 (e.g., 12 is divisible by 4 and by 3, but 16 is divisible by 4 and NOT by 3) — insufficient alone.

Q7. [Hard] "What is the value of y ?" Statement I: $x + y = 10$. Statement II: $x - y = 2$.

A. I alone sufficient B. II alone sufficient C. Both together sufficient, neither alone D. Not sufficient even together

Answer: C. Why: Each equation alone has infinitely many (x,y) solutions. Together: adding the equations gives $2x = 12$, $x = 6$, then $y = 4$ — a unique value. Neither alone works, but together they do.

Q8. [Hard] [Trap] "What is the value of x ?" Statement I: $x^2 - 5x + 6 = 0$. Statement II: x is a positive number.

A. I alone sufficient B. II alone sufficient C. Both together sufficient D. Not sufficient even together

Answer: D. Why: Statement I factors to $(x-2)(x-3)=0$, so $x = 2$ or $x = 3$ — not unique. Statement II only says x is positive; both 2 and 3 are positive, so combining still leaves two possible values. Not sufficient even together — the quadratic trap survives combination.

Q9. [Hard] "Is $x > y$?" Statement I: $x > 0$. Statement II: $y < 0$.

A. I alone sufficient B. II alone sufficient C. Both together sufficient, neither alone D. Each alone is sufficient

Answer: C. Why: Statement I alone doesn't fix y 's sign (y could be positive and larger than x). Statement II alone doesn't fix x 's sign similarly. Together: x is positive and y is negative, so $x > 0 > y$ is guaranteed — a definite YES. Sufficient only together.

Q10. [Hard] Decoded chain: $A > B = C > D \geq E > F$. Which conclusion is definitely true?

A. $A > F$ B. $B > D$ C. $D > F$ D. $A = F$

Answer: B. Why: Path $B \rightarrow C \rightarrow D$ has links $(=, >)$, both strict/equality — so $B > D$ definitely follows. (Check A: path $A \rightarrow \dots \rightarrow F$ crosses the non-strict $D \geq E$ link, so only $A \geq F$ is guaranteed, not strict $A > F$ — eliminates option A.)

Q11. [Hard] "What is the perimeter of a rectangle?" Statement I: its length is 10 cm. Statement II: its area is 80 sq cm.

A. I alone sufficient B. II alone sufficient C. Both together sufficient, neither alone D. Not sufficient even together

Answer: C. Why: Length alone doesn't give width. Area alone allows many length-width pairs (e.g., 10×8 , 20×4 , each with different perimeter) — insufficient alone. Together: length=10 and area=80 gives width=8 uniquely, so perimeter $= 2(10+8)=36$ is fixed. Sufficient only together.

Part D — Non-Verbal, Spatial & Analytical Reasoning

- D1. Non-Verbal Reasoning — Image Analysis, Mirror & Water Images
- D2. Non-Verbal Reasoning — Grouping, Shape Construction, Paper Folding & Cutting
- D3. Non-Verbal Reasoning — Series, Classification, Embedded Figures, Figure Matrix & Pattern Completion
- D4. Analytical Reasoning — Counting Figures (Squares, Rectangles, Triangles, Lines)
- D5. Cubes & Dice

D1. Non-Verbal Reasoning — Image Analysis, Mirror & Water Images

D1.1 Mirror Images [Level: Basic]

- A **mirror image** is the reflection of a figure/letter/word across a **VERTICAL** line (mirror held upright, to the left or right of the object).
- Effect: **LEFT-RIGHT reversal** (horizontal flip). Top and bottom stay the same; what was on the left moves to the right and vice versa.
- For a **word**, both things happen together: (a) the **ORDER** of letters reverses (last letter becomes first), **AND** (b) each individual letter is itself mirrored.

Uppercase letters that look IDENTICAL after a mirror (vertical-flip) reflection:

A H I M O T U V W X Y

(all of these have a vertical line of symmetry)

- **TRICK:** Memorize this exact 11-letter list ("AHIMOTUVWXY" — spell it as a nonsense word to recall it fast). Any letter NOT in this list changes appearance when mirrored (e.g., B, C, D, E, F, G, J, K, L, N, P, Q, R, S, Z all look different or flip into a different/no valid letter).

D1.2 Water Images [Level: Basic]

- A **water image** is the reflection of a figure across a **HORIZONTAL** line (as if the object sits above a still water surface and you see its reflection below).
- Effect: **TOP-BOTTOM reversal** (vertical flip). Left and right stay the same; what was on top moves to the bottom.

Uppercase letters that look IDENTICAL after a water (horizontal-flip) reflection:

B C D E H I K O X

(all of these have a horizontal line of symmetry)

- **TRAP:** The mirror-invariant list (AHIMOTUVWXY) and the water-invariant list (BCDEHIKOX) are **DIFFERENT** — only H, I, O, X are common to both (they have BOTH a vertical and horizontal symmetry line). Do not assume a letter unchanged by a mirror is also unchanged by a water reflection, or vice versa — check each list separately.
- For a word's water image: the letters do NOT reorder (reading direction left-to-right is preserved), but each letter individually flips top-to-bottom, and the whole word appears "below" as if stacked upside-down vertically (the word is typically also shown upside-down/rotated 180° in many textbook conventions — always check what the specific question's convention states, but the core rule tested is the per-letter top-bottom flip).

D1.3 Mirror/Water Images of Digits [Level: Basic-Medium]

Digit mirror-image behavior (vertical flip, standard digital/seven-segment style font):

0, 1, 8 → look the same (vertical line of symmetry)

2, 3, 4, 5, 6, 7, 9 → look different / distorted

Note: exact appearance is FONT-DEPENDENT for digits — always go with what the specific question's font/convention shows rather than assuming a universal rule.

- **TRICK:** For digit-clock and digit-string mirror questions, 0, 1, 8 are the safe "always symmetric" anchors — use them to quickly check if a proposed mirror answer is even plausible.

D1.4 Mirror & Water Image of Clock Time [Level: Medium]

A very common combination question (bridges this chapter with the Clocks chapter):

Mirror image of a clock time = subtract the given time from 11:60 (i.e., 12:00)

Water image of a clock time = subtract the given time from 5:60 (i.e., 6:00)

[applies when the clock's 12-6 axis is vertical / 9-3 axis is horizontal, the standard clock face]

- Worked example: Time shown = 4:20.
- Mirror image = 11:60 – 4:20 = **7:40**.
- Water image = 5:60 – 4:20 = **1:40**.
- **TRAP:** Students often subtract from 12:00 for the water image too — remember mirror uses 12:00, water uses 6:00, because the mirror reflects across the vertical 12-6 axis (swapping left/right of the dial) while the water image reflects across the horizontal 9-3 axis (swapping top/bottom).

D1.5 Rotation of Figures [Level: Medium-Hard]

- Rotating a figure by 90°/180°/270° (clockwise or anticlockwise) about its center.
- Systematic method: pick 2-3 clearly identifiable reference points/corners of the figure (e.g., "the notch," "the longest edge"), mentally track where EACH one moves under the stated rotation, then redraw/compare.
- Shortcut equivalences: a 180° rotation (either direction) = flipping BOTH left-right AND top-bottom simultaneously (i.e., mirror + water image combined). A 90° clockwise rotation is the same end-state as a 270° (=90°×3) anticlockwise rotation — use whichever is easier to visualize.
- **TRAP:** Rotation is NOT the same as reflection — a rotated letter/figure and its mirror image usually look different (a rotation preserves "handedness"/chirality, a reflection reverses it). Do not substitute one for the other.

Questions

(ordered Easy → Hard; [Trick] tags the shortcut-list application, [Trap] tags mirror/water confusion)

Q1. [Easy] Which of these letters looks the SAME as its own mirror image (vertical flip)?

A. B B. E C. H D. S

Answer: C. **Why:** H is in the mirror-invariant list (AHIMOTUVWXY); B, E, S are not.

Q2. [Easy] Which of these letters looks the SAME as its own water image (horizontal flip)?

A. F B. D C. N D. Z

Answer: B. **Why:** D is in the water-invariant list (BCDEHIKOX); F, N, Z are not.

Q3. [Medium] [Trap] Which letter looks unchanged under BOTH a mirror image AND a water image?

A. A B. O C. T D. C

Answer: B. Why: O appears in both lists (AHIMOTUVVWXY for mirror, BCDEHIKOX for water) — only H, I, O, X have this dual property; A, T are mirror-only, C is water-only.

Q4. [Medium] [Trick] What is the mirror image (letter order and letter shapes) of the word "TOOT"?

A. TOOT B. TUOT C. TOOT reversed as "TOOT" D. Cannot be determined

Answer: A. Why: Reverse the letter order: T-O-O-T reversed is still T-O-O-T (it's a palindrome), and every individual letter (T, O) is in the mirror-invariant list, so each letter also looks unchanged. Net result: identical to the original — "TOOT".

Q5. [Medium] A clock shows 3:50. What is its mirror image time?

A. 8:10 B. 7:10 C. 8:70 D. 9:10

Answer: A. Why: Mirror image = $11:60 - 3:50 = 8:10$.

Q6. [Medium] A clock shows 2:15. What is its water image time?

A. 4:45 B. 3:45 C. 3:15 D. 4:15

Answer: B. Why: Water image = $5:60 - 2:15 = 3:45$.

Q7. [Hard] [Trap] A clock shows 9:20. A student computes both its mirror and water image but mixes up the two base times. What are the CORRECT mirror and water images respectively?

A. Mirror = 2:40, Water = 8:40 B. Mirror = 2:40, Water = 3:40 C. Mirror = 3:40, Water = 2:40 D. Mirror = 8:40, Water = 2:40

Answer: A. Why: Mirror = $11:60 - 9:20 = 2:40$. Water = $5:60 - 9:20 \rightarrow$ since $9:20 > 5:60$, add 12 hours worth of minutes conceptually: treat as $(17:60 - 9:20) = 8:40$ (i.e., wrap using a 12-hour water-axis calculation: $5:60 + 12:00 - 9:20 = 8:40$). The key trap is using 12:00 for both instead of 12:00 (mirror) vs 6:00 (water).

Q8. [Hard] The word "FOX" is written and its water image is taken. Which letters (if any) change appearance, and does the letter order reverse?

A. Order reverses; F, O, X all change B. Order preserved; O and X unchanged, F changes C. Order preserved; all three unchanged D. Order reverses; only O unchanged

Answer: B. Why: Water image preserves letter ORDER (unlike a mirror image, which reverses order). Checking each letter against the water-invariant list (BCDEHIKOX): O and X are in the list (unchanged), F is NOT in the list (changes appearance).

Q9. [Hard] A figure is rotated 180° . Which single combined operation from this chapter produces the identical end result?

A. Mirror image only B. Water image only C. Mirror image followed by water image (or vice versa) D. No combination of mirror/water reproduces a 180° rotation

Answer: C. Why: A 180° rotation flips both left-right AND top-bottom simultaneously — exactly the combined effect of applying a mirror flip and a water flip together (in either order).

Q10. [Hard] [Trick] Without drawing, determine: does the letter "N" look the same after (a) a mirror image, (b) a water image, (c) a 180° rotation?

A. Same in all three B. Same only in (c) C. Different in all three D. Same in (a) and (b), different in (c)

Answer: B. Why: N is in neither the mirror list (AHIMOTUVVWXY) nor the water list (BCDEHIKOX), so (a) and (b) both change its appearance — but N has 180° -rotational symmetry (point symmetry), so rotating it 180° reproduces the same N. This is a good example of a letter that is NOT mirror- or water-invariant but IS rotation-invariant, showing the three properties are independent.

D2. Non-Verbal Reasoning — Grouping, Shape Construction, Paper Folding & Cutting

D2.1 Grouping of Images / Classification by Shape Property [Level: Basic]

- Given a set of figures, find which ones share a common structural rule (and/or spot the one that breaks it).
- General method:** systematically check each figure against a fixed checklist of attributes — number of sides, number of enclosed/closed regions, number of line segments used, symmetry (does it have a line of symmetry?), open vs closed figure, curved vs straight edges — and identify which attribute is CONSTANT across most figures and which one differs for the odd one(s) out.
- TRICK: Work the checklist in the same order every time (sides → regions → symmetry → curve/straight) rather than randomly — this prevents missing the one attribute that actually varies.

D2.2 Shape Construction (Figure Assembly) [Level: Basic-Medium]

- Given several small piece-shapes, determine which combination assembles into a stated target shape (or which target shape CANNOT be formed from the given pieces).
- Method:** treat it as a jigsaw-matching problem — check that (a) edge LENGTHS of adjoining pieces match, (b) ANGLES at each join are compatible with the target's corners, and (c) the TOTAL AREA of the pieces equals the target's area (a quick elimination check: if total piece area $\neq$ target area, that combination is immediately wrong, without needing to check the geometry at all).
- TRICK: The area-matching check is the fastest elimination filter — compute total area of candidate pieces first before trying to mentally "assemble" them.

D2.3 Paper Folding [Level: Medium]

- A sheet of paper is folded one or more times (each fold is described, e.g., "folded in half vertically," "folded diagonally," "folded again in half"), then a hole is punched through all the folded layers at a stated location, and the paper is unfolded — find the resulting hole pattern.
- Core rule:** the punched hole is reproduced, as a MIRROR reflection, across every fold crease that was made. Each layer that was folded over the punch point gets its own reflected copy of the hole when unfolded.
- Systematic method (work backwards):**
 - Note the LAST fold made (the one closest to the punch).
 - Reflect the punch point across that fold's crease line to get a second hole.
 - Move to the SECOND-to-last fold; reflect BOTH holes found so far across that crease.
 - Repeat for every fold, working backwards to the very first fold, doubling the hole-count at each step (n folds → up to 2^n holes, fewer if some reflected positions coincide).
- TRAP:** A very common error is reflecting the punch point across only the LAST fold and stopping — you must reflect across **every** fold line, working backwards through the entire fold sequence, or you will under-count the holes (typically getting 2 holes instead of the correct 4, 8, etc.).

D2.4 Paper Cutting [Level: Medium]

- Identical logic to paper folding, but a SHAPE (not just a point) is cut from the folded edge. The cut shape reappears as a mirror-repeated pattern across each fold line, using the same backwards-reflection method as D2.3.
- Worked example: A square sheet is folded in half vertically (crease down the middle), then folded in half again horizontally (crease across the middle), giving a smaller square with 2 fold lines. A small triangular notch is cut from the corner that is diagonally opposite both folded edges (the only "free" corner). When unfolded:
- Reflecting the notch across the horizontal crease (last fold) gives 2 notches (one above, one below that crease).
- Reflecting both of those across the vertical crease (first fold) gives 4 notches total — one in each quadrant of the original square, all at mirror-symmetric positions relative to both crease lines.
- TRICK: With n folds each roughly halving the paper, the final unfolded sheet generally shows 2^n copies of the cut/hole (unless the punch/cut sits exactly ON a fold line, in which case some copies coincide and you get fewer than 2^n distinct marks).

D2.5 Dot Situation Problems [Level: Medium-Hard]

- A figure is described as several overlapping regions (e.g., a circle, a triangle, and a square, overlapping in various described ways). A dot is placed satisfying stated conditions (e.g., "inside the circle and the triangle, but outside the square"). Task: identify which of several alternative figure-descriptions is consistent with that same dot condition.
- Method:** this is a SET-MEMBERSHIP matching problem —
 - From the original figure's description, determine every overlapping region (which shapes overlap with which, and the region where all three overlap, if any).
 - Note precisely which regions the dot occupies (a set of "inside"/"outside" flags, one per shape).
 - For each alternative figure, check whether a region with that EXACT same set of "inside"/"outside" flags actually exists in that figure's overlap structure — only an alternative containing a genuinely matching region is correct.
- TRAP:** A region satisfying "inside A and B" is not the same as "inside A, B, and C" — always match the dot's condition on ALL stated shapes simultaneously, not just the ones explicitly mentioned as "inside" (an unmentioned shape may still need to be checked as "outside").

Questions

(ordered Easy → Hard; [Trick] tags a shortcut, [Trap] tags a common error)

Q1. [Easy] Five figures are shown: a pentagon, a hexagon, a heptagon, an octagon, and a triangle with one side replaced by a curve. Which is the odd one out?

A. Pentagon B. Hexagon C. The triangle-with-a-curved-side D. Octagon

Answer: C. **Why:** The other four are all closed figures made ENTIRELY of straight sides (polygons); the fifth includes a curved edge, breaking the "all straight sides" pattern shared by the rest.

Q2. [Easy] [Trick] Four piece-shapes have areas 4, 6, 9, and 5 square units. A target shape has area 20 square units. Can all four pieces combine to exactly form the target (assuming perfect edge-matching were otherwise possible)?

A. Yes, since $4+6+9+5=24$ which is more than enough B. No, since $4+6+9+5=24 \neq 20$ C. Cannot be determined D. Yes, since some pieces can overlap

Answer: B. **Why:** Total piece area = $4+6+9+5 = 24$, which does NOT equal the target's 20 — the fast area-matching elimination check immediately rules this out, no need to check edge-matching at all.

Q3. [Medium] A square sheet is folded in half once (one vertical crease) and a hole is punched away from the crease. How many holes appear when unfolded?

A. 1 B. 2 C. 4 D. Depends on hole size

Answer: B. Why: One fold $\rightarrow$ reflect the punch across that single crease $\rightarrow 2^n = 2^1 = 2$ holes.

Q4. [Medium] [Trap] A sheet is folded twice (two creases), and a student reflects the punch hole across only the second (last) fold, getting 2 holes, and stops. What is the error, and how many holes should actually appear?

A. No error; 2 is correct B. Must also reflect across the first fold; correct answer is 4 holes C. Must reflect 3 times; correct answer is 8 holes D. Folding twice always gives exactly 3 holes

Answer: B. Why: Every fold line must be reflected across, working backwards through the whole sequence — 2 folds $\rightarrow 2^2 = 4$ holes, not 2.

Q5. [Medium] A rectangular sheet is folded in half (vertical crease), then in half again (horizontal crease), then a hole is punched at the exact center of the resulting small rectangle — a point that lies ON both crease lines when unfolded conceptually would coincide. How many DISTINCT holes appear when fully unfolded?

A. 4 B. 2 C. 1 D. 8

Answer: C. Why: A punch exactly at the intersection of both fold lines reflects onto itself at every step (its reflected position across each crease is itself, since it lies exactly on both creases) — so all "copies" coincide at a single point, giving only 1 distinct hole, despite 2 folds normally suggesting $2^2=4$.

Q6. [Hard] A figure consists of a circle and a square overlapping partially, plus a triangle overlapping only the square (not the circle). A dot is placed inside both the circle and the square, but outside the triangle. Which region does the dot occupy?

A. Circle-only region B. Square-only region C. The circle-square overlap region, outside the triangle D. The triangle-square overlap region

Answer: C. Why: The dot must be inside BOTH circle and square simultaneously (their overlap region) while being outside the triangle — this uniquely identifies the circle-square (but not triangle) region.

Q7. [Hard] [Trap] Using the same figure as Q6, an alternative answer figure shows a region satisfying only "inside circle and square" without checking the triangle. Why is this alternative NOT necessarily a correct match?

A. It is always correct since circle and square are the shapes explicitly mentioned B. The triangle condition ("outside triangle") must also be verified, since the triangle overlaps the square — part of the circle-square region could also fall inside the triangle C. Triangles never affect circle-square regions D. Dot situation problems never involve three shapes

Answer: B. Why: Since the triangle overlaps the square (per the setup), the circle-square region may be split into a part that's also inside the triangle and a part that isn't — an alternative must match the FULL condition (inside circle, inside square, AND outside triangle), not just the two explicitly-named shapes.

Q8. [Hard] A square sheet is folded into quarters (2 perpendicular creases, giving 4 layers), and a triangular notch is cut from the single free corner of the folded quarter-square. How many notches appear on the fully unfolded sheet, and what is their arrangement?

A. 2 notches, diagonally opposite B. 4 notches, one per quadrant, each mirror-symmetric about both crease lines C. 8 notches D. 1 notch, since cutting is different from punching

Answer: B. Why: Two fold lines $\rightarrow 2^2 = 4$ reflected copies of the cut, one landing in each quadrant of the original square, each positioned as the mirror image of the original notch across both crease lines (same mechanism as paper folding, D2.3/D2.4 worked example).

D3. Non-Verbal Reasoning — Series, Classification, Embedded Figures, Figure Matrix & Pattern Completion

D3.1 The Universal Attribute Checklist [Level: Basic — apply this to every sub-topic below]

Every non-verbal reasoning problem in this chapter is really the same underlying skill: spot what changes vs what stays constant across a set of figures. Always scan attributes in this fixed order so nothing is missed:

1. Number of sides / number of elements (dots, lines, shapes)
2. Size (growing / shrinking)
3. Shading or color
4. Rotation / orientation
5. Position (of a sub-element within the figure)
6. Shape/type change (e.g., circle becomes square)

- TRICK: Go down this checklist top-to-bottom every single time, even if the answer seems obvious after step 1 — many "hard" questions have TWO simultaneously-changing attributes, and stopping early after spotting the first one is the most common error in this entire chapter.

D3.2 Non-Verbal Series / Pattern Recognition [Level: Basic-Medium]

- A sequence of figures changes step-by-step by one (or more) consistent rule(s): rotation by a fixed angle each step, a shape gaining/losing an element, a shading pattern cycling, or a size trend.
- **Method:** compare each CONSECUTIVE pair of figures (not just first-vs-last) to isolate the exact rule, confirm it holds for every consecutive pair in the given sequence, then apply it one more time to predict the next figure.
- TRICK: If the "obvious" rule from figures 1→2 doesn't also hold for 2→3, look for a rule that alternates every other step (a common way to make a series look harder than it is) — treat odd-numbered and even-numbered figures as two interleaved sub-series if a single simple rule doesn't fit all steps.

D3.3 Classification (Odd-One-Out) [Level: Basic-Medium]

- Same attribute-checklist method as D3.1/D2.1, but applied to find the ONE figure that breaks a rule shared by all the others.
- **TRAP:** The "odd one out" is sometimes odd in a SUBTLE way (e.g., 4 figures have an even number of sides and 1 has an odd number, even though all 5 otherwise look visually similar) — don't classify purely on "looks similar/different at a glance"; always run the full checklist.

D3.4 Embedded / Hidden Figures [Level: Medium]

- A simple target figure is hidden as a sub-part within one or more complex figures; find which complex figure(s) contain it.
- **Method:** count the target figure's exact vertex count and line-segment count first, then scan each candidate complex figure for a subset of lines matching that same vertex/segment count AND the same relative proportions/angles (not just "a similar-looking shape somewhere in there") — a genuine match must preserve the target's exact structure, not just a rough resemblance.
- TRICK: Trace the target shape's outline with a finger (or mentally, one continuous path) and try to find that exact continuous path within the complex figure — random shapes that only superficially resemble the target will fail this continuous-path test.

D3.5 Figure Matrix (Figure Analogy in Grid Form) [Level: Medium-Hard]

- A 2×2 or 3×3 grid of figures follows a rule ACROSS EACH ROW and (often independently) DOWN EACH COLUMN simultaneously. One cell (usually bottom-right) is missing; find it.
- **Method:** first determine the row-wise rule using the two complete rows, confirm the same TYPE of rule applies down the columns (rotation/addition/shading, etc. — the row-rule and column-rule need not be identical in kind, but each must be internally consistent), then apply BOTH the row-rule (using the two known cells in the target's row) and the column-rule (using the two known cells in the target's column) — a correct answer must satisfy both simultaneously.
- **TRAP:** Checking only the row (or only the column) and picking the first option that fits is a common shortcut that produces wrong answers when the two rules interact — always verify the candidate answer against BOTH the row and the column before finalizing.

D3.6 Rule Detection [Level: Medium]

- The general skill of identifying which single logical rule governs a set of figures — this is really D3.1's checklist applied as its own named category, often tested by asking "which rule best describes this set" as a standalone question rather than asking for a missing/odd figure directly.

D3.7 Pattern Completion / Figure Completion [Level: Medium-Hard]

- A single figure has a symmetric or repeating internal pattern, with one piece missing; find the piece that correctly completes it.
- **Method:** identify the figure's symmetry axis (or its repeat unit, if the pattern tiles/repeats rather than mirrors), then determine exactly what the missing piece must contain so the full figure satisfies that same symmetry/repetition when reassembled.
- TRICK: If the figure has rotational symmetry (not just mirror symmetry), the missing piece is found by rotating an existing piece by the pattern's rotation angle — don't default to assuming mirror symmetry when the pattern is actually rotational.

D3.8 Visual Reasoning — General Transformation Types [Level: Hard]

Spatial-aptitude questions across this entire chapter draw on a fixed set of named transformation types — recognizing WHICH transformation(s) a question is testing narrows down the method to use immediately:

Translation (shift position, shape/size/orientation unchanged)
 Scaling (grow/shrink, shape unchanged)
 Rotation (turn about a center point)
 Mirroring (reflect across an axis)
 Assembling (combine pieces into a whole – D2.2)
 Grouping (classify by shared property – D2.1/D3.3)
 Paper Cutting / Paper Folding (D2.3/D2.4)
 Pattern in 2D/3D (repeating/symmetric structure – D3.7)

- **TRICK:** Before attempting to solve, first name which ONE (or two) of these 8 transformation types the question is actually testing — this instantly tells you which method from this chapter to apply, rather than re-deriving an approach from scratch each time.

Questions

(ordered Easy → Hard; [Trick] tags the checklist/method shortcut, [Trap] tags a common error)

Q1. [Easy] A series of figures: a triangle, then a triangle with 1 dot inside, then a triangle with 2 dots inside, then a triangle with 3 dots inside. What comes next?

A. A square with 4 dots B. A triangle with 4 dots inside C. A triangle with 3 dots (repeat) D. A triangle with 0 dots

Answer: B. **Why:** The shape (triangle) stays constant; only the dot-count increases by 1 each step (0,1,2,3,...) — next is 4 dots, same triangle.

Q2. [Easy] Five shapes: square, rectangle, rhombus, trapezium, circle. Which is the odd one out?

A. Square B. Rectangle C. Circle D. Rhombus

Answer: C. **Why:** The first four are all quadrilaterals (4 straight sides); the circle has no straight sides or vertices at all.

Q3. [Medium] A figure rotates 90° clockwise at each step across 4 given figures, and also gains one additional shaded corner each step. A student identifies only the rotation and picks an answer that rotates correctly but has the wrong number of shaded corners. What error did they make?

A. No error, rotation is the only rule that matters B. They stopped at step 1 of the attribute checklist and missed the simultaneously-changing shading attribute C. Rotation and shading cannot both change in a valid series D. The answer should have MORE shaded corners than predicted

Answer: B. **Why:** This is exactly the D3.1 trap — two attributes (rotation AND shading count) change simultaneously; verifying only the first one found is insufficient.

Q4. [Medium] [Trick] In a number/letter analogy-style embedded-figure question, a target shape has 5 vertices and 5 line segments (a pentagon outline). A candidate complex figure contains a 5-sided region, but tracing it shows its 5 sides are not all connected in one continuous path (one "side" is actually two separate disconnected segments). Does this candidate contain the embedded target?

A. Yes, since it has 5 sides B. No — the continuous-path test fails, so it is not a genuine embedding of the pentagon outline C. Cannot be determined without more information D. Yes, since vertex count alone is sufficient

Answer: B. **Why:** A genuine embedded figure must trace as ONE continuous path matching the target's structure exactly — a region that merely has the right side-count without a continuous connected outline is not a true match (D3.4 method).

Q5. [Medium] A 2×2 figure matrix: top-left = 1 circle, top-right = 2 circles, bottom-left = 1 square, bottom-right = ?. The row rule is "add one more of the same shape going left to right." What is the missing cell?

A. 1 square B. 2 circles C. 2 squares D. 1 circle

Answer: C. **Why:** Applying the row rule (established from the top row: left→right adds one more of the same shape) to the bottom row (1 square → 2 squares) gives 2 squares.

Q6. [Hard] [Trap] In a 3×3 figure matrix, a student finds an option that correctly continues the row pattern and immediately selects it without checking the column. The grid's actual rule requires BOTH row and column consistency. What is the risk?

A. None — row-consistency alone always guarantees correctness B. The chosen option might satisfy the row rule but violate the independent column rule, making it wrong despite "looking right" for the row C. Columns are never tested in figure matrices D. Only the middle column ever matters

Answer: B. **Why:** This is the D3.5 trap explicitly — a matrix answer must be verified against BOTH the row rule and the column rule; satisfying only one is not sufficient.

Q7. [Hard] A pattern-completion figure is a circle divided into 6 equal sectors, alternately shaded and unshaded going around (shaded, unshaded, shaded, unshaded, shaded, unshaded), with one sector's shading state hidden/missing. The missing sector is the 4th one going clockwise from a fixed start. What should it be?

A. Shaded B. Unshaded C. Half-shaded D. Cannot be determined

Answer: B. **Why:** The pattern alternates strictly: sectors 1,3,5 = shaded and 2,4,6 = unshaded (or vice versa depending on the start) — sector 4 falls in the "unshaded" parity group, so it must be unshaded to preserve the alternating (rotationally-patterned) rule.

Q8. [Hard] A question describes a shape moving to a new position on a grid without changing its size, orientation, or shading. Which single transformation type (from the D3.8 list) does this test?

A. Rotation B. Scaling C. Translation D. Mirroring

Answer: C. **Why:** A pure position-shift with no change to size/orientation/shading is, by definition, a Translation — naming the transformation type immediately tells you no angle or size calculation is needed, only position-tracking.

Q9. [Hard] [Trick] A figure series shows a shape's size shrinking by half at each step while ALSO rotating 45° clockwise each step. Naming the D3.8 transformation types involved, which TWO are simultaneously at play, and what is the correct approach?

A. Only Rotation; size is irrelevant B. Only Scaling; rotation is irrelevant C. Both Scaling and Rotation are simultaneously active — track each independently (halve the size AND add 45° rotation) before comparing to the answer options D. This combination is impossible in a valid series

Answer: C. **Why:** Naming both active transformation types (per D3.8's method) makes clear that BOTH the size-halving rule and the 45°-rotation rule must be applied together to the last given figure to predict the next one — treating them independently avoids missing either change (same principle as the D3.1/Q3 checklist trap).

D4. Analytical Reasoning — Counting Figures (Squares, Rectangles, Triangles, Lines)

(No images are reproducible from the source for this topic — every method below is taught as a fully systematic, describable procedure so it can be applied to any figure described in words.)

D4.1 Counting Squares in a Grid [Level: Basic-Medium]

- A grid of size $n \times n$ (unit cells) contains squares of every size from 1×1 up to $n \times n$.

Number of $(n-k+1) \times (n-k+1)$... simpler form:

Count of 1×1 squares = n^2

Count of 2×2 squares = $(n-1)^2$

Count of 3×3 squares = $(n-2)^2$

...

Count of $n \times n$ squares = 1^2

Total squares = $n^2 + (n-1)^2 + (n-2)^2 + \dots + 1^2 = \frac{n(n+1)(2n+1)}{6}$

- TRICK: You don't need to derive the sum-of-squares formula each time — just recall $\frac{n(n+1)(2n+1)}{6}$.

Worked Example

A 4×4 grid of unit squares. Total squares?

Sizes: $1 \times 1 \rightarrow 16$, $2 \times 2 \rightarrow 9$, $3 \times 3 \rightarrow 4$, $4 \times 4 \rightarrow 1$. Sum = $16+9+4+1 = 30$.

Check via formula: $\frac{4(5)(9)}{6} = 180/6 = 30$. ✓

- TRAP:** This formula only applies to a **square** grid ($n \times n$). For a rectangular grid ($m \times n$, $m \neq n$), you must count square-sizes up to $\min(m, n)$ individually: size $k \times k$ count = $(m-k+1)(n-k+1)$, summed for $k=1$ to $\min(m, n)$.

D4.2 Counting Rectangles in a Grid [Level: Medium]

- Core insight: any rectangle in a grid is uniquely determined by choosing **2 horizontal lines** and **2 vertical lines** from the grid's line network.

For an $m \times n$ grid of unit cells:

Horizontal lines available = $m + 1$

Vertical lines available = $n + 1$

Total rectangles = $C(m+1, 2) \times C(n+1, 2)$

- TRICK: This "choose 2 lines from each direction" insight also re-derives the square-counting formula as a special case (restricting to equal horizontal/vertical spacing), but it's easier to just use the two formulas separately.

Worked Example

A grid with 3 rows and 4 columns of unit cells ($m=3$, $n=4$). Total rectangles (including squares)?

Horizontal lines = 4, Vertical lines = 5. Total = $C(4, 2) \times C(5, 2) = 6 \times 10 = 60$.

- TRAP:** "Rectangles" **includes** squares (a square is a special rectangle) — if a question asks for "rectangles that are NOT squares," subtract the square-count (from D4.1) from the total rectangle-count.

D4.3 Counting Triangles [Level: Medium-Hard]

- No single universal formula exists (unlike squares/rectangles) — instead use this **systematic strategy**:

- Count the smallest ("basic") triangles first — the ones not made up of any smaller triangle.
- Count triangles formed by combining exactly 2 adjacent basic regions.
- Continue with combinations of 3, 4, ... basic regions, until no more distinct triangles can be formed.
- Sum all counts across every size category.

- TRICK for the common "triangle divided by cevians from vertices" figure (e.g., a triangle with lines drawn from each vertex to a point on the opposite side, or a triangle with its 3 medians/altitudes drawn): systematically list, for each pair/triple of drawn lines, whether their intersection region forms a valid triangle — this is tedious but mechanical if done size-by-size as above rather than trying to "see" all triangles at once.
- TRAP:** The most common error is **double-counting** a triangle that looks different when traced in two directions, or **missing** a large triangle formed by outer boundary lines plus one internal line — always do a final pass checking the largest possible triangle (using only the figure's outer boundary) separately.

D4.4 Counting Straight Lines / Intersecting Lines [Level: Basic-Medium]

If n straight lines are drawn such that no 2 are parallel and no 3 meet at a single point:

Number of intersection points = $C(n, 2) = \frac{n(n-1)}{2}$

- Adjustment for parallel lines: each pair of PARALLEL lines contributes **zero** intersection points (they never meet) — subtract $C(k, 2)$ from the total for each group of k mutually parallel lines.
- Adjustment for concurrent lines: if 3 or more lines all pass through one common point, that point is counted only ONCE, not once per pair — subtract the "extra" duplicate counts: if k lines meet at one point, they would naively contribute $C(k, 2)$ intersection points but actually contribute only 1, so subtract $[C(k, 2) - 1]$.

Worked Example

6 straight lines are drawn, no two parallel, no three concurrent. How many intersection points?

$C(6, 2) = \frac{6 \times 5}{2} = 15$.

Same 6 lines, but now 3 of them all pass through a common point (and no other coincidences). How many distinct intersection points?

Naive total (all $C(6,2)=15$) – extra duplicates from the concurrent triple: those 3 lines would contribute $C(3,2)=3$ pairs but only 1 actual point, so subtract $(3-1)=2$.

Distinct points = $15 - 2 = 13$.

Questions

Q1. [Easy] How many total squares are there in a 3×3 grid of unit squares?

A. 9 B. 12 C. 14 D. 16

Answer: C. Why: $1 \times 1: 9, 2 \times 2: 4, 3 \times 3: 1$. Total = $9+4+1 = 14$. (Check: $3(4)(7)/6=14$.)

Q2. [Easy] How many total squares (of all sizes) are in a 5×5 grid?

A. 40 B. 55 C. 30 D. 25

Answer: B. Why: Formula $n(n+1)(2n+1)/6 = 5(6)(11)/6 = 55$.

Q3. [Medium] A grid has 2 rows and 3 columns of unit cells. How many total rectangles (including squares) are there?

A. 12 B. 15 C. 18 D. 24

Answer: C. Why: Horizontal lines = 3, vertical lines = 4. Total = $C(3,2) \times C(4,2) = 3 \times 6 = 18$.

Q4. [Medium] [Trap] Using the grid from Q3 (2×3 unit cells), how many rectangles are NOT squares?

A. 8 B. 10 C. 14 D. 18

Answer: B. Why: Squares in a 2×3 grid: $1 \times 1 \rightarrow 6, 2 \times 2 \rightarrow (2-1)(3-1)=2$. Total squares = 8. Non-square rectangles = 18 (from Q3) – $8 = 10$.

Q5. [Medium] 5 straight lines are drawn on a plane such that no two are parallel and no three meet at a point. How many points of intersection are formed?

A. 5 B. 10 C. 15 D. 20

Answer: B. Why: $C(5,2) = 5 \times 4/2 = 10$.

Q6. [Medium] [Trap] Out of 7 lines drawn (no coincidences), 3 of them are mutually parallel. How many intersection points result?

A. 21 B. 18 C. 15 D. 12

Answer: B. Why: Naive total $C(7,2)=21$. The 3 parallel lines contribute 0 intersections among themselves instead of $C(3,2)=3$, so subtract 3: $21-3=18$.

Q7. [Hard] A rectangular grid has 4 rows and 6 columns of unit cells. Find the total number of squares of all sizes.

A. 24 B. 40 C. 50 D. 55

Answer: C. Why: $\min(m,n)=4$. Sizes: $1 \times 1: 4 \times 6=24; 2 \times 2: 3 \times 5=15; 3 \times 3: 2 \times 4=8; 4 \times 4: 1 \times 3=3$. Total = $24+15+8+3 = 50$.

Q8. [Hard] In triangle ABC, a cevian is drawn from vertex A to a point D on side BC, and a second cevian is drawn from vertex B to a point E on side AC, with AD and BE crossing at an interior point F. How many total triangles are visible in the resulting figure?

A. 4 B. 6 C. 8 D. 10

Answer: C. Why: Systematically list every 3-sided region using only the 5 drawn lines (AB, BC, CA, AD, BE): the 3 basic regions **ABF, BFD, AFE** (the 4th basic region, quadrilateral FDCE, is not a triangle); the 2 half-splits from each single cevian, **ABD, ACD** (from AD) and **ABE, CBE** (from BE) — but ABD is a triangle formed by AB+BD(part of BC)+DA, so counting only NEW ones beyond the halves already listed: ABD, ACD, ABE, CBE (4 more); plus the outer triangle **ABC** itself (1 more). Total = 3 (basic) + 4 (single-cevian halves) + 1 (outer) = **8**. **TRAP:** it's easy to stop after counting only the "obvious" pieces (the 3 basic regions + outer = 4) and miss that each cevian ALSO independently creates 2 more valid half-triangles using the opposite side — always check triangles formed by just ONE cevian at a time, not only the fully-subdivided basic pieces.

Q9. [Hard] 8 straight lines are drawn such that no two are parallel, but there is one point where exactly 4 of them are concurrent (and no other coincidences). How many distinct intersection points are there?

A. 22 B. 25 C. 28 D. 23

Answer: D. Why: Naive total = $C(8,2) = 28$. The 4 concurrent lines would naively contribute $C(4,2)=6$ points but actually give only 1, so subtract $(6-1)=5$. Distinct points = $28-5 = 23$.

D5. Cubes & Dice

D5.1 Standard Dice Basics [Level: Basic]

- A standard (ordinary) six-faced die has faces numbered 1–6.
- Default assumption unless stated otherwise:** opposite faces always sum to **7** — so the three opposite pairs are **(1,6), (2,5), (3,4)**.
- A cube always has exactly **3 pairs of opposite faces** (commonly called Top-Bottom, Front-Back, Left-Right) — no matter how it's numbered or oriented.
- TRAP:** This "sum to 7" rule applies only to a **standard** die. Many reasoning questions use a **non-standard** die (arbitrary numbers) and expect you to *derive* the opposite pairs from given views — never assume sum-to-7 unless the question says "standard dice."

D5.2 Finding Opposite Faces from Multiple Views [Level: Medium]

- Core rule: any two faces shown **simultaneously visible** in the same view can **never** be opposite each other (you can never see two opposite faces of a cube at the same time).
- Ways to solve** — two complementary techniques:
 - Direct elimination:** for each view, rule out every pair of simultaneously-visible faces as "not opposite."
 - Elimination by remainder:** since there are exactly 3 opposite pairs among 6 numbers, once enough pairs are ruled out, the numbers left unpaired must be opposite each other **by process of elimination** — this is usually what actually solves the question, not the direct rule alone.

Worked Example

Two views of the same (non-standard) die are shown:

View 1: Top = 2, Front = 3, Right = 4

View 2: Top = 2, Front = 4, Right = 6

Find the face opposite to 3.

From View 1: {2,3,4} are mutually visible → none of these pairs are opposite: $2 \neq \text{opp}(3)$, $2 \neq \text{opp}(4)$, $3 \neq \text{opp}(4)$.

From View 2: {2,4,6} are mutually visible → $2 \neq \text{opp}(4)$ [already known], $2 \neq \text{opp}(6)$, $4 \neq \text{opp}(6)$.

So: $\text{opp}(2)$ must be from the untested remainder {1,5}. Likewise $\text{opp}(4)$ must be from {1,5} too (since 4 is ruled out against 2,3,6).

Since 2 and 4 cannot share the same opposite number, one of {2,4} pairs with 1 and the other with 5 — using up all four numbers {1,2,4,5} in two pairs.

That leaves exactly **3 and 6** unpaired → **by elimination, 3 is opposite 6**.

D5.3 Rolling / Rotating a Die [Level: Medium]

- Track all six faces (Top/Bottom/North/South/East/West) and update them after each roll:

Roll to the RIGHT: Top → East → Bottom → West → Top (North/South faces stay fixed)
 Roll FORWARD: Top → North → Bottom → South → Top (East/West faces stay fixed)

- TRICK: Only the two faces perpendicular to the rolling axis stay fixed each time — track just the other four.

D5.4 Dice Nets (Open Dice / Unfolding) [Level: Medium-Hard]

- A net is a flat, unfolded pattern of 6 squares that folds into a cube.
- TRICK (master rule):** in the common "cross" or "T" shaped net, two faces that lie in the **same row or column, exactly 2 squares apart (with exactly one square between them)** become **opposite** faces when folded.

Worked Example

Net layout (row and column of a cross-shaped net):

```

      [2]
[3] [1] [4] [6]
      [5]
```

Row (3,1,4,6): squares 2 apart are (3,4) and (1,6) → **3 & 4 are opposite; 1 & 6 are opposite**.

Column (2,1,5): squares 2 apart are (2,5) → **2 & 5 are opposite**.

(Sanity check: this is exactly the standard 1-6/2-5/3-4 pairing — nets built around a standard die will always reproduce it.)

D5.5 Cubes: The Painted-Cube Formula Set [Level: Basic-Hard]

- Setup: a large cube made of $n \times n \times n$ unit cubes has its entire outer surface painted, then is disassembled into unit cubes.
- This single reference block is the highest-yield content in the whole chapter — memorize it cold:**

```

Unit cubes with 3 faces painted (corners)      = 8                (always 8, for any n >= 2)
Unit cubes with 2 faces painted (edges)        = 12 × (n-2)
Unit cubes with 1 face painted (face centers)   = 6 × (n-2)^2
Unit cubes with 0 faces painted (fully inside) = (n-2)^3
```

Check: $8 + 12(n-2) + 6(n-2)^2 + (n-2)^3 = n^3$ (always — verify for $n=4$: $8+24+24+8=64=4^3$ (checked))

- TRAP:** These formulas require $n \geq 2$. For $n=1$ (a single unit cube), the entire cube is one piece with all 6 faces painted — none of the four categories above apply.
- TRICK for "at least" / "at most" questions: compute each category separately from the reference block, then **add** the relevant ones — e.g., "at least 2 faces painted" = (3-face count) + (2-face count).
- Variant: if different **pairs of opposite faces** are painted with different colors, the position-counting logic (corner/edge/face/internal) is identical — only the color labels on each category change, not the counts.

Questions

(ordered Easy → Hard; [Trick] tests a shortcut, [Trap] tests a classic misconception)

Q1. [Easy] On a standard die, if 1 is on top, what number is at the bottom?

A. 2 B. 5 C. 6 D. 4

Answer: C. **Why:** Standard opposite pairs sum to 7: opposite of 1 is 6.

Q2. [Easy] [Numerical] A cube of side 3 units ($n=3$) is painted on all faces and cut into 27 unit cubes. How many unit cubes have exactly 2 faces painted?

A. 8 B. 12 C. 6 D. 1

Answer: B. **Why:** Edge count = $12(n-2) = 12 \times 1 = 12$.

Q3. [Medium] [Numerical] A cube of side 5 units is painted and cut into 125 unit cubes. How many have exactly 1 face painted?

A. 24 B. 36 C. 54 D. 64

Answer: C. **Why:** Face count = $6(n-2)^2 = 6 \times (3)^2 = 6 \times 9 = 54$.

Q4. [Medium] [Numerical] A cube of side 6 units is painted and cut into unit cubes. How many have NO paint at all?

A. 27 B. 64 C. 16 D. 8

Answer: B. **Why:** Internal count = $(n-2)^3 = 4^3 = 64$.

Q5. [Medium] [Trap] A cube of side 4 units is painted and cut into 64 unit cubes. How many have AT LEAST one face painted?

A. 8 B. 24 C. 32 D. 56

Answer: D. **Why:** "At least 1" = total – (0-face count) = $64 - (4-2)^3 = 64 - 8 = 56$. (Or directly: corners+edges+faces = $8 + 12(2) + 6(2)^2 = 8 + 24 + 24 = 56$.)

Q6. [Medium] [Numerical] A cube of side 5 units is painted and cut into unit cubes. How many have AT MOST 1 face painted (i.e., 0 or 1 face)?

A. 54 B. 27 C. 81 D. 62

Answer: C. **Why:** 0-face = $(3)^3 = 27$; 1-face = $6 \times (3)^2 = 54$. Sum = $27 + 54 = 81$.

Q7. [Hard] Two views of a non-standard die are shown: View 1: Top=2, Front=3, Right=4. View 2: Top=2, Front=4, Right=6. What number is opposite to 3?

A. 1 B. 5 C. 6 D. Cannot be determined

Answer: C. **Why:** {2,3,4} mutually visible in View 1 and {2,4,6} in View 2 rule out all pairings among 2,3,4,6. Numbers 2 and 4 must each pair with one of the remaining {1,5}, leaving 3 and 6 — the only unpaired numbers — opposite each other by elimination.

Q8. [Hard] [Trick] A cube of side 10 units is painted on all faces and cut into 1000 unit cubes. How many unit cubes have exactly 2 faces painted?

A. 8 B. 96 C. 480 D. 64

Answer: B. **Why:** Edge count = $12(n-2) = 12 \times 8 = 96$.

Q9. [Hard] A cube's net (unfolded pattern) is laid out with squares labeled, in a row, as P-Q-R-S (four in a straight line), and squares T and U attached above and below Q respectively. Which faces become opposite when folded?

A. P & R, Q & S, T & U B. P & S, Q & U, R & T C. P & Q, R & S, T & U D. P & R only; T,U,Q,S indeterminate

Answer: A. **Why:** In the row P-Q-R-S, the pairs exactly 2 apart are (P,R) and (Q,S). In the column T-Q-U, the pair 2 apart is (T,U). All three opposite pairs are determined by the "2-apart in a straight line" net rule.

Q10. [Hard] [Trap] A cube of side 2 units is painted on all faces and cut into 8 unit cubes. How many have exactly 1 face painted?

A. 0 B. 6 C. 8 D. 4

Answer: A. **Why:** For $n=2$, $(n-2)=0$, so the 1-face formula $6(n-2)^2 = 6 \times 0 = 0$. In fact for $n=2$, every unit cube is a corner cube (all 8 have exactly 3 faces painted) — a boundary case worth checking formulas against directly rather than assuming they always give a "typical" spread.

Q11. [Hard] A large cube is painted and cut into 216 unit cubes. What was the value of n , and how many unit cubes have exactly 2 faces painted?

A. $n=6$; 96 B. $n=6$; 48 C. $n=5$; 54 D. $n=6$; 60

Answer: B. **Why:** $216 = 6^3$, so $n=6$. Edge count = $12(n-2) = 12 \times 4 = 48$.